Chapter 1

Visualization of the Silicon Crystal

1.1 (a) Please refer to Figure 1-2. The 8 corner atoms are shared by 8 unit cells and therefore contribute 1 atom. Similarly, the 6 face atoms are each shared by 2 unit cells and contribute 3 atoms. And, 4 atoms are located inside the unit cell. Hence, there are total 8 silicon atoms in each unit cell.

(b) The volume of the unit cell is

$$V_{unit\ cell} = (5.43\ A)^3 = (5.43\times10^{-8}cm)^3 = 1.60\times10^{-22}cm^3,$$

and one unit cell contains 8 silicon atoms. The atomic density of silicon is

$$N_{Si} = \frac{8\ silicon\ atoms}{V_{unit\ cell}} = 5.00\times10^{22}\ (silicon\ atoms)\ cm^{-3}.$$

Hence, there are 5.00×10^{22} silicon atoms in one cubic centimeter.

(c) In order to find the density of silicon, we need to calculate how heavy an individual silicon atom is

$$Mass_{1\ Si\ atom} = \frac{28.1\ (g/mole)}{6.02\times10^{23}\ (atoms/mole)} = 4.67\times10^{-23}(g/atom).$$

Therefore, the density of silicon (ρ_{Si}) in g/cm^3 is

$$\rho_{Si} = N_{Si}\times \mathrm{Mass}_{1\ Si\ atom} = 2.33\ g/cm^3.$$

Fermi Function

1.2 (a) Assume $E = E_f$ in Equation (1.7.1), f(E) becomes ½. Hence, the probability is ½.

(b) Set $E = E_c + kT$ and $E_f = E_c$ in Equation (1.7.1):

$$f(E) = \frac{1}{1+e^{[(E_c+kT)-E_c]/kT}} = \frac{1}{1+e^1} = 0.27.$$

The probability of finding electrons in states at $E_c + kT$ is 0.27.

* For Problem 1.2 Part (b), we cannot use approximations such as Equations (1.7.2) or (1.7.3) since $E-E_f$ is neither much larger than kT nor much smaller than -kT.

(c) f(E) is the probability of a state being filled at E, and 1-f(E) is the probability of a state being empty at E. Using Equation (1.7.1), we can rewrite the problem as

$$f(E = E_c + kT) = 1 - f(E = E_c + 3kT)$$
$$\frac{1}{1+e^{(E_c+kT-E_f)/kT}} = 1 - \frac{1}{1+e^{(E_c+3kT-E_f)/kT}}$$

where

$$1 - \frac{1}{1+e^{(E_c+3kT-E_f)/kT}} = \frac{1+e^{(E_c+3kT-E_f)/kT}-1}{1+e^{(E_c+3kT-E_f)/kT}} = \frac{e^{(E_c+3kT-E_f)/kT}}{1+e^{(E_c+3kT-E_f)/kT}}$$
$$= \frac{1}{1+e^{-(E_c+3kT-E_f)/kT}}.$$

Now, the equation becomes

$$\frac{1}{1+e^{(E_c+kT-E_f)/kT}} = \frac{1}{1+e^{-(E_c+3kT-E_f)/kT}}.$$

This is true if and only if

$$E_c + kT - E_f = -(E_c + 3kT - E_f).$$

Solving the equation above, we find

$$E_f = E_c + 2kT.$$

1.3 (a) Assume $E = E_f$ and $T > 0K$ in Equation (1.7.1). f(E) becomes ½. Hence, the probability is ½.

(b) f(E) is the probability of a state being filled at E, and 1-f(E) is the probability of a state being empty at E. Using Equation (1.7.1), we can rewrite the problem as

$$f(E = E_c) = 1 - f(E = E_v)$$
$$\frac{1}{1+e^{(E_c-E_f)/kT}} = 1 - \frac{1}{1+e^{(E_v-E_f)/kT}}$$

where

$$1-\frac{1}{1+e^{(E_c-E_f)/kT}}=\frac{1+e^{(E_v-E_f)/kT}-1}{1+e^{(E_v-E_f)/kT}}=\frac{e^{(E_v-E_f)/kT}}{1+e^{(E_v-E_f)/kT}}=\frac{1}{1+e^{-(E_v-E_f)/kT}}.$$

Now, the equation becomes

$$\frac{1}{1+e^{(E_c-E_f)/kT}}=\frac{1}{1+e^{-(E_v-E_f)/kT}}.$$

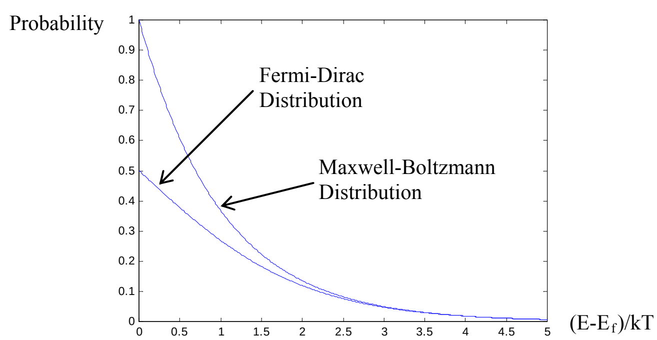
This is true if and only if

$$E_c-E_f=-(E_v-E_f).$$

Solving the equation above, we find

$$E_f=\frac{E_c+E_v}{2}.$$

(c) The plot of the Fermi-Dirac distribution and the Maxwell-Boltzmann distribution is shown below.



The Boltzmann distribution considerably overestimates the Fermi distribution for small $(E-E_f)/kT$. If we set $(E-E_f)/kT = A$ in Equations (1.7.1) and (1.7.2), we have

$$e^{-A}\le 1.10\left[\frac{1}{1+e^{A}}\right].$$

Solving for A, we find

$$e^{A}\ge\frac{1+e^{A}}{1.10}\quad\rightarrow\quad e^{A}\ge 10.11\quad\rightarrow\quad A\ge\ln(10.11)=2.31.$$

Therefore, the Boltzmann approximation is accurate to within 10% for $(E-E_f)/kT \ge 2.31$.

1.4 (a) Please refer to the example in Sec. 1.7.2. The ratio of the nitrogen concentration at 10 km above sea level to the nitrogen concentration at sea level is given by

$$\frac{N(N_2)_{10\ km}}{N(N_2)_{Sea\ Level}} = \frac{e^{-E_{10\ km}/kT}}{e^{-E_{Sea\ Level}/kT}} = e^{-(E_{10\ km} - E_{Sea\ Level})/kT}$$

where

$$\begin{aligned} E_{10\ km} - E_{Sea\ Level} &= altitude \times mass\ of\ N_2\ molecule \times acceleration\ of\ gravity \\ &= 10^6\ cm \times 28 \times 1.66 \times 10^{-24}\ g \times 980\ cm \cdot s^{-2} = 4.56 \times 10^{-14}\ erg. \end{aligned}$$

The ratio is

$$\frac{N(N_2)_{10\ km}}{N(N_2)_{Sea\ Level}} = e^{-(4.56\times10^{-14}\ erg)/(1.38\times10^{-16}\ erg\cdot K^{-1}\times 273\ K)} = e^{-1.21} = 0.30\ .$$

Since nitrogen is lighter than oxygen, the potential energy difference for nitrogen is smaller, and consequently the exponential term for nitrogen is larger than 0.25 for oxygen. Therefore, the nitrogen concentration at 10 km is more than 25% of the sea level N_2 concentration.

(b) We know that

$$\frac{N(O_2)_{10\ km}}{N(O_2)_{Sea\ Level}} = 0.25\ ,\ \frac{N(N_2)_{10\ km}}{N(N_2)_{Sea\ Level}} = 0.30\ ,\ \text{and}\ \frac{N(N_2)_{Sea\ Level}}{N(O_2)_{Sea\ Level}} = 4\ .$$

Then,

$$\begin{aligned} \frac{N(N_2)_{10\ km}}{N(O_2)_{10\ km}} &= \frac{N(N_2)_{10\ km}}{N(N_2)_{Sea\ Level}} \times \frac{N(N_2)_{Sea\ Level}}{N(O_2)_{Sea\ Level}} \times \frac{N(O_2)_{Sea\ Level}}{N(O_2)_{10\ km}} \\ &= 0.30 \times 4 \times \frac{1}{0.25} = 4.8\ . \end{aligned}$$

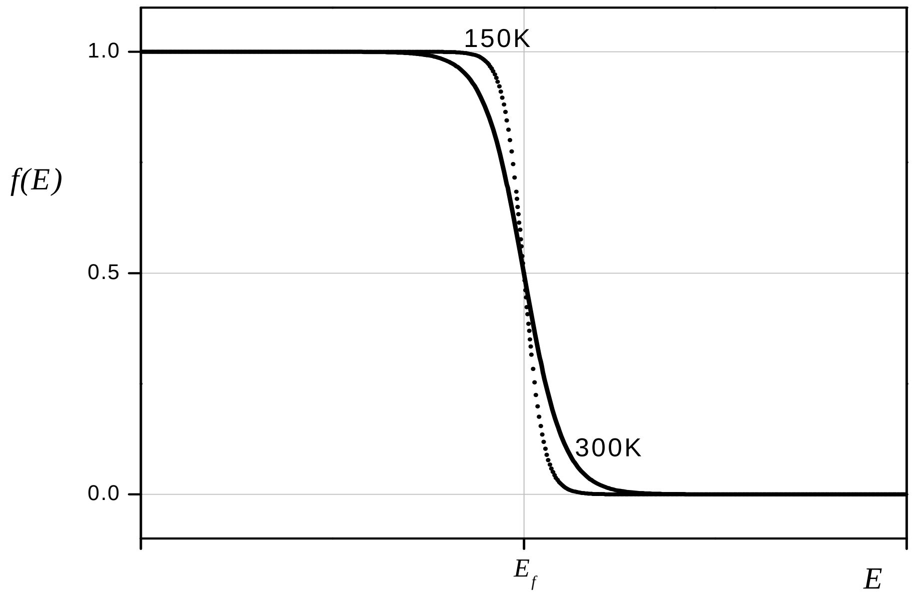
It is more N_2-rich than at sea level.

1.5

$$\begin{aligned} 1 - f\left(E_f - \Delta E\right) &= 1 - \frac{1}{1 + e^{\left(E_f - \Delta E - E_f\right)/kT}} \\ &= \frac{e^{\left(E_f - \Delta E - E_f\right)/kT}}{1 + e^{\left(E_f - \Delta E - E_f\right)/kT}} \\ &= \frac{1}{1 + e^{-\left(E_f - \Delta E - E_f\right)/kT}} \end{aligned}$$

$$= \frac{1}{1+e^{-\left(E_f - \Delta E - E_f\right)/kT}}$$
$$= \frac{1}{1+e^{\left(E_f + \Delta E - E_f\right)/kT}}$$
$$= f\left(E_f + \Delta E\right)$$

1.6 (a)



(b) *At 0K, the probability of a state below the Fermi level being filled is 1 and a state above the Fermi level being filled is 0. So a total of 7 states are filled which means there are 14 electrons (since 2 electrons can occupy each state) in the system.*

Density of States

1.7 Since the semiconductor is assumed to be, We are asked to use Equations (1.7.2) and (1.7.4) to approximate the Fermi distribution. (This means that the doping concentration is low and E_f is not within a few kTs from E_c or E_v. A lightly doped semiconductor is known as a non-degenerate semiconductor.) The carrier distribution as a function of energy in the conduction band is proportional to

$$\textit{Distribution(E)} \propto \left(E - E_c\right)^{1/2} e^{-\left(E - E_f\right)/kT},$$

where $e^{-(E-E_f)/kT}$ is from Equation (1.7.2). Taking the derivative with respect to E and setting it to zero, we obtain

$$\frac{d}{dE}\left[(E-E_c)^{1/2}e^{-(E-E_f)/kT}\right]=\frac{1}{2}\frac{1}{\sqrt{E-E_c}}e^{-(E-E_f)/kT}+\sqrt{E-E_c}\left(-\frac{1}{kT}\right)e^{-(E-E_f)/kT}=0$$

The exponential terms cancel out. Solving the remaining equation yields

$$\frac{1}{2}(E-E_c)^{-1/2}=\frac{1}{kT}(E-E_c)^{1/2}\quad\rightarrow\quad(E-E_c)=\frac{kT}{2}\quad\rightarrow\quad E=E_c+\frac{kT}{2}.$$

So, the number of carriers in the conduction band peaks at E_c+kT/2.
Similarly, in the valence band, the carrier distribution as a function of energy is proportional to

$$\textit{Distribution(E)}\propto(E_v-E)^{1/2}e^{-(E_f-E)/kT},$$

where $e^{-(E_f-E)/kT}$ is Equation (1.7.2). Taking the derivative and setting it to zero, we obtain

$$\frac{d}{dE}\left[(E_v-E)^{1/2}e^{-(E_f-E)/kT}\right]=\frac{1}{2}(E_v-E)^{-1/2}e^{-(E_f-E)/kT}+(E_v-E)^{1/2}\left(\frac{1}{kT}\right)e^{-(E_f-E)/kT}=0.$$

Again, the exponential terms cancel out, and solving the remaining equation yields

$$\frac{1}{2}(E_v-E)^{-1/2}=\frac{1}{kT}(E_v-E)^{1/2}\quad\rightarrow\quad(E_v-E)=\frac{kT}{2}\quad\rightarrow\quad E=E_v-\frac{kT}{2}.$$

Therefore, the number of carriers in the valence band peaks at E_v-kT/2.

1.8 Since it is given that the semiconductor is non-degenerate (not heavily doped), E_f is not within a few kTs from E_c or E_v. We can use Equations (1.7.2) and (1.7.4) to approximate the Fermi-Dirac distribution.

(a) The electron concentration in the conduction band is given by

$$n=\int_{C.B.}D_c(E)\,f(E)dE\approx\int_{E_c}^{\infty}A\sqrt{E-E_c}\,e^{-(E-E_f)/kT}\,dE\,.$$

In order to simplify the integration, we make the following substitutions:

$$\frac{E-E_c}{kT}=x\rightarrow E=kT\,x+E_c,\quad\frac{1}{kT}dE=dx\rightarrow dE=kT\,dx,\quad \textit{and } x:\ \textit{from } 0 \textit{ to } \infty\,.$$

Now the equation becomes

$$n = \int_0^{\infty} A\sqrt{kTx}\, e^{-\left(kTx+E_c-E_f\right)/kT}\, kT\, dx = A\left(kT\right)^{3/2} e^{-\left(E_c-E_f\right)/kT} \int_0^{\infty} \sqrt{x}\, e^{-x} dx$$

where

$$\int_0^{\infty} \sqrt{x}\, e^{-x} dx = \int_0^{\infty} \left(x\right)^{\frac{3}{2}-1} e^{-x} dx = \Gamma\left(3/2\right) = \frac{\sqrt{\pi}}{2}. \text{ (Gamma function)}$$

Hence, the electron concentration in the conduction band is

$$n = \frac{\sqrt{\pi}}{2} A\left(kT\right)^{3/2} e^{-\left(E_c-E_f\right)/kT}.$$

Similarly, the hole concentration is given by

$$p = \int_{V.B.} D_v(E)\left[1\text{-}f(E)\right] dE \approx \int_{-\infty}^{E_v} B\sqrt{E_v - E}\, e^{-\left(E_f-E\right)/kT}\, dE.$$

Again, we make the following substitutions to simplify the integration:

$$\frac{E_v - E}{kT} = x \rightarrow E = -kT\,x + E_v, -\frac{1}{kT} dE = dx \rightarrow dE = -kT\,dx, \text{ and } x: \text{ from } \infty \text{ to } 0.$$

Now the equation becomes

$$p = \int_{\infty}^{0} -B\sqrt{kTx}\, e^{-\left(E_f+kTx-E_v\right)/kT}\, kT\, dx = B\left(kT\right)^{3/2} e^{-\left(E_f-E_v\right)/kT} \int_0^{\infty} \sqrt{x}\, e^{-x} dx$$

where

$$\int_0^{\infty} \sqrt{x}\, e^{-x} dx = \int_0^{\infty} \left(x\right)^{\frac{3}{2}-1} e^{-x} dx = \Gamma\left(3/2\right) = \frac{\sqrt{\pi}}{2}. \text{ (Gamma function)}$$

Therefore, the hole concentration in the conduction band is

$$p = \frac{\sqrt{\pi}}{2} B\left(kT\right)^{3/2} e^{-\left(E_f-E_v\right)/kT}.$$

(b) The word "Intrinsic" implies that the electron concentration and the hole concentration are equal. Therefore,

$$n = p \rightarrow \frac{\sqrt{\pi}}{2} A\left(kT\right)^{3/2} e^{-\left(E_c-E_i\right)/kT} = \frac{\sqrt{\pi}}{2} B\left(kT\right)^{3/2} e^{-\left(E_i-E_v\right)/kT}.$$

This simplifies to

$$Ae^{-(E_c - E_i)/kT} = Be^{-(E_i - E_v)/kT} .$$

Solving for E_i yields

$$E_i = \frac{E_c + E_v}{2} + \frac{kT}{2}\ln\left(\frac{1}{2}\right) = \frac{E_c + E_v}{2} - 0.009\,eV;\, k = 8.62\times10^{-5}\,eV\,K^{-1}, T = 300\,K .$$

Hence, the intrinsic Fermi level (E_i) is located at 0.009 eV below the mid-bandgap of the semiconductor.

1.9 The unit step functions set the integration limits. $D_c(E)$ is zero for $E < E_c$, and $D_v(E)$ is zero for $E > E_v$. Since it is given that the semiconductor is non-degenerate (not heavily doped), E_f is not within a few kTs from E_c or E_v. We can use Equations (1.7.2) and (1.7.4) to approximate the Fermi-Dirac distribution.

(a) The electron concentration in the conduction band is given by

$$n = \int_{C.B.} D_c(E)\, f(E) dE \approx \int_{E_c}^{\infty} A(E - E_c)e^{-(E-E_f)/kT}\, dE .$$

In order to simplify the integration, we make the following substitutions:

$$\frac{E - E_c}{kT} = x \rightarrow E = kT\,x + E_c, \quad \frac{1}{kT}dE = dx \rightarrow dE = kT\,dx, \quad and\ x: from\ 0\ to\ \infty .$$

Now the equation becomes

$$n = \int_0^{\infty} AkTx\,e^{-(kTx + E_c - E_f)/kT}\, kT\,dx = A\,(kT)^2\, e^{-(E_c - E_f)/kT}\int_0^{\infty} x\,e^{-x}dx$$

where

$$\int_0^{\infty} x\,e^{-x}dx = 1 .$$

Hence, the electron concentration in the conduction band is

$$n = A(kT)^2\, e^{-(E_c - E_f)/kT} .$$

Similarly, the hole concentration is given by

$$p = \int_{V.B.} D_v(E)[1\text{-}f(E)]dE \approx \int_{-\infty}^{E_v} B(E_v - E)e^{-(E_f - E)/kT}\, dE .$$

Again, we make the following substitutions to simplify the integration:

$$\frac{E_v - E}{kT} = x \rightarrow E = -kT\,x + E_v,\ -\frac{1}{kT}dE = dx \rightarrow dE = -kT\,dx,\ and\ x:\ from\ \infty\ to\ 0\,.$$

Now the equation becomes

$$p = \int_{\infty}^{0} -BkTxe^{-(E_f + kTx - E_v)/kT}\,kT\,dx = B\,(kT)^2\,e^{-(E_f - E_v)/kT}\int_0^{\infty} xe^{-x}dx$$

where

$$\int_0^{\infty} xe^{-x}dx = 1\,.$$

Therefore, the hole concentration in the conduction band is

$$p = B(kT)^2\,e^{-(E_f - E_v)/kT}\,.$$

(b) The word "Intrinsic" implies that the electron concentration and the hole concentration are equal. Therefore,

$$n = p \rightarrow A(kT)^2\,e^{-(E_c - E_i)/kT} = B(kT)^2\,e^{-(E_i - E_v)/kT}\,.$$

This simplifies to

$$Ae^{-(E_c - E_i)/kT} = Be^{-(E_i - E_v)/kT}\,.$$

If we solve for E_i, we obtain

$$E_i = \frac{E_c + E_v}{2} + \frac{kT}{2}\ln\left(\frac{1}{2}\right) = \frac{E_c + E_v}{2} - 0.009\,eV; k = 8.62\times10^{-5}\,eV\,K^{-1}, T = 300\,K.$$

Hence, the intrinsic Fermi level (E_i) is located at 0.009 eV below the mid-bandgap of the semiconductor.

1.10 (a) The carrier distribution as a function of energy in the conduction band is proportional to

$$Distribution(E) \propto (E - E_c)^{1/2}\,e^{-(E - E_f)/kT},$$

where $e^{-(E-Ef)/kT}$ is from Equation (1.7.2). Taking the derivative and setting it to zero, we obtain

$$\frac{d}{dE}\left[(E - E_c)^{1/2}\,e^{-(E - E_f)/kT}\right] = \frac{1}{2}\frac{1}{\sqrt{E - E_c}}e^{-(E - E_f)/kT} + \sqrt{E - E_c}\left(-\frac{1}{kT}\right)e^{-(E - E_f)/kT} = 0\,.$$

The exponential terms cancel out. Solving the remaining equation yields

$$\frac{1}{2}(E-E_c)^{-1/2}=\frac{1}{kT}(E-E_c)^{1/2} \quad \rightarrow \quad (E-E_c)=\frac{kT}{2} \quad \rightarrow \quad E=E_c+\frac{kT}{2}.$$

Hence, the number of carriers in the conduction band peaks at E_c+kT/2.

(b) The electron concentration in the conduction band is given by

$$n=\int_{C.B.} D_c(E)\, f(E) dE=\int_{E_c}^{Top\ of\ the\ Conduction\ Band} D_c(E)\, f(E) dE .$$

We assume that the function f(E) falls off rapidly such that

$$\frac{\int_{Top\ of\ the\ Conduction\ Band}^{\infty} D_c(E)\, f(E) dE}{\int_{E_c}^{Top\ of\ the\ Conduction\ Band} D_c(E)\, f(E) dE}\approx 0 .$$

Now we may change the upper limit of integration from the Top of the Conduction Band to ∞:

$$n=\int_{E_c}^{\infty} A\sqrt{E-E_c}\, e^{-(E-E_f)/kT}\, dE .$$

Also, in order to simplify the integration, we make the following substitutions:

$$\frac{E-E_c}{kT}=x \rightarrow E=kT\,x+E_c, \quad \frac{1}{kT}dE=dx \rightarrow dE=kT\,dx, \quad and\ x:\ from\ 0\ to\ \infty .$$

The equation becomes

$$n=\int_0^{\infty} A\sqrt{kTx}\, e^{-(kTx+E_c-E_f)/kT}\, kT\, dx=A\,(kT)^{3/2}\, e^{-(E_c-E_f)/kT}\int_0^{\infty}\sqrt{x}\, e^{-x} dx$$

where

$$\int_0^{\infty}\sqrt{x}\, e^{-x} dx=\int_0^{\infty}(x)^{\frac{3}{2}-1}\, e^{-x} dx=\Gamma(3/2)=\frac{\sqrt{\pi}}{2}. \text{ (Gamma function)}$$

Therefore, the electron concentration in the conduction band is

$$n=\frac{\sqrt{\pi}}{2}A(kT)^{3/2}\, e^{-(E_c-E_f)/kT} .$$

(b) The ratio of the peak electron concentration at E = E_c+(1/2)kT to the electron concentration at E = E_c+40kT is

$$\frac{n(E_c+40kT)}{n(E_c+\frac{1}{2}kT)}=\frac{A\left(E_c+40kT-E_C\right)^{1/2}e^{-\left(E_c+40kT-E_f\right)/kT}}{A\left(E_c+0.5kT-E_c\right)^{1/2}e^{-\left(E_c+0.5kT-E_f\right)/kT}}=$$

$$=\left(40kT/0.5kT\right)^{1/2}e^{\left[-\left(E_c+40kT-E_f\right)+\left(E_c+0.5kT-E_f\right)\right]/kT}=(40/0.5)e^{-39.5}=5.60\times10^{-16}.$$

The ratio is very small, and this result justifies our assumption in Part (b).

(c) The kinetic energy of an electron at E is equal to E-E_C. The average kinetic energy of electrons is

$$\langle K.E.\rangle=\frac{sum\ of\ the\ kinetic\ energy\ of\ all\ electrons}{total\ number\ of\ electrons}=$$

$$=\frac{\int_{C.B.}(E-E_c)D_c(E)\,f(E)dE}{\int_{C.B.}D_c(E)\,f(E)dE}$$

$$\approx\frac{\int_{E_c}^{\infty}(E-E_c)A\sqrt{E-E_c}\,e^{-\left(E-E_f\right)/kT}\,dE}{\int_{E_c}^{\infty}A\sqrt{E-E_c}\,e^{-\left(E-E_f\right)/kT}\,dE}.$$

In order to simplify the integration, we make the following substitutions:

$$\frac{E-E_c}{kT}=x\rightarrow E=kT\,x+E_c,\quad \frac{1}{kT}dE=dx\rightarrow dE=kT\,dx,\quad and\ x:\ from\ 0\ to\ \infty.$$

Now the equation becomes

$$\frac{\int_0^{\infty}A\left(kTx\right)^{3/2}e^{-\left(kTx+E_c-E_f\right)/kT}\,kT\,dx}{\int_0^{\infty}A\left(kTx\right)^{1/2}e^{-\left(kTx+E_c-E_f\right)/kT}\,kT\,dx}=\frac{A\left(kT\right)^{5/2}e^{-\left(E_c-E_f\right)/kT}\int_0^{\infty}\left(x\right)^{3/2}e^{-x}dx}{A\left(kT\right)^{3/2}e^{-\left(E_c-E_f\right)/kT}\int_0^{\infty}\left(x\right)^{1/2}e^{-x}dx}$$

where

$$\int_0^{\infty}\left(x\right)^{3/2}e^{-x}dx=\int_0^{\infty}\left(x\right)^{\frac{5}{2}-1}e^{-x}dx=\Gamma\left(5/2\right)=\frac{3\sqrt{\pi}}{4}\quad\text{(Gamma functions)}$$

and

$$\int_0^{\infty}\left(x\right)^{1/2}e^{-x}dx=\int_0^{\infty}\left(x\right)^{\frac{3}{2}-1}e^{-x}dx=\Gamma\left(3/2\right)=\frac{\sqrt{\pi}}{2}.\quad\text{(Gamma functions)}$$

Hence, the average kinetic energy is (3/2)kT.

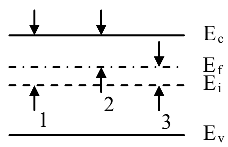
Electron and Hole Concentrations

1.11 (a) We use Equation (1.8.11) to calculate the hole concentration:

$$n \times p = n_i^2 \rightarrow p = n_i^2 / n = \left(10^{10}\right)^2 / 10^5 cm^{-3} = 10^{15} cm^{-3}.$$

(b) Please refer to Equations (1.9.3a) and (1.9.3b). Since N_d-N_a >> n_i and all the impurities are ionized, $n = N_d - N_a$, and $p = (n_i)^2/(N_d - N_a)$.

(c) Since the Fermi level is located 0.26 eV above E_i and closer to E_c, the sample is n-type. If we assume that E_i is located at the mid-bandgap (~ 0.55 eV), then E_c-E_f = 0.29 eV.



1: E_c-E_i =0.55 eV
2: E_c-E_f =0.29 eV
3: E_f-E_i =0.26 eV

Using Equations (1.8.5) and (1.8.11), we find

$$n = N_c e^{-\left(E_c - E_f\right)/kT} = 4.01 \times 10^{14} cm^{-3} \ and \ p = n_i^2 / n = 2.49 \times 10^5 cm^{-3}.$$

Therefore, the electron concentration is 4.01×10^{14} cm^{-3}, and the hole concentration is 2.49×10^{5} cm^{-3}.

* There is another way to solve this problem:

$$n = n_i e^{\left(E_f - E_i\right)/kT} = 2.20 \times 10^{14} cm^{-3} \ and \ p = n_i^2 / n = 4.55 \times 10^5 cm^{-3}.$$

(d) If T = 800 K, there is enough thermal energy to free more electrons from silicon-silicon bonds. Hence, using Equation (1.8.12), we first calculate the intrinsic carrier density n_i at 800 K:

$$n_i = \sqrt{N_c(800K)N_v(800K)}\, e^{-\left(E_g\right)/(2kT)} = 2.56 \times 10^{16} cm^{-3}.$$

where

$$N_c(T = 800K) = 2\left[\frac{2\pi m_{dn} kT}{h^2}\right]^{3/2} = 2.8 \times 10^{19} \times \left(\frac{T}{300K}\right)^{3/2} cm^{-3} = 1.22 \times 10^{20} cm^{-3}$$

and

$$N_v(T = 800K) = 2\left[\frac{2\pi m_{dp} kT}{h^2}\right]^{3/2} = 1.04 \times 10^{19} \times \left(\frac{T}{300K}\right)^{3/2} cm^{-3} = 4.53 \times 10^{19} cm^{-3}.$$

Clearly, n_i at 800K is much larger than N_d-N_a (which is equal to n from the previous part). Hence the electron concentration is $n \approx n_i$, and the hole concentration is $p = (n_i)^2/n \approx n_i$. The semiconductor is intrinsic at 800K, and E_f is located very close to the mid-bandgap.

Nearly Intrinsic Semiconductor

1.12 Applying Equation (1.8.11) to this problem yields

$$n_i^2 / p = n = 2p \rightarrow 2p^2 = n_i^2 \rightarrow p = \frac{1}{\sqrt{2}} n_i = 7.07\times10^{12} cm^{-3} \; and \; n = 1.41\times10^{13} cm^{-3}.$$

1.13 (a) B is a group III element. When added to Si (which belongs to Group IV), it acts as an acceptor producing a large number of holes. Hence, this becomes a P-type Si film.

(b) At T = 300 K, since the intrinsic carrier density is negligible compared to the dopant concentration, $p=N_a=4\times10^{16}$ cm^{-3}, and $n = (n_i=10^{10}cm^{-3})^2/p = 2500$ cm^{-3}.
At T = 600 K,

$$n_i = \sqrt{N_c(600K)N_v(600K)}e^{-(E_g)/(2kT)} = 1.16\times10^{15} cm^{-3}$$

where

$$N_c(T=600K) = 2\left[\frac{2\pi m_{dn} kT}{h^2}\right]^{3/2} = 2.8\times10^{19}\times\left(\frac{T}{300K}\right)^{3/2} cm^{-3} = 7.92\times10^{19} cm^{-3}$$

and

$$N_v(T=600K) = 2\left[\frac{2\pi m_{dp} kT}{h^2}\right]^{3/2} = 1.04\times10^{19}\times\left(\frac{T}{300K}\right)^{3/2} cm^{-3} = 2.94\times10^{19} cm^{-3}.$$

The intrinsic carrier concentration is no more negligible compared to the dopant concentration. Thus, we have

$$p = N_a + n_i = \left(4\times10^{16} + 1.16\times10^{15}\right) cm^{-3} = 4.12\times10^{16} cm^{-3}, \text{ and}$$
$$n = n_i^2/p = \left(1.16\times10^{15} cm^{-3}\right)^2 / 4.12\times10^{16} cm^{-3} = 3.27\times10^{13} cm^{-3}.$$

The electron concentration has increased by many orders of magnitude.

(c) At high temperatures, there is enough thermal energy to free more electrons from silicon-silicon bonds, and consequently, the number of intrinsic carriers increases.

(d) Using Equation 1.8.8, we calculate the position of the Fermi level with respect to E_v.

$$E_f - E_v = kT \ln\left[N_v(T)/p(T)\right] = 0.34\, eV, \quad T = 600K.$$

At 600 K, the Fermi level is located 0.34 eV above the valence band.

Incomplete Ionization of Dopants and Freeze-out

1.14 From Equation (1.9.1), we know that $n + N_a^- = p + N_d^+$. Since N_d^+ is much larger than N_a^-, all the samples are n-type, and $n \approx N_d^+ - N_a^- = 3\times10^{15}$ /cm^3. This value is assumed to be constant. Using the Equations (1.8.10) and (1.9.3b),

$$p = n_i^2 / \left(N_d^+ - N_a^-\right) = N_c N_v \exp\left(-E_g / kT\right) = CT^3 \exp\left(-E_g / kT\right),$$

where C is a temperature independent constant. Using the sensitivity of p defined by $\partial p/\partial T$,

$$\partial p / \partial T = \left(3 + E_g / kT\right) \times CT^2 \exp\left(-E_g / kT\right)$$

Therefore, the larger the energy gap is the less sensitive to temperature the minority carrier is. For the definition of the sensitivity of *p*,

$$\left(\partial p / \partial T\right) / p = \left(3 + E_g / kT\right) / T$$

The temperature sensitivity of the minority carrier is greater for larger E_g.

1.15 (a) Let us first consider the case of n-type doping. The dopant atoms are located at energy E_d inside the bandgap, near the conduction band edge. The problem states that we are considering the situation in which half the impurity atoms are ionized, i.e. $n=N_d/2$. In other words, the probability of dopant atoms being ionized is ½, or conversely, the probability that a state at the donor energy E_D is filled is ½.

From Problem 1.2 part (a), we know that if f $(E_D)=1/2$, then $E_D=E_f$. From Equation 1.8.5,

$$n = N_c e^{-\left(E_c - E_f\right)/kT}.$$

We also know that $E_f=E_D$ and $E_c-E_D=0.05$eV.

$$N_c(T) = 2\left[\frac{2\pi\, m_{dn} kT}{h^2}\right]^{3/2} = 2.8\times10^{19} \times \left(\frac{T}{300K}\right)^{3/2} cm^{-3}.$$

$$N_c(T) e^{-\left(E_c - E_f\right)/kT} = N_c(T) e^{-\left(E_c - E_D\right)/kT} = \frac{N_D}{2} \quad\rightarrow\quad \frac{2N_c(T)}{N_d} = e^{\left(E_c - E_D\right)/kT}.$$

This equation can be solved iteratively. Starting with an arbitrary guess of 100K for T, we find T converges to 84.4 K.

Similarly, for boron

$$N_v(T) = 2\left[\frac{2\pi\, m_{dp} kT}{h^2}\right]^{3/2} = 1.04\times10^{19} \times \left(\frac{T}{300K}\right)^{3/2} cm^{-3}.$$

$$N_v(T) e^{-\left(E_f - E_v\right)/kT} = N_v(T) e^{-\left(E_a - E_v\right)/kT} = \frac{N_a}{2} \quad\rightarrow\quad \frac{2N_v(T)}{N_a} = e^{\left(E_a - E_v\right)/kT}.$$

Starting from T =100K, we find T converges to 67.7K.

(b) We want to find T where n_i is $10N_d$. This can be written as

$$n_i = \sqrt{N_c(T)N_v(T)}e^{-(E_g)/(2kT)} = 1.71\times10^{19}\times\left(\frac{T}{300K}\right)^{3/2} e^{-(E_g)/(2kT)} = 10N_d$$

where

$$N_c(T) = 2\left[\frac{2\pi m_{dn}kT}{h^2}\right]^{3/2} = 2.8\times10^{19}\times\left(\frac{T}{300K}\right)^{3/2} cm^{-3} \text{ and}$$

$$N_v(T) = 2\left[\frac{2\pi m_{dp}kT}{h^2}\right]^{3/2} = 1.04\times10^{19}\times\left(\frac{T}{300K}\right)^{3/2} cm^{-3}.$$

We need to solve the equation iteratively, as in part (a) for $n_i=10N_d=10^{17}cm^{-3}$. Starting from T=300K, we get T=777 K for $n_i=10N_d$.

For $n_i=10N_a$, we simply replace N_d in the equation above with N_a. Starting from T =300K, we find T=635 K.

(c) If we assume full ionization of impurities at T = 300 K,

For arsenic: $n \cong N_d = 10^{16}cm^{-3} >> n_i,\ p = \dfrac{n_i^2}{N_d} = 2.1\times10^4 cm^{-3}$

For boron: $p \cong N_a = 10^{15}cm^{-3} >> n_i,\ n = \dfrac{n_i^2}{N_a} = 2.1\times10^5 cm^{-3}$

(d) Please refer to the example in Section 2.8. For arsenic,

$$E_f - E_v = kT\ln\frac{N_v}{p} = \frac{1.04\times10^{19}cm^{-3}}{2.1\times10^4 cm^{-3}} = 0.88eV.$$

For boron,

$$E_f - E_v = kT\ln\frac{N_v}{p} = \frac{1.04\times10^{19}cm^{-3}}{10^{15}cm^{-3}} = 0.24eV.$$

(e) In case of arsenic + boron,

$$n \cong N_d - N_a = 9\times10^{15}cm^{-3}, \text{ and } \quad p = \frac{n_i^2}{n} = \frac{\left(10^{10}cm^{-3}\right)^2}{9\times10^{15}cm^{-3}} = 1.11\times10^4 cm^{-3}, \text{ and}$$

$$E_f - E_v = kT\ln\left(\frac{N_v}{p}\right) = 0.026(eV)\ln\left(\frac{1.04\times10^{19}cm^{-3}}{1.11\times10^4 cm^{-3}}\right) = 0.90\ eV.$$

1.16 (a) If we assume full ionization of impurities, the electron concentration is n ≈ N_d = $10^{17}cm^{-3}$. The hole concentration is $p=(n_i)^2/n=(10^{10}\ cm^{-3})^2/10^{17}cm^{-3}=10^3\ cm^{-3}$. The Fermi level position, with respect to E_c, is

$$E_c - E_f = kT\ln[N_c/n] = 0.026\ln\left[2.8\times10^{19}cm^{-3}/10^{17}cm^{-3}\right] = 0.15eV .$$

E_f is located 0.15 eV below E_c.

(b) In order to check the full ionization assumption with the calculated Fermi level, we need to find the percentage of donors occupied by electrons.

$$E_D - E_f = (E_c - E_f) - (E_c - E_D) = 0.1eV \text{ , and}$$

$$n_D = N_d \frac{1}{1+e^{(E_D - E_f)/kT}} = \frac{10^{17}(cm^{-3})}{1+e^{(0.1eV/0.026eV)}} = 2.09\times10^{15}cm^{-3} \equiv 2\%\ of\ N_d .$$

Since only 2% of dopants are not ionized, it is fine to assume that the impurities are fully ionized.

(c) We assume full ionization of impurities, the electron concentration is n ≈ N_d = $10^{19}cm^{-3}$. The hole concentration is $p=(n_i)^2/n = (10^{10}cm^{-3})^2/10^{19}\ cm^{-3} = 10\ cm^{-3}$. The Fermi level position, with respect to E_c, is

$$E_c - E_f = kT\ln[N_c/n] = 0.026\times\ln\left[2.8\times10^{19}cm^{-3}/10^{19}cm^{-3}\right] = 0.027eV .$$

It is located 0.027 eV below E_c.

Again, we need to find the percentage of donors occupied by electrons in order to check the full ionization assumption with the calculated Fermi level.

$$E_D - E_f = (E_c - E_f) - (E_c - E_D) = -0.023eV \text{ , and}$$

$$n_D = N_d \frac{1}{1+e^{(E_D - E_f)/kT}} = \frac{10^{19}(cm^{-3})}{1+e^{(-0.023eV/0.026eV)}} = 7.08\times10^{18}cm^{-3} \equiv 71\%\ of\ N_d .$$

Since 71% of dopants are not ionized, the full ionization assumption is not correct.

(d) For T=30 K, we need to use Equation (1.10.2) to find the electron concentration since the temperature is extremely low. First, we calculate N_c and N_v at T=30K:

$$N_c(T=30K) = 2\left[\frac{2\pi\, m_{dn} kT}{h^2}\right]^{3/2} = 2.8\times10^{19}\times\left(\frac{T}{300K}\right)^{3/2} cm^{-3} = 8.85\times10^{17}cm^{-3}$$

and

$$N_v(T=30K) = 2\left[\frac{2\pi\, m_{dp} kT}{h^2}\right]^{3/2} = 1.04\times10^{19}\times\left(\frac{T}{300K}\right)^{3/2} cm^{-3} = 3.29\times10^{17}cm^{-3}.$$

The electron concentration is

$$n=\sqrt{\frac{N_c(30\,K)N_d}{2}}\,e^{-(E_c-E_D)/(2kT)}=8.43\times10^{8}\,cm^{-3}.$$

And, the hole concentration is

$$p=n_i^2/n\approx 0$$

where

$$n_i=\sqrt{\frac{N_c(30\,K)N_v(30\,K)}{2}}\,e^{-(E_g)/(2kT)}=2.32\times10^{-75}\,cm^{-3}.$$

Since n_i is extremely small, we can assume that all the electrons are contributed by ionized dopants. Hence,

$$n=N_d\left(1-\frac{1}{1+e^{(E_D-E_f)/kT}}\right)=8.43\times10^{8}\,cm^{-3}\rightarrow\frac{8.43\times10^{8}\,cm^{-3}}{10^{17}\,cm^{-3}}=8.43\times10^{-9}.$$

The full ionization assumption is not correct since only 8.43×10^{-7}% of N_d is ionized.
To locate the Fermi level,

$$E_D-E_f=kT\ln\left(\left(1-\frac{n}{N_d}\right)^{-1}-1\right)=-0.048\,eV.$$

E_c-E_f = 0.05-0.048 = 0.002 eV. Therefore, the Fermi level is positioned 0.002 eV below E_c, between E_c and E_D.

1.17 (a) We assume full ionization of impurities, the electron concentration is $n\approx N_d=10^{16}cm^{-3}$. The hole concentration is $p=(n_i)^2/n=(10^{10}cm^{-3})^2/10^{16}cm^{-3}=10^4\ cm^{-3}$. The Fermi level position, with respect to E_c, is

$$E_c-E_f=kT\ln[N_c/n]=0.026\ln\left[2.8\times10^{19}\,cm^{-3}/10^{16}\,cm^{-3}\right]=0.21eV.$$

It is located 0.21 eV below E_c.

We need to find the percentage of donors occupied by electrons in order to check the full ionization assumption with the calculated Fermi level.

$$E_D-E_f=(E_c-E_f)-(E_c-E_D)=0.16eV\text{, and}$$

$$n_D=N_d\frac{1}{1+e^{(E_D-E_f)/kT}}=\frac{10^{16}\left(cm^{-3}\right)}{1+e^{(0.16eV/0.026eV)}}=2.12\times10^{13}\,cm^{-3}\equiv2.12\times10^{-1}\%\ of\ N_d.$$

Since only 0.21% of dopants are not ionized, the full ionization assumption is correct.

(b) We assume full ionization of impurities, the electron concentration is $n \approx N_d = 10^{18}cm^{-3}$. The hole concentration is $p=(n_i)^2/n = (10^{10}cm^{-3})^2/10^{18}cm^{-3} = 10^2\ cm^{-3}$. The Fermi level position with respect to E_c is

$$E_c - E_f = kT\ln[N_c/n] = 0.026\ln[2.8\times10^{19}cm^{-3}/10^{18}cm^{-3}] = 0.087eV\ .$$

It is located 0.087 eV below E_c.

We need to find the percentage of donors occupied by electrons in order to check the full ionization assumption with the calculated Fermi level.

$$E_D - E_f = (E_c - E_f) - (E_c - E_D) = 0.037eV\ \text{, and}$$

$$n_D = N_d\frac{1}{1+e^{(E_D-E_f)/kT}} = \frac{10^{18}(cm^{-3})}{1+e^{(0.037eV/0.026eV)}} = 1.94\times10^{17}cm^{-3} \equiv 19\%\ of\ N_d\ .$$

Since 19% of dopants are not ionized, the full ionization assumption is not accurate but acceptable.

(c) We assume full ionization of impurities, the electron concentration is $n \approx N_d = 10^{19}cm^{-3}$. The hole concentration is $p=(n_i)^2/n = (10^{10}cm^{-3})^2/10^{19}cm^{-3} = 10\ cm^{-3}$. The Fermi level position, with respect to E_c, is

$$E_c - E_f = kT\ln[N_c/n] = 0.026\ln[2.8\times10^{19}cm^{-3}/10^{19}cm^{-3}] = 0.027eV\ .$$

It is located 0.027 eV below E_c.

Again, we need to find the percentage of donors occupied by electrons in order to check the full ionization assumption with the calculated Fermi level.

$$E_D - E_f = (E_c - E_f) - (E_c - E_D) = -0.023eV\ \text{, and}$$

$$n_D = N_d\frac{1}{1+e^{(E_D-E_f)/kT}} = \frac{10^{19}(cm^{-3})}{1+e^{(-0.023eV/0.026eV)}} = 7.08\times10^{18}cm^{-3} \equiv 71\%\ of\ N_d\ .$$

Since 71% of dopants are not ionized, the full ionization assumption is not correct.

Since N_d is not fully ionized and N_d(ionized) << N_d(not-ionized),

$$n = N_d[1 - f(E_D)] \approx N_d e^{(E_D-E_f)/kT} = N_c e^{-(E_c-E_f)/kT}\ .$$

Solving the equation above for E_f yields

$$E_f = \frac{(E_D + E_c)}{2} + \frac{kT}{2}\ln\left(\frac{N_d}{N_c}\right).$$

Chapter 2

Mobility

2.1 (a) The mean free time between collisions using Equation (2.2.4b) is

$$\mu_n = \frac{q\tau_{mn}}{m_n} \quad \rightarrow \quad \tau_{mn} = \frac{\mu_n m_n}{q} = 2.85 \times 10^{-13} \sec$$

where μ_n is given to be 500 cm^2/Vsec (= 0.05 m^2/Vsec), and m_n is assumed to be m_0.

(b) We need to find the drift velocity first:

$$v_d = \mu_n \mathcal{E} = 50000\, cm / \sec .$$

The distance traveled by drift between collisions is

$$d = v_d \tau_{mn} = 0.14\, nm .$$

2.2 From the thermal velocity example, we know that the approximate thermal velocity of an electron in silicon is

$$v_{th} = \sqrt{\frac{3kT}{m}} = 2.29 \times 10^7\, cm / \sec .$$

Consequently, the drift velocity (v_d) is (1/10)v_{th} = 2.29×10^6 cm/sec, and the time it takes for an electron to traverse a region of 1 μm in width is

$$t = \frac{10^{-4}\, cm}{2.29 \times 10^6\, cm / \sec} = 4.37 \times 10^{-11} \sec .$$

Next, we need to find the mean free time between collisions using Equation (2.2.4b):

$$\mu_n = \frac{q\tau_{mn}}{m_n} \quad \rightarrow \quad \tau_{mn} = \frac{\mu_n m_n}{q} = 2.10 \times 10^{-13} \sec$$

where μ_n is 1400 cm^2/Vsec (=0.14 m^2/Vsec, for lightly doped silicon, given in Table 2-1), and m_n is 0.26m_0 (given in Table 1-3). So, the average number of collision is

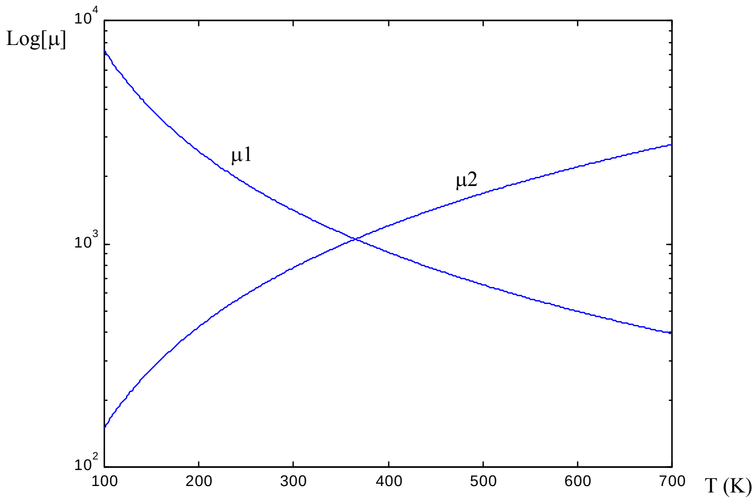
$$\frac{t}{\tau_{mn}} = 207.7\ collision \quad \Rightarrow \quad 207\ collisions\ .$$

In order to find the voltage applied across the region, we need to calculate the electric field using Equation (2.2.3b):

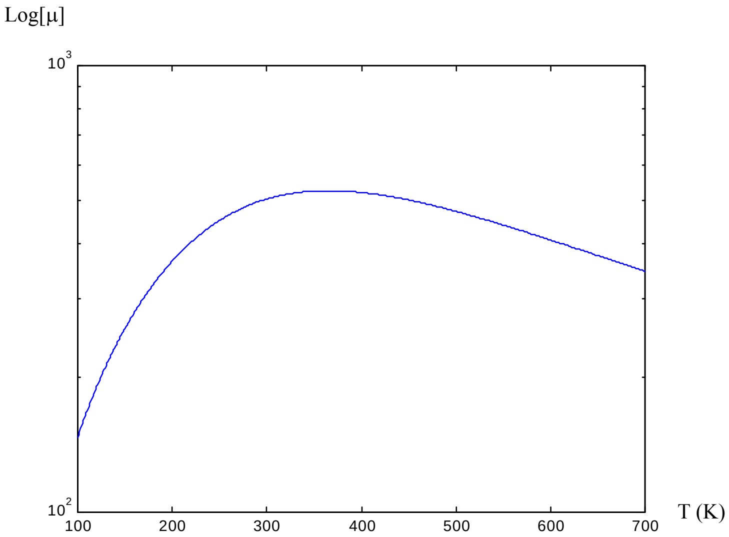
$$v_d = -\mu_n \mathcal{E} \quad \rightarrow \quad \mathcal{E} = \frac{v_d}{\mu_n} = \frac{2.29 \times 10^6\, cm/\sec}{1400\, cm^2/V\sec} = 1635.71\, Vcm^{-1}\ .$$

Then, the voltage across the region is
$V = \mathcal{E} \times width = 1635.71\, Vcm^{-1} \times 10^{-4}\, cm = 0.16V$.

2.3 (a)



(b) If we combine μ_1 and μ_2,



The total mobility at 300 K is

$$\mu_{TOTAL}(300K) = \left(\frac{1}{\mu_1(300K)} + \frac{1}{\mu_2(300K)} \right)^{-1} = 502.55\,cm^2/V\sec.$$

(c) The applied electric field is

$$\mathcal{E} = \frac{V}{l} = \frac{1V}{1mm} = 10V/cm.$$

The current density is

$$J_{ndrift} = q\mu_n n\mathcal{E} = q\mu_n N_d \mathcal{E} = 80.41A/cm^2.$$

Drift

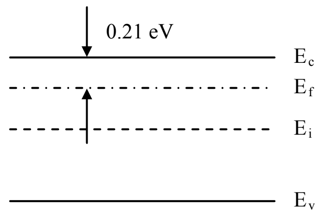
2.4 (a) From Figure 2-8 on page 45, we find the resistivity of the N-type sample doped with 1×10^{16}cm^{-3} of phosphorous is 0.5 Ω-cm.

(b) The acceptor density (boron) exceeds the donor density (P). Hence, the resulting conductivity is P-type, and the net dopant concentration is $N_{net} = |N_d - N_a| = p = 9\times10^{16}$cm^{-3} of holes. However, the mobilities of electrons and holes depend on the total dopant concentration, $N_T=1.1\times10^{17}$cm^{-3}. So, we have to use Equation (2.2.14) to calculate the resistivity. From Figure 2-5, $\mu_p(N_T=1.1\times10^{17}\text{cm}^{-3})$ is 250 cm^2/Vsec. The resistivity is

$$\rho = \frac{1}{\sigma} = \frac{1}{qN_{net}\mu_p} = \frac{1}{q\times9\times10^{16}cm^{-3}\times(250cm^2/V\sec)} = 0.28\,\Omega cm.$$

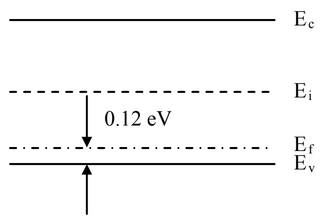
(c) For the sample in part (a),

$$E_c - E_f = kT\ln\left(\frac{N_c}{N_d}\right) = 0.026V\ln\left(\frac{2.8\times10^{19}cm^{-3}}{10^{16}cm^{-3}}\right) = 0.21eV.$$



For the sample in part (b),

$$E_f - E_v = kT \ln\left(\frac{N_v}{N_{net}}\right) = 0.026V \ln\left(\frac{1.04 \times 10^{19} cm^{-3}}{9 \times 10^{16} cm^{-3}}\right) = 0.12 eV$$



2.5 (a) Sample 1: N-type Holes are minority carriers.
$p = n_i^2/N_d = (10^{10}cm^{-3})^2/10^{17}cm^{-3} = 10^2\ cm^{-3}$

Sample 2: P-type Electrons are minority carriers.
$n = n_i^2/N_a = (10^{10}cm^{-3})^2/10^{15}cm^{-3} = 10^5\ cm^{-3}$

Sample 3: N-type Holes are minority carriers.
$p = n_i^2/N_{net} = (10^{10}cm^{-3})^2/(9.9\times10^{17}cm^{-3}) \approx 10^2\ cm^{-3}$

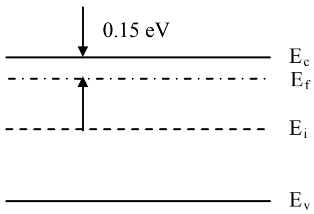
(b) Sample 1: $N_d = 10^{17}cm^{-3}$
$\mu_n(N_d = 10^{17}cm^{-3}) = 750\ cm^2/Vsec$ (from Figure 2-4)
$\sigma = qN_d\mu_n = 12\ \Omega^{-1}cm^{-1}$

Sample 2: $N_a = 10^{15}cm^{-3}$
$\mu_p(N_a = 10^{15}cm^{-3}) = 480\ cm^2/Vsec$ (from Figure 2-4)
$\sigma = qN_a\mu_p = 12\ \Omega^{-1}cm^{-1}$

Sample 3: $N_T = N_d+N_a = 1.01\times10^{17}cm^{-3}$
$\mu_n(N_T = 1.01\times10^{17}cm^{-3}) = 750\ cm^2/Vsec$ (from Figure 2-4)
$N_{net} = N_d-N_a = 0.99\times10^{17}cm^{-3}$
$\sigma = qN_{net}\mu_n = 11.88\ \Omega^{-1}cm^{-1}$

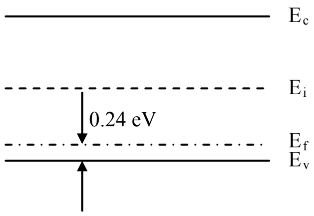
(c) For Sample 1,

$$E_c - E_f = kT \ln\left(\frac{N_c}{N_d}\right) = 0.026V \ln\left(\frac{2.8 \times 10^{19} cm^{-3}}{10^{17} cm^{-3}}\right) = 0.15 eV .$$



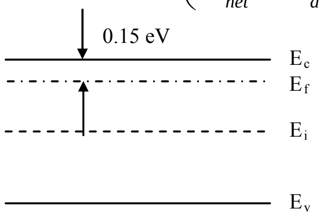
For Sample 2,

$$E_f - E_v = kT\ln\left(\frac{N_v}{N_a}\right) = 0.026V\ln\left(\frac{1.04\times10^{19}cm^{-3}}{10^{15}cm^{-3}}\right) = 0.24eV\ .$$



For Sample 3,

$$E_c - E_f = kT\ln\left(\frac{N_c}{N_{net} = N_d - N_a}\right) = 0.026V\ln\left(\frac{2.8\times10^{19}cm^{-3}}{9.9\times10^{16}cm^{-3}}\right) = 0.15eV\ .$$



2.6 (a) From Figure 2-5, $\mu_n(N_d = 10^{16}cm^{-3}$ of As) is 1250 cm^2/Vs. Using Equation (2.2.14), we find

$$\rho = \frac{1}{\sigma} = \frac{1}{qn\mu_n} = 0.5\ \Omega cm\,.$$

(b) The mobility of electrons in the sample depends not on the net dopant concentration but on the total dopant concentration N_T:

$$N_T = N_d + N_a = 2\times10^{16}cm^{-3}\,.$$

From Figure 2-5,

$$\mu_n(N_T) = 1140\,cm^2/Vs \quad and \quad \mu_p(N_T) = 390\,cm^2/Vs\,.$$

$N_{net} = N_d\text{-}N_a = 0$. Hence, we can assume that there are only intrinsic carriers in the sample. Using Equation (2.2.14),

$$\rho = \frac{1}{\sigma} = \frac{1}{qn_i\mu_n + qp_i\mu_p} = \frac{1}{qn_i(\mu_n + \mu_p)}$$
$$= \frac{1}{q \times 1 \times 10^{10} cm^{-3} \times (1140 + 390)(cm^2 / V\sec)}.$$

The resistivity is 4.08×10^5 Ω-cm.

(c) Now, the total dopant concentration (N_T) is 0. Using the electron and hole mobilities for lightly doped semiconductors (from Table 2.1), we have

$$\mu_n = 1400\, cm^2 / V\sec \quad and \quad \mu_p = 470\, cm^2 / V\sec .$$

Using Equation (2.2.14),

$$\rho = \frac{1}{\sigma} = \frac{1}{qn_i\mu_n + qp_i\mu_p} = \frac{1}{qn_i(\mu_n + \mu_p)}$$
$$= \frac{1}{q \times 1 \times 10^{10} cm^{-3} \times (1400 + 470)(cm^2 / V\sec)}.$$

The resistivity is 3.34×10^5 Ω-cm. The resistivity of the doped sample in part (b) is higher due to ionized impurity scattering.

2.7 It is given that the sample is *n*-type, and the applied electric field ε is1000V/cm. The hole velocity υ_{dp} is 2×10^5cm/s.

(a) From the velocity and the applied electric field, we can calculate the mobility of holes:

$\upsilon_{dp} = \mu_p\varepsilon$, $\mu_p = \upsilon_{dp}/\varepsilon =$ $2\times10^5/1000 = 200\text{cm}^2/\text{V}\cdot\text{s}$.

From Figure 2-5, we find N_d is equal to $4.5\times10^{17}/\text{cm}^3$. Hence,

$n = N_d = 4.5\times10^{17}/\text{cm}^3$, and $p = n_i^2/n = n_i^2/ N_d = 10^{20} / 4.5\times10^{17} = 222/\text{cm}^3$.

Clearly, the minority carriers are the holes.

(b) The Fermi level with respect to E_c is

$E_f = E_c - kT\,ln(N_d/N_c) = E_c - 0.107$ eV.

(c) R = ρL/A. Using Equation (2.2.14), we first calculate the resistivity of the sample:

$\sigma = q(\mu_n n + \mu_p p) \approx q\mu_n n = 1.6\times10^{-19} \times 400 \times 4.5\times10^{17} = 28.8/\Omega\text{-cm}$, and
$\rho = \sigma^{-1} = 0.035\ \Omega\text{-cm}$.

Therefore, R = (0.035) × 20μm / (10μm× 1.5μm) = 467 Ω.

Diffusion

2.8 (a) Using Equation (2.3.2),

$J = qn\upsilon = qD(dn/dx).$

Therefore,

$\upsilon = D(1/n)(dn/dx) = -D/\lambda.$ **(**constant**)**

(b) $J = q\mu_n n\mathcal{E} = qn\upsilon$ and $\upsilon = \mu_n\mathcal{E}.$

Therefore, $\mathcal{E} = -D/\mu_n \lambda = -(kT/q)/\lambda.$

(c) $\mathcal{E}$ = -1000V/cm = -0.026/λ. Solving for λ yields 0.25μm.

2.9 (a) $\mathcal{E} = -\frac{dV}{dx} = \frac{1}{q}\frac{dE_v}{dx} = \frac{1}{q}\frac{\Delta}{L} = \frac{\Delta}{qL}.$

(b) E_c is parallel to E_v. Hence, we can calculate the electron concentration in terms of E_c.

$$n(x) = n_0 e^{-(E_c(x)-E_c(0))/kT} \quad where \quad E_c(x) - E_c(0) = (\Delta / L)x.$$

Therefore, $n(x) = n_0 e^{-x\Delta / LkT}$.

(c) $J_n q n \mu_n \mathcal{E} + qD_n \frac{dn}{dx} = 0$

$$qn_i e^{-\Delta x / LkT} \mu_n \frac{\Delta}{qL} + qD_n n_i e^{-\Delta x / LkT}\left(-\frac{\Delta}{LkT}\right) = 0$$

Therefore,

$$\frac{\mu_n}{q} = \frac{D_n}{kT} \Rightarrow D_n = \frac{kT}{q}\mu_n.$$

Chapter 3

Terminology and knowledge

3.1 Integrated semiconductor companies
Companies that design and fabricate integrated circuits

Fabless
With no fabrication/processing facility

Foundries
Companies that specialize in processing wafers to produce silicon devices

Wafer fab
Fabrication facility where wafers are processed to produce silicon devices

Integrated circuits
System of transistors manufactured on silicon wafers

CPU
Central processor unit that is an active part of computer containing the datapath and control

DRAM
Dynamic random access memory

Flat-panel displays
Display panels with flat screens

MEMS
Micro-electro-mechanical-system

DNA chips
Silicon chips used for DNA screening

Dry oxidation
Growth of SiO_2 using oxygen gas

Wet oxidation
Growth of SiO_2 using water vapor

Horizontal furnace
Horizontally oriented oxidation furnace

Vertical furnace
Vertically oriented oxidation furnace

Photolithography/Optical lithography
Process in which the resist is optically patterned and selectively removed from designated areas on a wafer

Wafer stepper
Equipment used in lithography process

Photoresist
Ultraviolet-light sensitive material

Photomask
Quartz photo-plate containing a copy of pattern to be transferred to Si or SiO_2 surface

Negative resist
Photoresist that becomes polymerized and resistant to a developer when exposed to an UV light

Positive resist
Photoresist whose stabilizer breaks down when exposed to an UV light, leading to the preferential removal of exposed regions in a developer

Strip
Removal of photoresist

Asher
System that removes the resist on a wafer by oxidizing it in oxygen plasma or UV ozone system

Lithography field
Small area exposed to an UV light during the exposure through a photomask and an optical reduction system

Stepper
Another name for lithography equipment

Step-and-repeat action
The process of exposing different parts of a wafer until the whole wafer has been exposed

Phase-shift photomask
Photomask that produces 180 degree phase difference in neighboring clear features so that their diffraction fringes cancel each other

Optical Proximity Correction (OPC)
Printing a slightly different shape on the photomask to correct distortions resulting from an exposure process

Overlay
Alignment between 2 separate lithography steps

Extreme UV lithography (EUVL)
Lithography that uses 13nm wavelength and is expected to result in much higher resolution

Soft-x-ray lithography
Old name of EUVL

Electron-beam lithography (EBL)
Lithography using a focused stream of electrons

Electron projection lithography (EPL)
EBL that exposes a complex pattern simultaneously using a mask and a reduction electron lens system

Wet etching
Removal of SiO_2 using hydrofluoric acid

Isotropic
Without preference in direction

Dry etching (also known as plasma etching or reactive-ion etching (RIE))
Removal of SiO_2 using plasma and reactive ions

Anisotropic
With preference in direction

Selectivity
The extent in which an etching process distinguishes between different materials

End-point detector
Detector that monitors the emission of characteristic light from the etching products so as to signal when etching should end

Plasma process induced damage / Wafer charging damage
Damage to devices on wafers due to the use of plasma

Antenna Effect
Sensitivity of the damage to the size of the conductor

Ion implantation
Method of doping in which ions of impurity are accelerated and shot into the semiconductor surface

Gas-source doping
Method of doping in which a gas reacts with silicon and liberates phosphorus so that phosphorous diffuses into the silicon substrate

Solid-source diffusion
Method of doping in which the dopants from the thin film coated on the silicon surface diffuse into silicon

Anneal
Heating of wafers for dopant activation and damage removal

Dopant activation
Making dopants behave as donors and acceptors by heating the wafers

Implantation dose
Total number of implanted ions/cm^2

Depth/ Range
The location of peak concentration below the surface of silicon

Straddle
Spread of dopant concentration profile

3.2 Diffusion
The movement of molecules from an area of high concentration to an area of low concentration

Junction depth
Thickness of diffusion layer

Diffusivity
Constant that describes how quickly a given impurity diffuses in silicon for a given furnace temperature

Predeposition
Portion of diffusion process step with the source present

Drive-in
Portion of diffusion process step without the source

Furnace annealing
Heating of wafers in a furnace for dopant activation and damage removal

Rapid thermal annealing (RTA)
Annealing process in which a wafer is rapidly heated to high temperature and cooled quickly down to the room temperature

Rapid thermal oxidation
Oxidation process in which a wafer is heated to the designated temperature quickly, oxidized, and then cooled rapidly down to the room temperature

Rapid thermal chemical vapor deposition (CVD)
Chemical vapor deposition process in which a wafer is heated to the designated temperature quickly, the material is deposited on the wafer, and the wafer is cooled rapidly down to the room temperature

Laser annealing
Annealing process in which a silicon wafer is heated with a pulsed laser

Transient enhanced diffusion (TED)
Diffusion phenomena in which diffusion rate is increased by crystal damage due to ion implantation

Interconnect
Metal connection between devices in integrated circuits

Inter-metal dielectrics
Materials used for electrical isolation between metal layers

Crystalline
Molecular structure with nearly perfect periodicity

Polycrystalline
Molecular structure composed of densely packed crystallites or grains of single-crystals

Amorphous
Structure with no atomic or molecular ordering

Grain boundary
Interface between crystal grains

Thin-film transistors (TFT)
Transistors made of amorphous or polycrystalline silicon, widely used in flat-panel displays

Sputtering target
The source material for sputtering

Reactive sputtering
Sputtering accompanied by chemical reaction of sputtered ions. For example, Ti sputtered in nitrogen gas forms TiN film on the Si wafer

Physical vapor deposition (PVD)
Another name for sputtering

Step coverage problem
The inability of sputtering to deposit uniform films in small holes or vertical features on wafer

High-temperature oxide (HTO)
Very conformal oxide formed by a CVD process at a high temperature

Low-pressure chemical vapor deposition (LPCVD)
CVD process at low pressure, which yields good thickness uniformity and low gas consumption

Plasma-enhanced chemical vapor deposition (PECVD)
CVD using plasma; low deposition temperatures

In-situ doping
Doping process in which dopant species are introduced during the CVD deposition of polycrystalline silicon

Spin-on
Process in which liquid materials are spun onto the wafer

Epitaxy
Special type of thin-film deposition technology that produces a crystalline layer on silicon surfaces that is an extension of the underlying semiconductor crystal arrangement

Metallization
Interconnection of individual devices by metal lines

Via/ Plug
Electrical connection between adjacent metal layers

Electromigration
Migration of metal along the crystal-grain boundaries in a quasi-random manner, causing voids to occur in metal interconnects

Damascene process
Process used to form copper interconnect lines

Chemical-mechanical polishing (CMP)
Process in which a polishing pad and slurry are used to polish away material and leave a very flat surface

Back-end process
Metallization; last step of IC fabrication

Front-end processes
Oxidation, diffusion

Planarization
Process to obtain a flat surface to improve lithography and etching

Multi-chip modules
Multiple chips put into one package

Solder bumps
Electrical connection between chip and package

Flip-chip bonding
Melting pre-formed solder bumps on IC pads to make all connections simultaneously

Burn-in
Subjecting IC packages to higher than normal voltage and temperatures; identify potential failures

Qualification
Routine used to verify the quality of manufacturing and reliability

Operation life test
Part of qualification process; to find out if the devices last over one thousand operating hours

3.3 (a) Lithography field
A small area having the best optical resolution (The beam intensity is uniform within a lithography field.)
(b) Misalignment
Layer to layer mismatch (Each new mask level should be aligned to one of the previous levels)
(c) Selectivity

The ratio of etching rate of the film to be etched to that of the substrate film below

(d) End-point detection
Detecting the exposed substrate film after the removal of the desired film.

(e) Step coverage
The ratio of film thickness deposited on the flat surface to that deposited on the non-flat surface.

(f) Electromigration
The movement of atoms in a metal film due to momentum transfer from the electrons carrying the current.

3.4 (a) Wet. Otherwise, it would take too long to grow the thick oxide.

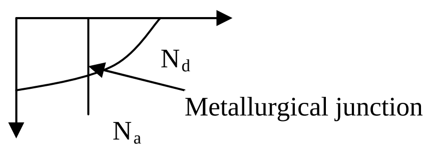
(b) Dry because it provides better control of the gate (channel) length and vertical wall profile.

(c) Arsenic because
- it is a donor ion (group V),
- it reduces R_p and ΔR_p, and
- it reduces diffusivity.

(d) PECVD

(e) Sputtering, dry etching, plasmas containing chlorine.

(f)



Oxidation

3.5 Wet oxidation is faster than dry oxidation because the solid solubility of H_2O steam into SiO_2 is higher than that of O_2 gas into SiO_2. This creates a very sharp concentration profile that causes H_2O to diffuse towards the SiO_2-Si interface much more effectively than O_2 under the same conditions.

3.6 (a) $\tau = \dfrac{0.2um * 0.2um + 0.165um * 0.2um}{0.0117um^2 / hr} = 6.239hr$

Solving the given equation for t_{ox} and using quadratic formula,

$$T_{ox} = \frac{-A + \sqrt{A^2 + 4B(t+\tau)}}{2} = 0.256um .$$

(b) Linear approximation: $T_{ox} \approx \dfrac{B}{A}(t+\tau)$ => 0.655um (156% error).

Quadratic approximation: $T_{ox}^2 \approx B(t+\tau)$ => 0.320um (25% error).

In this case $A \approx T_{ox}$, neither linear nor quadratic approximations would be valid.

Deposition

3.7 Poly Silicon:
$SiH_4 \rightarrow Si(s) + 2H_2$
Silicon Nitride:
$3SiH_2Cl_2 + 4NH_3 \rightarrow Si_3N_4 + 6HCl + 6H_2$
LTO:
$SiH_4 + O_2 \rightarrow SiO_2 + 2HCl + 2H_2$
HTO:
$SiH_2Cl_2 + 2H_2O \rightarrow SiO_2 + 2HCl + 2H_2$

Diffusion

3.8 (a) $x_j = \sqrt{Dt} \Rightarrow x_j + \Delta x_j = \sqrt{Dt + \Delta(Dt)}$

$$x_j^2 + 2x_j\Delta x_j + \left(\Delta x_j\right) = Dt + \Delta(Dt) = x_j^2 + \Delta(Dt)$$

$$2x_j\Delta x_j + \left(\Delta x_j\right) = \Delta(Dt)$$

Since $x_j >> \Delta x_j$,

$$2x_j\Delta x_j \approx \Delta(Dt) \quad \Rightarrow \quad \Delta x_j \approx \frac{\Delta(Dt)}{2x_j}.$$

(b) For boron,
D(500K) = 10.5×Exp[-3.69/(8.614×10-5×773)]cm2/sec = 9.0×10^{-24} cm^2/sec, and 500 years = 1.58×10^{10} sec.
Hence,

$$\Delta x_j = 1.42\times10^{-13}\,cm = 1.42\times10^{-9}\,\mu m.$$

Our assumption ($x_j >> \Delta x_j$) in part (a) is correct.

3.9 (a) Junction depth is distance from surface where the dopant concentration equals the substrate concentration.

$N_{sub} = N_d = N_{junction} = 10^{15}\,cm^{-3}$ (Required junction dopant concentration)

From Eq. (3.6.1),

$$N_{junction} = \frac{N_0}{\sqrt{\pi Dt}} e^{-x_j^2/4Dt} = 10^{15}\,cm^{-3}$$

Using

$$D = D_0 e^{-E_a/kT}, \quad D_0 = 10.5\,cm^2/\sec, \quad N_0 = 10^{15}\,cm^{-2}, \quad and\ E_a = 3.7eV,$$
$$X_j = 1.97\times 10^{-4}\,cm.$$

(b) X_j has the form

$$X_j = 2\sqrt{Dt}\left[\ln\left(\frac{k}{\sqrt{Dt}}\right)\right]^{1/2}.$$

Taking the derivative,

$$\frac{dX_j}{dDt} = \frac{1}{\sqrt{Dt}}\left\{\left[\ln\left(\frac{k}{\sqrt{Dt}}\right)\right]^{1/2} - \frac{1}{2\left[\ln\left(\frac{k}{\sqrt{Dt}}\right)\right]^{1/2}}\right\} = \frac{X_j}{2Dt} - \frac{1}{X_j}.$$

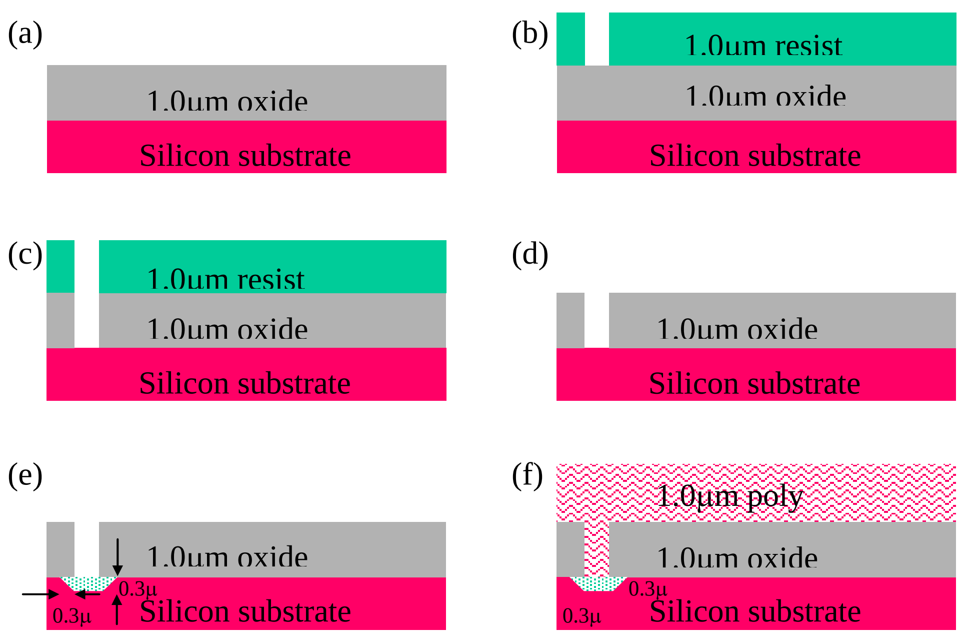
$$\frac{dX_j}{dDt} = \frac{1.98\times 10^{-4}}{2\times 10^{-9}} - \frac{1}{1.98\times 10^{-4}} = 9.39\times 10^4\,cm^{-1}$$

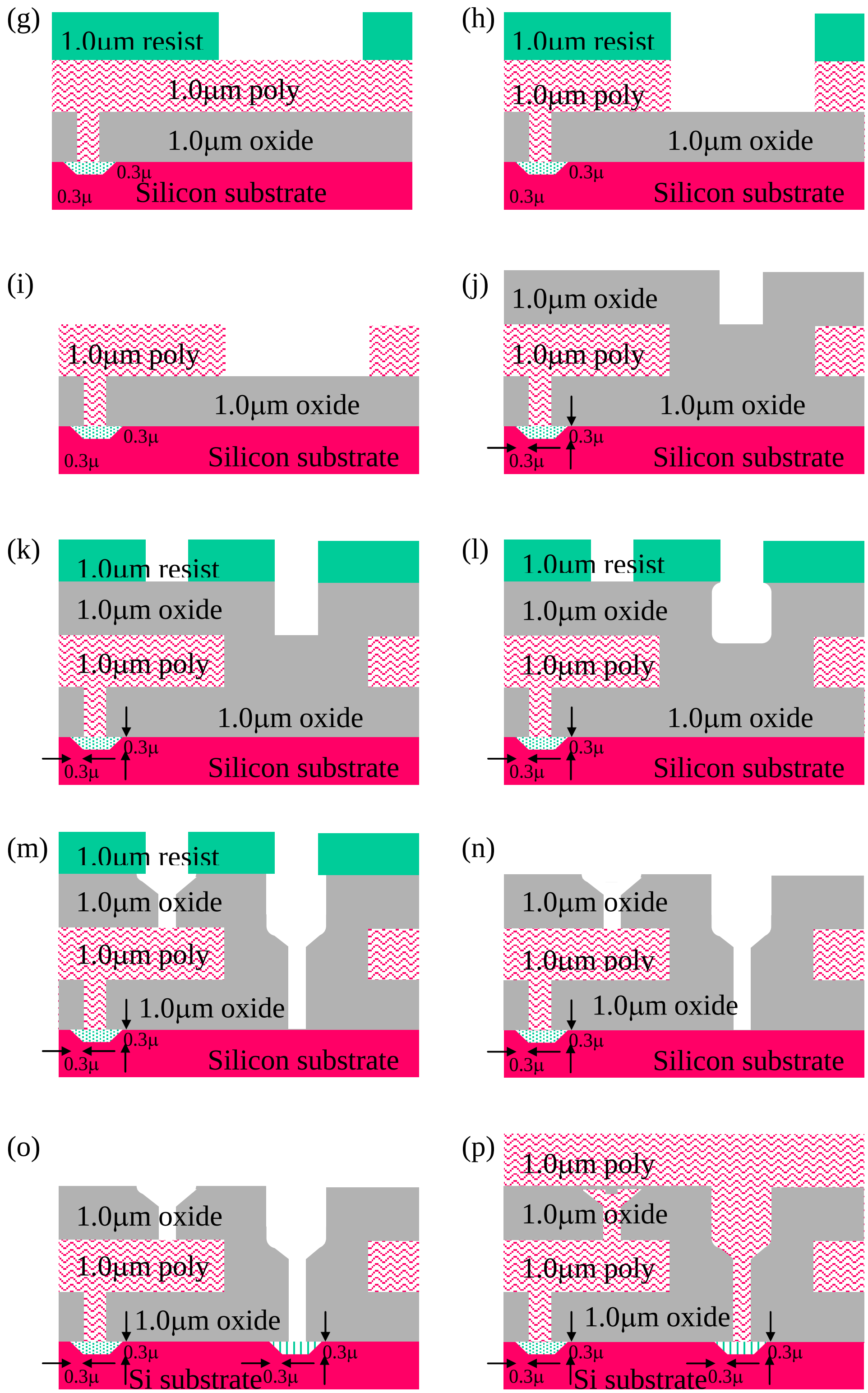
$$\Delta Dt = 10.5e^{\frac{-3.70}{373}}.\left(10^6\right)\left(3600\right) = 3.996\times 10^{-40}\,cm^2$$

Hence, $\Delta X_j = \frac{\partial X_j}{\partial Dt}.\Delta Dt = 3.75\times 10^{-35}\,cm \approx 0.$

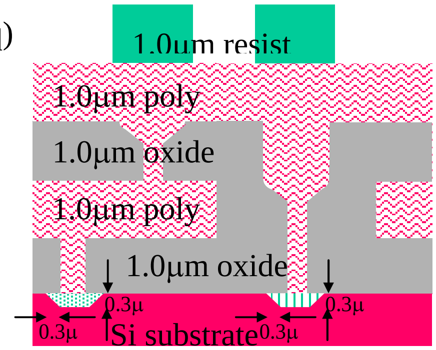
Visualization

3.10

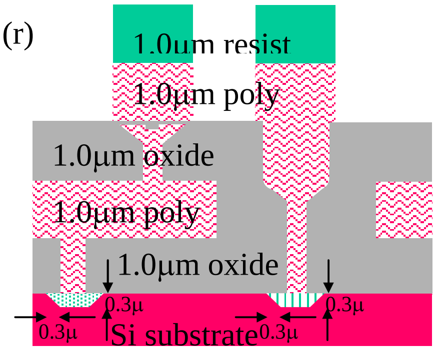



(g)
1.0μm resist
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Silicon substrate
(h)
1.0μm resist
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Silicon substrate
(i)
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Silicon substrate
(j)
1.0μm oxide
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Silicon substrate
(k)
1.0μm resist
1.0μm oxide
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Silicon substrate
(l)
1.0μm resist
1.0μm oxide
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Silicon substrate
(m)
1.0μm resist
1.0μm oxide
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Silicon substrate
(n)
1.0μm oxide
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Silicon substrate
(o)
1.0μm oxide
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Si substrate
0.3μ
0.3μ
(p)
1.0μm poly
1.0μm oxide
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Si substrate
0.3μ
0.3μ

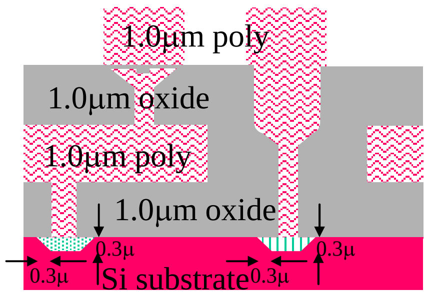
(q)


1.0μm resist
1.0μm poly
1.0μm oxide
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Si substrate
0.3μ
0.3μ

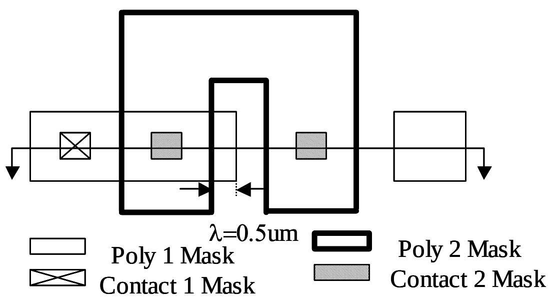
(r)


1.0μm resist
1.0μm poly
1.0μm oxide
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Si substrate
0.3μ
0.3μ

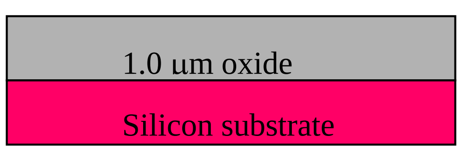
(s)


1.0μm poly
1.0μm oxide
1.0μm poly
1.0μm oxide
0.3μ
0.3μ
Si substrate
0.3μ
0.3μ

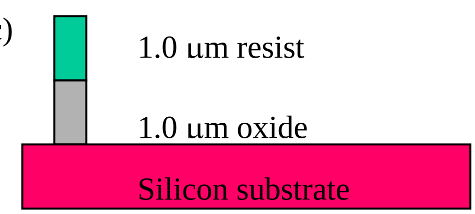
3.11


λ=0.5um
Poly 1 Mask
Contact 1 Mask
Poly 2 Mask
Contact 2 Mask

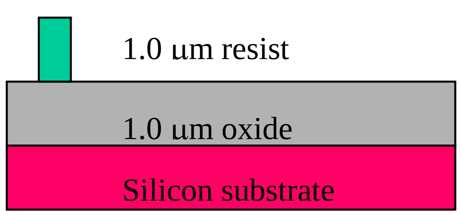
(a)


1.0 um oxide
Silicon substrate

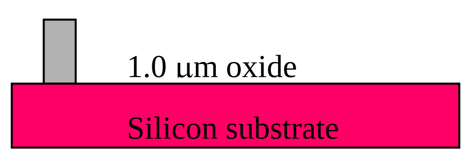
(c)


1.0 um resist
1.0 um oxide
Silicon substrate

(b)


1.0 um resist
1.0 um oxide
Silicon substrate

(d)


1.0 um oxide
Silicon substrate

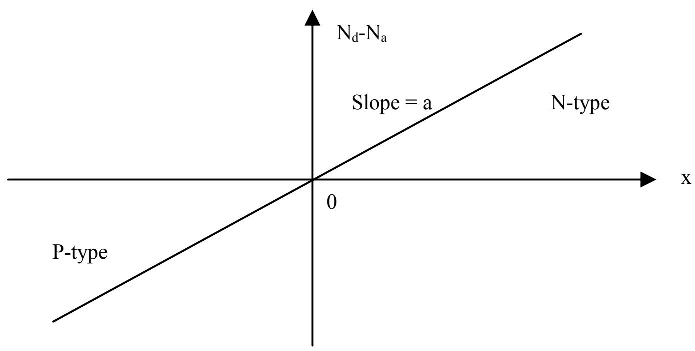
Changing the polarity of the contact 1 mask results in the same cross section as problem **3.10** (d). The same cross-section is obtained by using a negative resist and the reversed mask of contact 1, which is opaque in the inside of contact 1 area.

Chapter 4. Problems

Part I: PN Junction

Electrostatics of PN Junctions

4.1 The linearly graded junction is P-type for x<0 and n-type for x>0 with the junction at x = 0.



Since the doping levels are symmetric on either side of the junction, the depletion region widths into both sides must be the same (x_n=x_p=W_{dep}/2).

(a) To find the electric field distribution, we utilize the following relation:

$$\frac{d\mathcal{E}}{dx}=\frac{\rho}{\varepsilon_S}=\frac{q(N_d-N_a)}{\varepsilon_S}=\frac{q\times a\times x}{\varepsilon_S}.$$

Integrating the equation with the boundary conditions $\mathcal{E}$(x = x_n= x_p = W_{dep}/2) =0, we find

$$\mathcal{E}(x)=\frac{q\times a}{2\varepsilon_S}\left(x^2-\frac{W_{dep}^2}{4}\right).$$

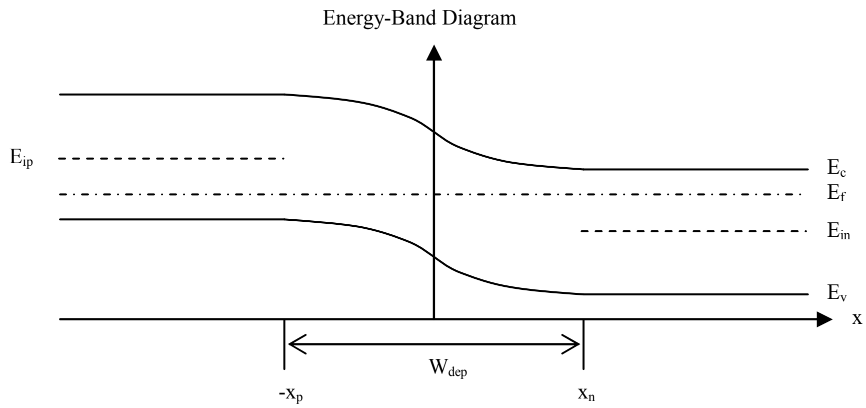
(b) The potential distribution is

$$\frac{dV}{dx}=-\mathcal{E}\quad\Rightarrow\quad V=-\frac{qa}{2\varepsilon_S}\left(\frac{x^3}{3}-\frac{W_{dep}^2x}{4}\right)+C.$$

For convenience, if we define V(x=0) =0 as a reference, we find C = 0. Hence,

$$\frac{dV}{dx}=-\mathcal{E}\quad\Rightarrow\quad V=-\frac{qa}{2\varepsilon_S}\left(\frac{x^3}{3}-\frac{W_{dep}^2x}{4}\right).$$

(c) The built-in potential is the same as $(E_{ip}-E_{in})/q$ at thermal equilibrium.



The carrier concentrations at the edges of the depletion region are

$$n(x_n) = n_i e^{\frac{(E_f - E_{in})}{kT}} \Rightarrow E_{in} = E_i(x = x_n) = E_f - kT\ln\left[\frac{n(x_n)}{n_i}\right] = E_f - kT\ln\left[\frac{aW_{dep}}{2n_i}\right]$$

and

$$p(x_p) = p_i e^{\frac{(E_{ip} - E_f)}{kT}} \Rightarrow E_{ip} = E_i(x = x_p) = E_f - kT\ln\left[\frac{p(x_p)}{n_i}\right] = E_f + kT\ln\left[\frac{aW_{dep}}{2n_i}\right].$$

Therefore, ϕ_{bi} is

$$\phi_{bi} = \frac{E_{ip} - E_{in}}{q} = 2\frac{kT}{q}\ln\frac{aW_{dep}}{2n_i}.$$

(d) The total voltage drop takes place in the depletion region. From part (b), we know that $|V(x_n)|=|V(x_p)|=qaW_{dep}^3/(24\varepsilon_S)$. Remember that $x_n=x_p=W_{dep}/2$. Hence, the potential drop is $\phi_{bi}-V_a = qaW_{dep}^3/(12\varepsilon_S)$. Solving for W_{dep} gives

$$W_{dep} = \left[\frac{12\varepsilon_S}{qa}(\phi_{bi} - V_a)\right]^{1/3}.$$

Note that if we substitute this in part (c), we can iteratively solve for ϕ_{bi}.

4.2 (a)

$$\phi_{bi} = \left(\frac{kT}{q}\right)\ln\left(\frac{N_a N_d}{n_i^2}\right) = (0.026V)\times\ln\left(\frac{(10^{16}cm^{-3})\times(5\times10^{15}cm^{-3})}{(10^{10}cm^{-3})^2}\right) = 0.7V$$

(b)

$$\begin{aligned} W_{dep} &= \sqrt{\frac{2\varepsilon_s\phi_{bi}}{q}\left(\frac{1}{N_a}+\frac{1}{N_d}\right)} \\ &= \sqrt{\frac{2\times12\times(8.85\times10^{-14})\times0.7}{1.6\times10^{-19}}\times\left(\frac{1}{10^{16}}+\frac{1}{5\times10^{15}}\right)} \\ &= 5.28\times10^{-5}cm \\ &= 0.528\mu m \\ &= 528nm \end{aligned}$$

In order to calculate x_n and x_p we need to use $W_{dep} = x_n+x_p$ and $x_n N_d=x_p N_a$.

$$x_n = W_{dep} - x_p = W_{dep} - \frac{N_d}{N_a}x_n$$

$$\therefore x_n = \left(\frac{N_a}{N_a+N_d}\right)W_{dep} = \left(\frac{5\times10^{15}cm^{-3}}{(5\times10^{15}cm^{-3})+(10^{16}cm^{-3})}\right)\times(0.528\mu m) = 0.176\mu m$$

Likewise,

$$x_p = \left(\frac{N_d}{N_a+N_d}\right)W_{dep} = \left(\frac{10^{16}cm^{-3}}{(5\times10^{15}cm^{-3})+(10^{16}cm^{-3})}\right)\times(0.528\mu m) = 0.352\mu m$$

(c)

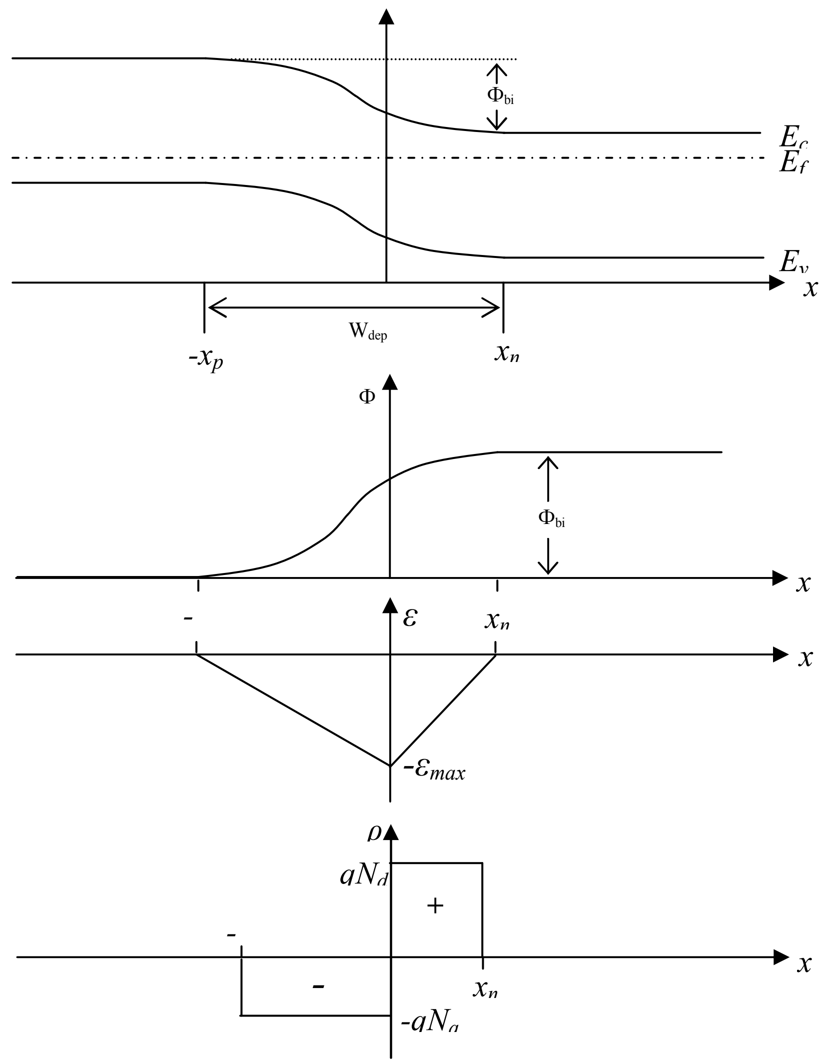
$$\mathcal{E}_{max} = \frac{qN_d x_n}{\varepsilon_s} = \frac{(1.6\times10^{-19}C)\times(10^{16}cm^{-3})\times(1.76\times10^{-5}cm)}{12\times(8.85\times10^{-14}F/cm)} = 2.652\times10^4 V/cm$$

Since the built-in potential is the integration of the electric field, the maximum electric field can also be calculated from the area of the triangle of the electric field profile.

$$\phi_{bi} = \frac{1}{2}W_{dep}\mathcal{E}_{max}$$

$$\therefore \mathcal{E}_{max} = \frac{2\phi_{bi}}{W_{dep}} = \frac{2\times(0.7V)}{5.28\times10^{-5}cm} = 2.652\times10^4 V/cm$$

(d)



(e)

$$\phi_{bi} = \left(\frac{kT}{q}\right)\ln\left(\frac{N_a N_d}{n_i^2}\right) = (0.026\,V)\times\ln\left(\frac{(10^{16}\,cm^{-3})\times(10^{18}\,cm^{-3})}{(10^{10}\,cm^{-3})^2}\right) = 0.838\,V$$

$$\begin{aligned} W_{dep} &= \sqrt{\frac{2\varepsilon_s\phi_{bi}}{q}\left(\frac{1}{N_a}+\frac{1}{N_d}\right)} \\ &= \sqrt{\frac{2\times 12\times(8.85\times10^{-14})\times 0.838}{1.6\times10^{-19}}\times\left(\frac{1}{10^{16}}+\frac{1}{10^{18}}\right)} \\ &= 3.35\times10^{-5}\,cm \\ &= 0.335\,\mu m \end{aligned}$$

$$x_n = \left(\frac{N_a}{N_a + N_d}\right) W_{dep} = \left(\frac{10^{18} cm^{-3}}{\left(10^{18} cm^{-3}\right) + \left(10^{16} cm^{-3}\right)}\right) \times (0.335 \mu m) = 0.332 \mu m$$

$$x_p = \left(\frac{N_d}{N_a + N_d}\right) W_{dep} = \left(\frac{10^{16} cm^{-3}}{\left(10^{18} cm^{-3}\right) + \left(10^{16} cm^{-3}\right)}\right) \times (0.366 \mu m) = 0.003 \mu m$$

$$\mathcal{E}_{max} = \frac{2\phi_{bi}}{W_{dep}} = \frac{2 \times (0.838 V)}{3.35 \times 10^{-5} cm} = 5.003 \times 10^4 V / cm$$

The depletion width has decreased due to a higher doping and since the acceptor doping is now 100 times greater than the donor concentration, most of the depletion region is on the n-side.

4.3 (a) If we apply reverse bias to the sample, the depletion width will increase. From Equation (4.2.5), we know that $N_a x_p = N_d x_n$. This means that the numbers of ionized dopants on the N and the P sides are equal. The side that has its dopants totally ionized is fully depleted. Hence, we need to find the total number of dopants per unit area on each side. The side with the smaller value will become depleted before the other.

$N_a \times W_P = 5 \times 10^{16} cm^{-3} \times 1.2 \times 10^{-4} cm = 6 \times 10^{12} cm^{-2}$ and
$N_d \times W_N = 1 \times 10^{17} cm^{-3} \times 0.4 \times 10^{-4} cm = 4 \times 10^{12} cm^{-2}$.

where W_P and W_N are the widths of the P-type region and N-type region, respectively. Since the N-type region contains less dopants per unit area, the N-type region will be fully depleted before the P-type region.

(b) Repeating the calculation for $N_d = 1 \times 10^{16} cm^{-3}$,

$$N_a \times W_P = 1 \times 10^{16} cm^{-3} \times 1.2 \times 10^{-4} cm = 1.2 \times 10^{12} cm^{-2}.$$

Hence, the P-type region will be fully depleted first.

(Note: You can also solve part (a) and (b) by finding the voltage at which each side depletes. However, this will make the problem unnecessarily complicated.)

(c) From Equation (4.4.1), we know that

$$C_{dep/unit\ area} = \frac{\varepsilon_S}{W_{dep}}.$$

First, we need to find the total depletion region width.

For part (a), we know that the N-type region is fully depleted. And, from Equation (4.2.5), $N_a x_p = N_d x_n$. Hence,

$$N_d W_N = N_a x_p \Rightarrow x_p = \frac{N_d W_N}{N_a} = 0.8\,\mu m \Rightarrow W_{dep} = 0.4\,\mu m + 0.8\,\mu m = 1.2\,\mu m.$$

$$C_{dep/unit\ area} = \frac{\varepsilon_S}{1.2\,\mu m} = 8.63\times10^{-9}\,F/cm^2.$$

For part (b), we know that the P-type region is fully depleted. And, from Equation (4.2.5), $N_a x_p = N_d x_n$. Hence,

$$N_d x_n = N_a W_P \Rightarrow x_n = \frac{N_a W_P}{N_d} = 0.12\,\mu m \Rightarrow W_{dep} = 0.12\,\mu m + 1.2\,\mu m = 1.32\,\mu m.$$

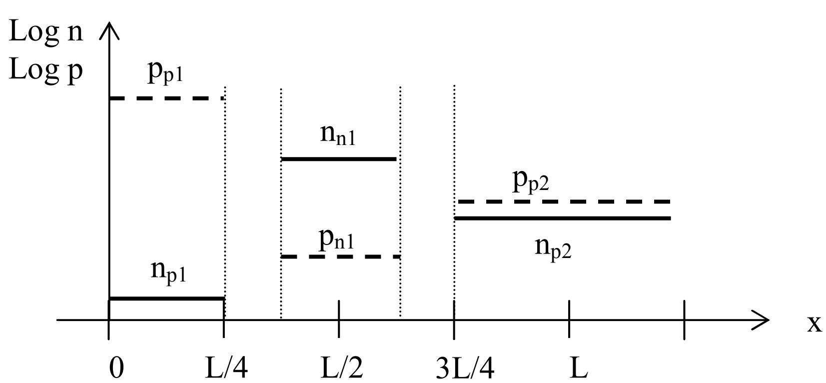
$$C_{dep/unit\ area} = \frac{\varepsilon_S}{1.32\,\mu m} = 7.85\times10^{-9}\,F/cm^2.$$

4.4 (a) At thermal equilibrium, the Fermi level is constant throughout the system. Since this is the case for the given figure, equilibrium conditions prevail.

(b) Carrier concentrations are given by:

$$n = N_c e^{-(E_c - E_f)/kT} \quad and \quad n = N_v e^{-(E_f - E_V)/kT}.$$

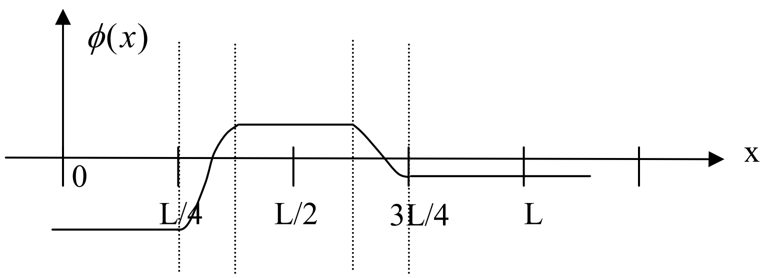
Therefore,



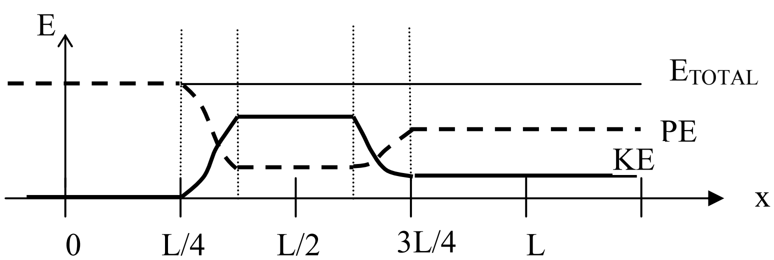
(c) The potential is the reverse of the band diagram. That is,

$$\phi = -\frac{1}{q}(E_i - E_f).$$

For this problem, sketching the qualitative shape is sufficient.



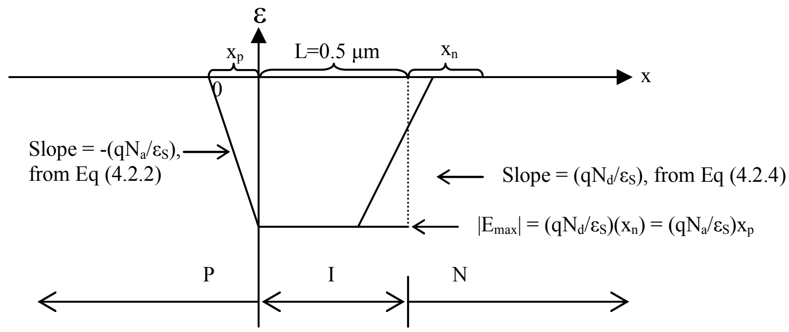
(d) We assume that the total energy is constant: $E_{TOTAL} = E_{KE} + E_{PE} =$ Constant.



4.5 (a) The intrinsic region has no dopants. Consequently, there is no charge exposed in the I-region. On the N-type side, the ionized donor atoms are exposed due to the diffusion of the electrons. The same situation prevails on the P-type region due to the diffusion of holes. The field lines that start from the ionized donor atoms on the N-type side do not have any negative charge to terminate on until they reach the P-type region where ionized acceptors are present. Thus, the field is constant in the I-region. Alternatively, you may note that $d\mathcal{E}/dx=0$ in the intrinsic region since $\rho = 0$. The free-carriers that are present in the I-region will be swept away by the built-in electric field. As a result, we may assume the I-region is fully depleted. To find the depletion-region width, we first need to calculate the built-in potential ϕ_{bi} using Equation (4.1.2):

$$\phi_{bi} = \frac{kT}{q}\ln\left(\frac{N_d N_a}{n_i^2}\right) = 0.78V .$$

The electric field distribution in the structure is



ϕ_{bi} is the voltage drop across the P-I-N structure, and is the sum of the area under the electric field in each region. That is,

$$\phi_{bi} = \frac{1}{2}\mathcal{E}_{\max} x_p + \mathcal{E}_{\max} L + \frac{1}{2}\mathcal{E}_{\max} x_n \quad \text{where } \mathcal{E}_{\max} = \frac{qN_d}{\varepsilon_S} x_n .$$

We also know from Equation (4.2.5) that

$$x_p N_a = x_n N_d \Rightarrow x_p = \frac{N_d}{N_a} x_n .$$

Hence, the equation becomes

$$\frac{qN_d}{2\varepsilon_S}\left(\frac{N_d}{N_a} + 1\right) x_n^2 + \frac{qN_d L}{\varepsilon_S} x_n - \phi_{bi} = 0 .$$

Solving for x_n yields 0.032 μm. Consequently, x_p is 0.019 μm. Then, the depletion-region width W_{dep} is $x_n + x_p + L = 0.551$ μm.

(b) We found the maximum electric field in part (a):

$$\left|\mathcal{E}_{\max}\right| = \frac{qN_d}{\varepsilon_S} x_n = \frac{qN_a}{\varepsilon_S} x_p = 148\, V/cm .$$

(c) We also found the built-in potential in part (a). It was 0.78 V.

(d) The breakdown voltage is the sum of the areas under the electric field distribution with $\mathcal{E}_{\max} = \mathcal{E}_{crit} = 2\cdot 10^5$ Vcm^{-1}. The depletion-region width in each region can be written in term of $\mathcal{E}_{crit}$ as

$$\left|\mathcal{E}_{crit}\right| = \frac{qN_d}{\varepsilon_S}x_n = \frac{qN_a}{\varepsilon_S}x_p \Rightarrow x_n = \frac{\left|\mathcal{E}_{crit}\varepsilon_S\right|}{qN_d} \text{ and } x_p = \frac{\left|\mathcal{E}_{crit}\varepsilon_S\right|}{qN_a}.$$

For P-N junction, the breakdown voltage is

$$V_B = \frac{1}{2}\mathcal{E}_{crit}x_p + \frac{1}{2}\mathcal{E}_{crit}x_n = \frac{\varepsilon_S}{2q}\mathcal{E}_{crit}^2\left(\frac{1}{N_d} + \frac{1}{N_a}\right) = 6.91\,V.$$

For P-I-N junction, the breakdown voltage is

$$V_B = \frac{1}{2}\mathcal{E}_{crit}x_p + \mathcal{E}_{crit}L + \frac{1}{2}\mathcal{E}_{crit}x_n =$$
$$= \frac{\varepsilon_S}{2q}\mathcal{E}_{crit}^2\left(\frac{1}{N_d} + \frac{1}{N_a}\right) + \mathcal{E}_{crit}0.5\mu m = 16.14\,V.$$

Therefore, the breakdown voltage of the P-I-N structure is ~ 10V higher than the P-N structure with the same doping levels.

Diffusion Equation

4.6 (a) At x = -∞, all the holes generated by illumination in the region x > 0 are recombined well before reaching x = -∞ due to the finite lifetime. Hence, there is no excess holes exist at x = -∞, and the hole concentration is

$$p = p_0 = \frac{n_i^2}{N_a} = 210.25\,cm^{-3}.$$

(b) At x = +∞, excess hole concentration won't have any position dependence. According to the continuity equations, photo generation will be balanced by thermal recombination (since the steady-state conditions prevail), therefore G_L = U = p'/τ.

$$p = p_0 + p' = 210.25\,cm^{-3} + G_L\tau \approx 10^9\,cm^{-3}.$$

(c) "Low-level injection" implies that the minority carrier concentration is much smaller than the majority carrier concentration. For this problem, since the maximum minority carrier concentration (p = 10^9 cm^{-3}) is much smaller than the majority carrier concentration (n = N_d = 10^{18} cm^{-3}), low-level injection conditions do prevail.

(d) The continuity equation is

$$\frac{dp'}{dt} = D_p \frac{\partial^2 p'(x)}{\partial x^2} + G_L - \frac{p'}{\tau} \quad and \quad \frac{dp'}{dt} = 0 \; at \; steady \; state .$$

For $x \geq 0$,

$$\frac{dp'}{dt} = D_p \frac{\partial^2 p'(x)}{\partial x^2} + G_L - \frac{p'}{\tau} \quad where \quad G_L = 10^{15} cm^{-3} \sec^{-1}$$

$$\Rightarrow p'(x) = A_1 e^{\left(x/\sqrt{D_p \tau}\right)} + A_2 e^{\left(-x/\sqrt{D_p \tau}\right)} + G_L \tau .$$

Since $p' \neq \infty$ as $x \to \infty$, $A_2 = 0$. Therefore,

$$p_1^{'}(x) = A_1 e^{\left(x/\sqrt{D_p \tau}\right)} + G_L \tau .$$

For $x < 0$,

$$p^{'}(x) = C_1 e^{\left(x/\sqrt{D_p \tau}\right)} + C_2 e^{\left(-x/\sqrt{D_p \tau}\right)} \quad (G_L = 0).$$

Similarly $C_1 = 0$. Therefore,

$$p_2'(x) = C_2 e^{\left(x/\sqrt{D_p \tau}\right)} .$$

Now, if we apply the boundary condition or continuity to the equations above, we find

$$p_1'(x=0) = p_2'(x=0) \; \& \; \frac{\partial p_1'(0)}{\partial x} = \frac{\partial p_2'(0)}{\partial x} \quad \Rightarrow \quad A_1 = C_2 + G_L \tau \; \& \; A_1 = -C_2$$

$$A_1 = \frac{G_L \tau}{2} = 5 \times 10^8 cm^{-3} \quad and \quad C_2 = -\frac{G_L \tau}{2} = -5 \times 10^8 cm^{-3} .$$

Therefore,

$$p_1'(x) = 10^9 cm^{-3} - 5 \times 10^8 e^{\left(-x/\sqrt{D_p \tau}\right)} \; \text{ for } x \geq 0 \; \text{ and}$$

$$p_2'(x) = 5 \times 10^8 e^{\left(x/\sqrt{D_p \tau}\right)} \quad for \quad x < 0 \; \text{ where } \; D_p \tau = 4.07 \times 10^{-6} cm^2 .$$

4.7 (a) Thermal equilibrium will be disturbed if any non-thermally generated excess carriers are present. If L (the distance from the origin to the excess carrier generation point) is much larger than the diffusion length of the holes (i.e., $L >> \sqrt{D_p \tau_p}$), then almost all the excess carriers generated due to the light would have recombined (i.e., $p_N' \approx 0$). In this case, we could assume thermal equilibrium. Otherwise, if L is not much larger than $\sqrt{D_p \tau_p}$, then significant number of excess carriers will be present at x=0 and the origin will not be at thermal equilibrium.

(b) $p_N'(x) = n_N'(x)$ everywhere in the semiconductor because of charge neutrality inside the semiconductor and all the excess carriers due to light are generated in pairs of electrons and holes. Excess concentrations at x = -L is γN_d. Total electron and hole concentrations at x=-L are $n(x = -L) = N_d + \gamma N_d$ and $p(x = -L) = n_i^2/N_d + \gamma N_d$.

(c) The excess carriers are generated at $x = -L$ and $x = L$. In the rest of the semiconductor the excess carriers decay due to recombination. That is,

$p_N'(-L) > p_N'(x)$ for $-L < x < L$.

So the maximum excess carrier concentration is at $x = -L$ and $x = L$. Now, $p_N'(-L) = p_N'(L) = \gamma N_d$. At $\gamma = 10^{-3}$, $p_N'(-L) << N_d$ in which N_d is the majority carrier concentration at thermal equilibrium. Therefore, $p_N'(x) << N_d$ for $-L \le x \le L$ which means the excess carrier concentration is always much smaller than the equilibrium majority carrier concentration. This is the condition for low-level injection.

(d) Inside the bar, carrier generation due to light is zero. In steady state, the continuity equation simplifies to terms involving diffusion and recombination:

$$0 = D_p \frac{d^2 p_N'}{dx^2} - \frac{p_N'}{\tau_n}.$$

Rearranging terms gives us

$$\frac{d^2 p_N'}{dx^2} = \frac{p_N'}{D_p \tau}.$$

(e) The general solution of the differential equation in part (d) is

$$\Delta p_N(x) = \mathrm{A} \cdot e^{-\frac{x}{L_p}} + B \cdot e^{\frac{x}{L_p}}$$

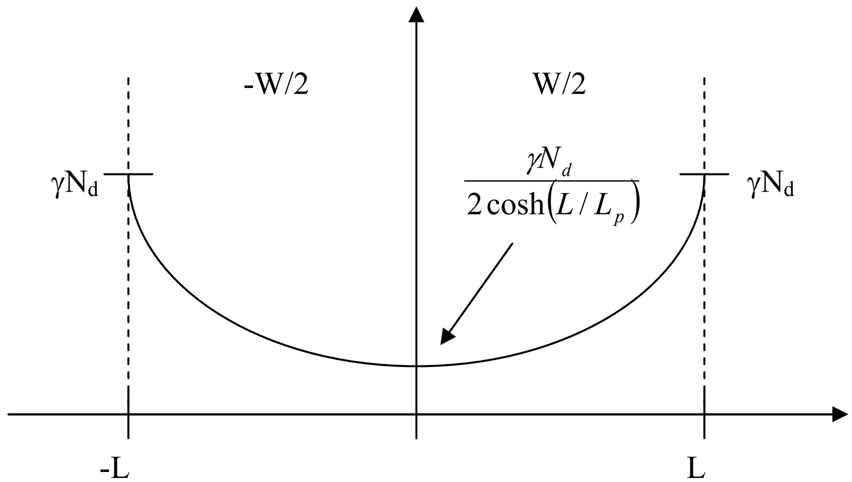
where $L_p = \sqrt{D_p \tau}$, and A and B are integration constants.

The boundary conditions for this problem are

$p_N'(-L) = p_N'(L) = \gamma N_d$

Although you do not have to solve the differential equation, the solution to this particular problem is

$$p_N'(x) = \frac{\gamma N_d}{2\cosh\left(\frac{L}{L_p}\right)} \cdot \cosh\left(\frac{x}{L_p}\right) \quad \text{(Please note that } cosh(z) = \frac{e^z + e^{-z}}{2}\text{.)}$$



This curve confirms what you would physically expect from carrier injection on both ends of a silicon bar.

4.8 (a) From Figure 2-5, μ_p is approximately 450 $cm^2V^{-1}sec^{-1}$. τ_p is given to be 1 μsec. Using Equation (2.5.6b) and (4.7.6), we calculate D_p and L_p, respectively:

$$D_p = \frac{kT}{q}\mu_p = 11.7 cm^2/\sec \quad and \quad L_p = \sqrt{D_p \tau_p} = 34.2 \mu m\,.$$

(b) Using Equation (4.6.3), we calculate the excess hole concentration:

$$p_N'(0) = p_{N0}\left[e^{qV/kT} - 1\right] = \frac{n_i^2}{N_d}\left[e^{qV/kT} - 1\right]$$

Hence, $p_N' = n_N' = 0$ for (i), and $p_N' = n_N' = 4.8 \times 10^{10}$ cm^{-3} for (ii).

4.9 (a) From Figure 2-8,

$$\rho = 0.2\ \Omega cm,\ n-type \quad \Rightarrow \quad N_d = 3 \times 10^{16} cm^{-3}, \text{ and}$$
$$\rho = 1\ \Omega cm,\ p-type \quad \Rightarrow \quad N_a = 1.5 \times 10^{16} cm^{-3}\,.$$

Therefore, using Equation (4.1.2), we find that the built-in voltage is

$$\phi_{bi} = kT \ln \frac{N_d N_a}{n_i^2} = 0.76\ V\,.$$

(b) Using Equation (4.8.2),

$$n_P(0) = \frac{n_i^2}{N_a}\left(e^{(qV/kT)} - 1\right) = 0.65\times10^{14}\,cm^{-3}, \text{ and}$$

$$p_N(0) = \frac{n_i^2}{N_d}\left(e^{(qV/kT)} - 1\right) = 3.25\times10^{13}\,cm^{-3}.$$

(c) P-type side:

$N_a = 1.5\times10^{16}cm^{-3} \Rightarrow \mu_n = 1150\ cm^2V^{-1}sec^{-1}$, $\tau_n=10^{-6}sec$, $D_n=29.9\ cm^2/sec$, and $L_n=5.47\times10^{-3}cm$.

Using Equation (4.9.2), we need to derive an expression for $J_n(x)$:

$$J_n(x) = -q\frac{D_n}{L_n}n_{P0}\left(e^{qV/kT} - 1\right)e^{x/L_n} =$$

$$= -q\frac{D_n}{L_n}\frac{n_i^2}{N_a}\left(e^{qV/kT} - 1\right)e^{x/L_n} = 54e^{x/L_n}\ mA/cm^2.$$

N-type side:

$N_d = 3\times10^{16}cm^{-3} \Rightarrow \mu_p = 400\ cm^2V^{-1}sec^{-1}$, $\tau_p=10^{-8}sec$, $D_p=10.4\ cm^2/sec$, and $L_p=3.22\times10^{-4}cm$.

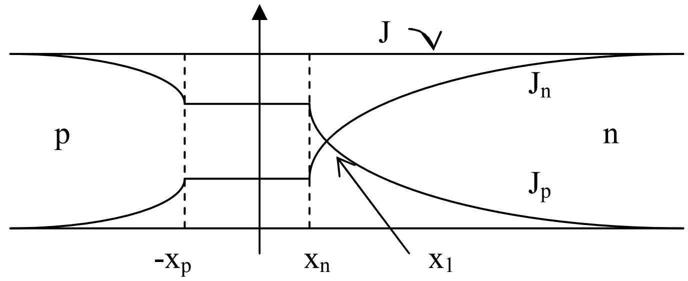
Using Equation (4.9.1), we need to derive an expression for $J_p(x)$:

$$J_p(x) = -q\frac{D_p}{L_p}p_{N0}\left(e^{qV/kT} - 1\right)e^{-x/L_p} =$$

$$= -q\frac{D_p}{L_p}\frac{n_i^2}{N_d}\left(e^{qV/kT} - 1\right)e^{-x/L_p} = 168e^{-x/L_p}\ mA/cm^2.$$

And the total current density J is

$$J = J_n(0) + J_p(0) = 0.222\ A/cm^2.$$



(d) At the point (x_1) where J_n and J_p cross each other, $J_n = J_p = ½ J$. Therefore,

$$J_p = \frac{0.222}{2} A/cm^2 = 0.111 A/cm^2 = 0.168 e^{-(x_1)/L_p} \Rightarrow x_1 = 1.38\,\mu m\,.$$

4.10 For this problem, please note that:

(1) $\Delta p_N = 0$ for $x_b \le x \le x_c$ because $\tau_p = 0$. This yields the boundary condition $\Delta p_N = 0$ and $x = x_b$.
(2) Because we have a P^+N diode, we need only deal with the N-side of the junction in establishing an expression for I. Moreover, the depletion width is all but totally on the N-side of the junction given a P^+N diode (i.e., $x_n \cong W_{dep}$).
(3) Because $\tau_p = \infty$ for $0 \le x \le x_b$, there will be no $I_{R\text{-}G}$ current. ($\tau_p = \infty$ implies that there are no R-G centers.) Thus, we only need to develop an expression for the diffusion current flowing in the diode.

Since we are interested in the static state, $\partial\Delta p_N/\partial t = 0$. Also, $G_L = 0$ (no light) and $\Delta p_N/\tau_p \rightarrow 0$ because $\tau_p = \infty$. Thus the minority carrier diffusion equation reduces to the form

$\partial^2\Delta p_N/\partial x^2 = 0$ for $W \le x \le x_b$

which is subject to the boundary conditions

$$\Delta p_N(x_b) = 0 \text{ and } \Delta p_N(W) = \frac{n_i^2}{N_d}\left(e^{\frac{qV_A}{kT}} - 1\right).$$

The general (narrow-base type) solution is $\Delta p_N(x) = A_1 + A_2 x$.
Applying the boundary conditions

$0 = A_1 + A_2 x_b$ and $\Delta p_N(W) = A_1 + A_2 W$

giving

$$\Delta p_N(W) = -A_2(x_b - W) \text{ or } A_2 = -\Delta p_N(W)/(x_b - W) \text{ and } A_1 = -A_2 x_b = \Delta p_N(W)\frac{x_b}{x_b - W}.$$

Thus $\Delta p_N(x) = \Delta p_N(W)\left(\dfrac{x_b - x}{x_b - W}\right) = \dfrac{n_i^2}{N_d}\left(\dfrac{x_b - x}{x_b - W}\right)\cdot\left(e^{\frac{qV_A}{kT}} - 1\right)$ for $W \le x \le x_b$,

$$J_p \cong -qD_p \frac{d\Delta p_n}{dx'} = q\frac{n_i^2}{N_d}\left(\frac{D_p}{x_b - W}\right)\cdot\left(e^{\frac{qV_A}{kT}} - 1\right) \text{and}$$

$$I \cong AJ_p = qA\frac{n_i^2}{N_d}\left(\frac{D_p}{x_b - W}\right)\cdot\left(e^{\frac{qV_A}{kT}} - 1\right).$$

Please note that since $\Delta p_N(x)$ is a linear function of x, J_p is constant throughout the narrow-base ($W_{dep} \le x_b \le x_b$) region.

Proof of Minority Drift Current Being Negligible

4.11 (a) 0.5 Ω-cm n-type Si => $N_d = 10^{16}\text{cm}^{-3}$, $D_n = 30\ \text{cm}^2\text{sec}^{-1}$, L_n=5.5 μm, $D_p = 10\ \text{cm}^2\text{sec}^{-1}$, L_p=3.2 μm

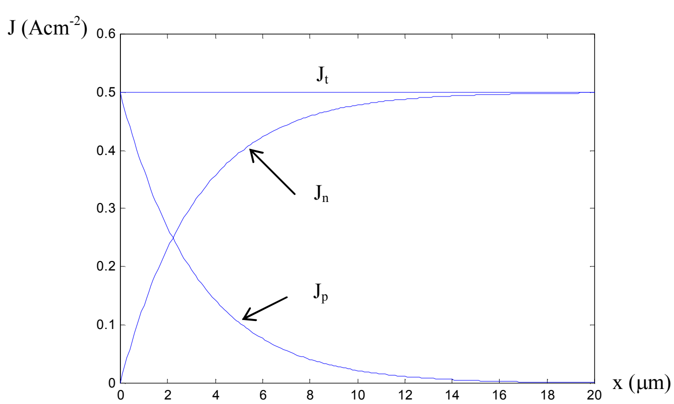
$$p(x) = p_{N0} + \frac{n_i^2}{N_d}\left(e^{qV_a/kT} - 1\right)e^{-x/L_p} \approx 10^{14}\, e^{-x/L_p}\, cm^{-3}.$$

(b) Minority current:

$$J_p(x) = -qD_p\frac{dp(x)}{dx} = q\frac{n_i^2}{N_d L_p}D_p\left(e^{qV_a/kT} - 1\right)e^{-x/L_p} = 0.5e^{-x/L_p}\, A/cm^2.$$

Majority current:

$$J_n(x) = J_t(x) - J_p(x) = J_p(0) - J_p(x) = 0.5\left(1 - e^{-x/L_p}\right)A/cm^2$$



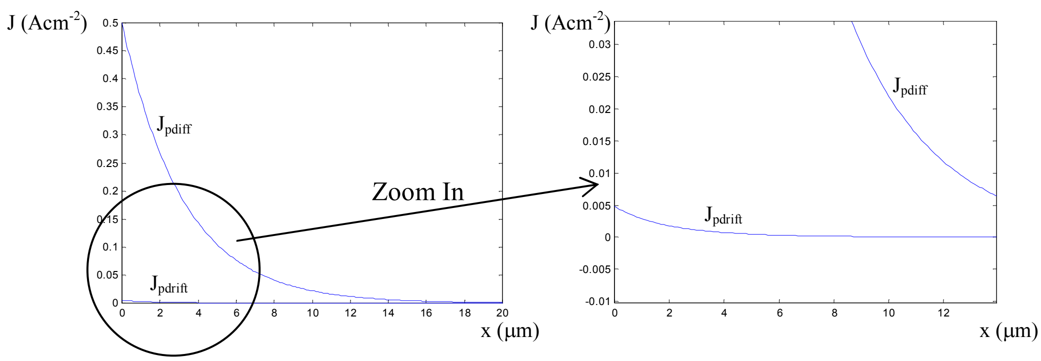
(c) $J_{ndiff}(x) = qD_n \dfrac{d(n_0 + n'(x))}{dx} = qD_n \dfrac{dn'(x)}{dx} = -J_p(x)\dfrac{D_n}{D_p} = 3J_p(x) = -1.5e^{-x/L_p}$

(d) $J_{ndrift}(x) = J_n(x) - J_{ndiff}(x) = 0.5 + 1 \times e^{-x/L_p}\ \ A/cm^2$.

$$E(x) = \frac{J_{ndrift}}{nq\mu_n} = 0.25 + 0.5 \times e^{-x/L_p}\ \ Vcm^{-1}.$$

$$J_{pdrift}(x) = pq\mu_p E(x) \approx p'q\mu_p E(x) = 1.6 \times 10^{-3} e^{-x/L_p}(1 + 2e^{-x/L_p})\ A/cm^2.$$

$$J_{pdrift} << J_{pdiff} \Rightarrow J_p \approx J_{pdiff}.$$



(e) From $E(x)$ in (d) and the Possion equation, Eq. (4.1.5), one the net charge density, ρ to be

$$1.7 \times 10^{-9}\ e^{-x/Lp}\ \ C/cm^{-3}.$$

This ρ is negligibly small compared with the excess hole charge density, qp' in (a), which is

$$1.6 \times 10^{-5}\ e^{-x/Lp}\ \ C/cm^{-3}.$$

This shows that $p' = n'$ is a good assumption because $\rho = qp'$ _ qn' .

Temperature Effect on IV

4.12 (a) From Equations (4.9.4) and (4.9.5),

$$I \approx I_0\left(e^{qV/kT}\right) \quad where \quad I_0 = AqN_cN_v e^{-E_g/kT}\left(\frac{D_p}{L_pN_d} + \frac{D_n}{L_nN_a}\right) \equiv \eta e^{-E_g/kT}.$$

For simplicity, we will ignore the weak temperature dependence of η.

$$V \approx \frac{kT}{q}\ln\left(\frac{I}{I_0}\right) \quad \Rightarrow \quad \frac{dV}{dT} \approx \frac{k}{q}\ln\left(\frac{I}{I_0}\right) + \frac{kT}{q}\frac{d}{dT}\left(\ln I - \ln \eta + \frac{E_g}{kT}\right)$$

$$\approx \frac{k}{q}\frac{qV}{kT} + \frac{kT}{q}\frac{d}{dT}\left(\frac{E_g}{kT}\right) \qquad .$$

$$= \frac{V}{T} - \frac{E_g}{kT}$$

(b) Assuming V=0.5V and T=300K,

$$\frac{dV}{dT} = \frac{0.5 - 1.1}{300} = 0.002 \text{ V/K}.$$

(c) In Fig. 4-21, start at T=300K, V=0.5V, or I = 240 μA:
At that fixed I, ΔV = -0.2 V between -25C and 75C (ΔT = 50 K). Hence, ΔV / ΔT ~ -0.2 V/ 100 ºK = -0.002 V/ºK, in excellent agreement with the value found in part (b).

4.13

$$I = Aq\left(\frac{n_i^2}{N_d}\frac{L_p}{\tau_p} + \frac{n_i^2}{N_a}\frac{L_n}{\tau_n}\right) = Aq\left(\frac{-p'}{\tau_p}L_p + \frac{-n'}{\tau_n}L_n\right).$$

4.14 (a) If we model the quasi-neutral regions as a resistor R, the voltage drop across the neutral regions is $V_{neutral} = IR$ and that across the junction is

$$V - V_{neutral} = V - IR \,.$$

Hence, Equation 4.9.4 becomes

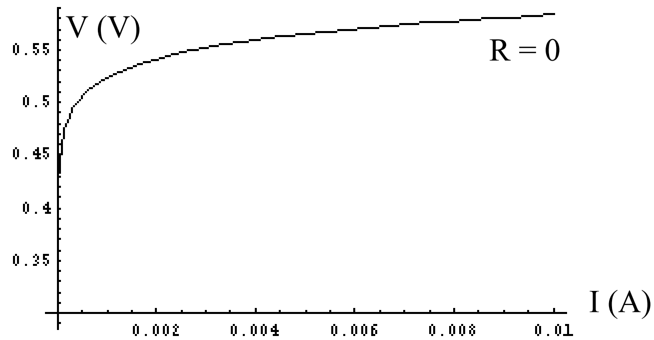
$$I = I_0\left(e^{q(V-IR)/kT} - 1\right).$$

(b) Solving the above equation for V yields

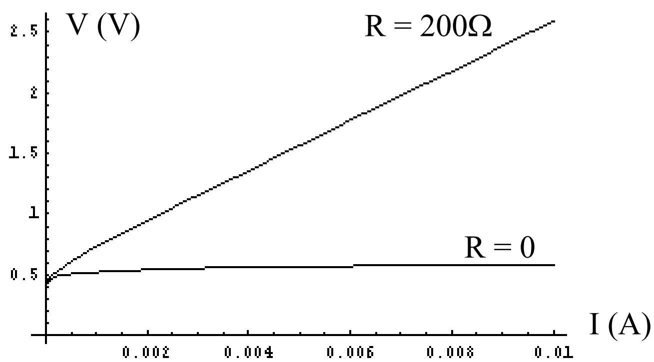
$$V = IR + \frac{kT}{q}\ln\left(\frac{I}{I_0} + 1\right).$$

(c) Let's use $I_0 = 1.8\times10^{-12}$A from Figure 4-22.

With R = 0,



With R = 200Ω,



Clearly for the case R = 200Ω, the first term IR becomes dominant over $\ln(I/I_0+1)$, and the IV curve becomes very linear.

Charge Storage

4.15 Please refer to Section 4.9, 4.10, 8.2, 8.3, and 8.7.

(a) First, we need to find an expression for Q_t:

$$Q_t = Aq\left[\int_{x_n}^{W_B} p_N'(x)dx + \int_{-W_E}^{-x_p} n_P'(x)dx\right] =$$

$$= Aq\,p_{N0}\left(e^{qV_a/kT}-1\right)\left(-\frac{W_B'}{2}\right)\left(1-\frac{x-x_n}{W_B'}\right)^2\Bigg|_{x_n}^{W_B}$$

$$+ Aqn_{P0}\left(e^{qV_a/kT}-1\right)\left(\frac{W_E'}{2}\right)\left(1+\frac{x+x_p}{W_E'}\right)^2\Bigg|_{-W_E}^{-x_p} =$$

$$= \frac{A\cdot q}{2}\left(e^{qV_A/kT}-1\right)\left(p_{N0}W_B'+n_{P0}W_E'\right) = \frac{1}{2}Aqn_i^2\left(e^{qV_a/kT}-1\right)\left(\frac{W_B'}{N_d}+\frac{W_E'}{N_a}\right).$$

W_B' and W_E' are the actual lengths of the base and emitter regions respectively excluding the depletion regions. I_t for a short-base diode is given by

$$I_t = Aqn_i^2\left(e^{qV_a/kT}-1\right)\left(\frac{D_p}{W_B'}\frac{1}{N_d}+\frac{D_n}{W_E'}\frac{1}{N_a}\right).$$

Hence,

$$\tau_S = \frac{Q_t}{I_t} = \frac{1}{2}\frac{\left(\frac{W_B'}{N_d}+\frac{W_E'}{N_a}\right)}{\left(\frac{D_p}{W_B'}\frac{1}{N_d}+\frac{D_n}{W_E'}\frac{1}{N_a}\right)} = \frac{(W_B'W_E')(N_aW_B'+N_dW_E')}{2(D_pW_E'N_a+D_nW_B'N_d)}.$$

(b) $s = 1.8\times10^{-10}$ sec ($<< \tau_S = \tau_S = 1\times10^{-6}$ sec)

(c) A short-based diode has a shorter storage time, thus it can operate at a higher frequency.

Part II: Application to Optoelectronic Devices

IV of Photodiode/Solar Cell

4.16 (a) Let us examine the minority carrier diffusion equation for hole. In general,

$$\frac{\Delta\partial p_N}{\partial t} = D_p\frac{\partial^2\Delta p_N}{\partial x^2} - \frac{\Delta p_N}{\tau_p} + G_L.$$

For the steady state problem at hand, $\partial\Delta p_N/\partial t = 0$. Also $\partial^2\Delta p_N/\partial x^2 = 0$ if one goes far from the junction on the N-side where carrier perturbation introduced by the junction has decayed to zero. Thus,

$$0 = -\frac{\Delta p_N(x\to\infty)}{\tau_p} + G_L \text{ or } \Delta p_N(x\to\infty) = G_L\tau_p. \Leftarrow \text{boundary condition}$$

(b) One simply parallels the ideal diode derivation to obtain the desired I-V_A relationship. Given a P^+N junction, however, we need consider only the lightly doped side of the junction. To simplify the mathematics, we set the origin of coordinates to the n-edge of the depletion region. Under steady state conditions and with x' as defined as above, we must solve

$$0 = D_p \frac{\partial^2 \Delta p_N}{\partial x^2} - \frac{\Delta p_N}{\tau_p} + G_L$$

subject to the boundary conditions

$$\Delta p_N(x'=0) \quad = \quad \frac{n_i^{\,2}}{N_d}\left(e^{\frac{qV_A}{kT}} - 1\right) \text{ and } \Delta p_N(x \to \infty) = G_L \tau_p .$$

The general solution is

$$\Delta p_N(x') = G_L \tau_p + A_1 e^{\frac{-x'}{L_p}} + A_2 e^{\frac{x'}{L_p}} .$$

Because $\exp(x'/L_p) \rightarrow \infty$ as $x' \rightarrow \infty$, the only way the second boundary condition can be satisfied is for A_2 to be identically zero. With $A_2 = 0$, the application of the first boundary condition yields

$$\Delta p_N(x'=0) = G_L \tau_p + A_1 = \frac{n_i^{\,2}}{N_d}\left(e^{\frac{qV_A}{kT}} - 1\right) \text{ or}$$

$$A_1 = \frac{n_i^{\,2}}{N_d}\left(e^{\frac{qV_A}{kT}} - 1\right) - G_L \tau_p \text{ and}$$

$$\Delta p_N(x') = G_L \tau_p + \left[\frac{n_i^{\,2}}{N_d}\left(e^{\frac{qV_A}{kT}} - 1\right) - G_L \tau_p\right] \cdot e^{\frac{-x'}{L_p}} .$$

The associated hole current density is then

$$J_p(x') = -qD_p \frac{d\Delta p_N}{dx'} = q\frac{D_p}{L_p}\left[\frac{n_i^{\,2}}{N_d}\left(e^{\frac{qV_A}{kT}} - 1\right) - G_L \tau_p\right] \cdot e^{\frac{-x'}{L_p}}$$

and for a P^+N diode,

$I = AJ = A[J_n(x = -x_p) + J_p(x = x_n)] \cong AJ_p(x' = 0)$ or

$$I = qA\frac{D_p}{L_p}\frac{n_i^{\,2}}{N_d}\left(e^{\frac{qV_A}{kT}} - 1\right) - qA\frac{D_p \tau_p}{L_p} G_L .$$

Finally noting $D_p \tau_p = L_p^2$, we conclude

$$I = I_0\left(e^{\frac{qV_A}{kT}} - 1\right) - I_L$$

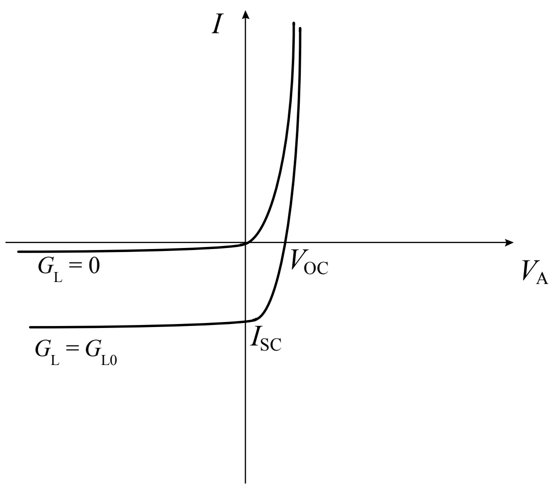
where

$$I_0 = qA\frac{D_p}{L_p}\frac{n_i^2}{N_d} \text{ and } I_L = qAL_pG_L .$$

(c) Voltage that appears when the illuminated diode is open circuit is

$$V_{OC} = \frac{kT}{q}\ln\left(1 + \frac{I_L}{I_0}\right).$$

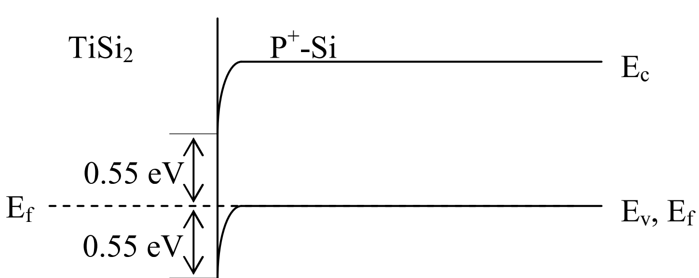
Current that flows when the illuminated diode is short-circuited is $I_{SC} = -I_L$.



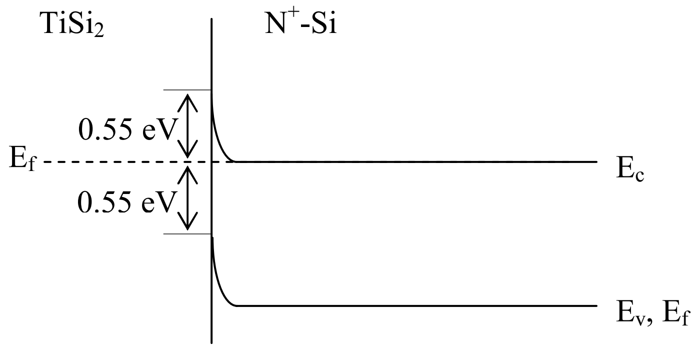
Part III: Metal-Semiconductor Junction

Ohmic Contacts and Schottky Diodes

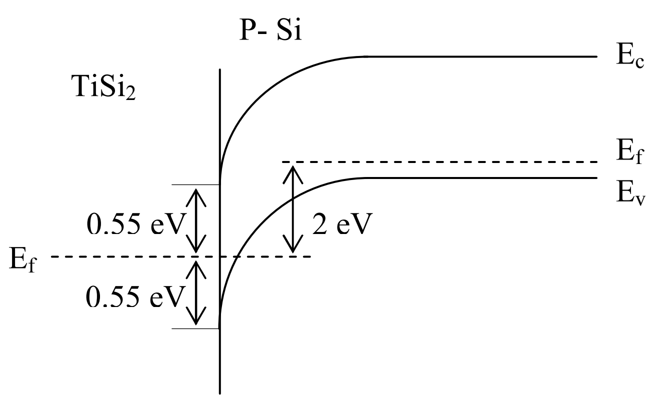
4.17 (a) A very heavy P^+ doping is important to forming an ohmic contact.



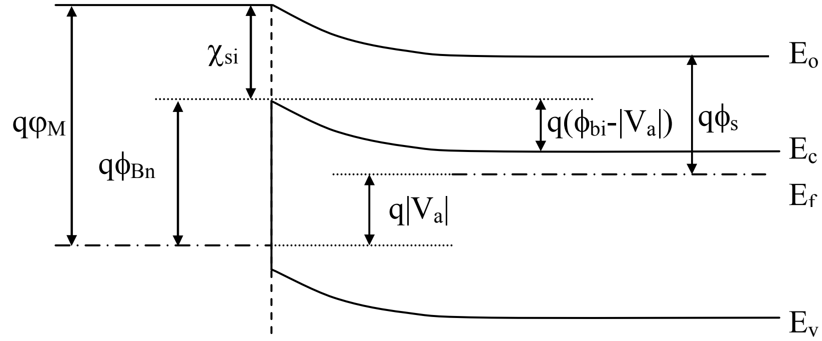
(b) A very heavy N^+ doping is important to forming an ohmic contact.



(c) A very heavy doping is unacceptable to forming a rectifying contact.

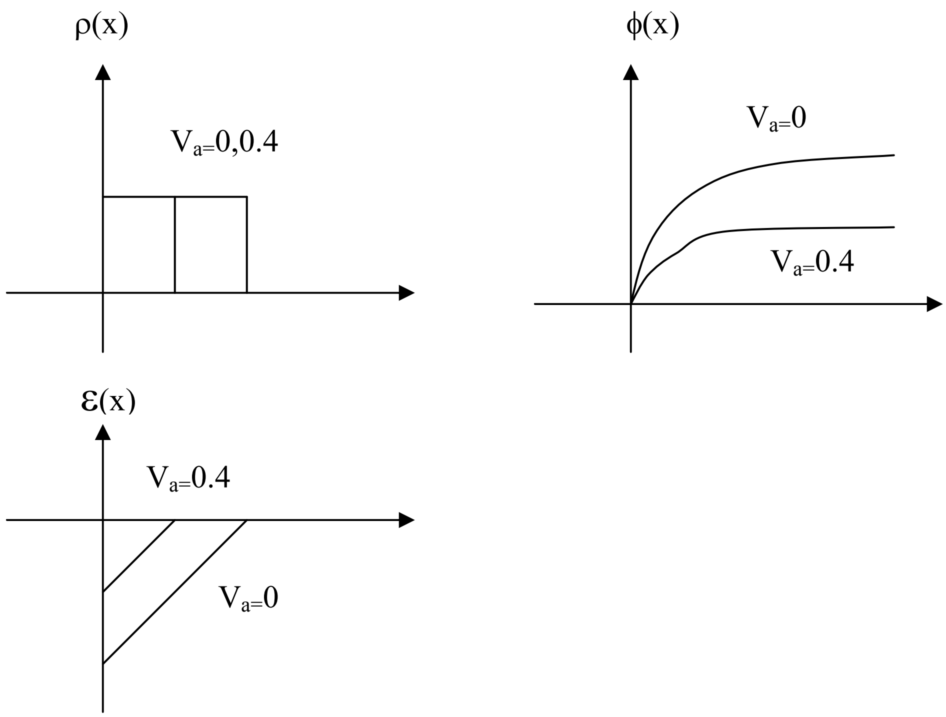


4.18 (a)

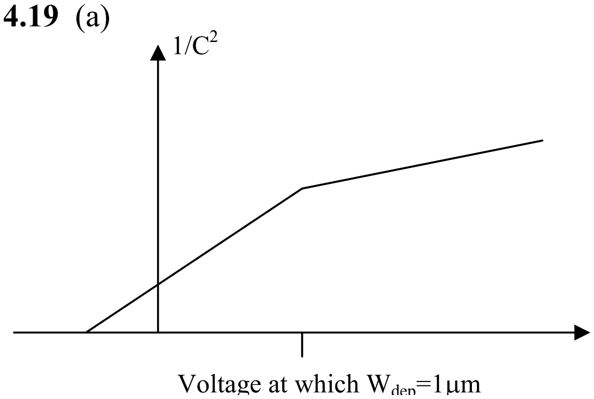


$$\varphi_M = 4.8V,\ \chi_{Si} = 4.05\,eV$$
$$\phi_s = \chi_{si} + 0.026\ln\frac{3.22\times10^{19}}{10^{16}} = 4.26V$$
$$\phi_{Bn} = 4.8 - 4.05 = 0.75V$$
$$\phi_{bi} = \phi_M - \phi_s = 0.54V$$
$$\phi_{bi} - V_a = 0.54 - 0.4 = 0.14V$$

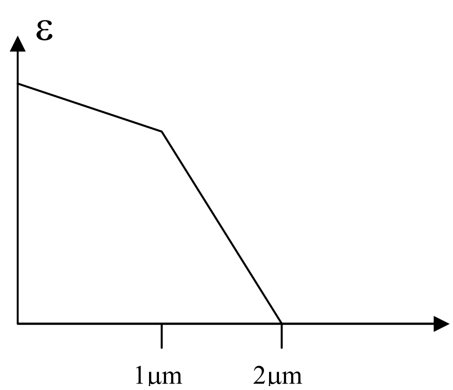
(b)



4.19 (a)



(b)



(c) $V=\int_0^{2\mu m}\mathcal{E}dx = 1.\frac{4}{5}\mathcal{E}_{\max}+\frac{1}{2}.1.\frac{1}{5}\mathcal{E}_{\max}+\frac{1}{2}.1.\frac{4}{5}\mathcal{E}_{\max}=(1.3\mu m)\mathcal{E}_{\max}$

To find $\mathcal{E}_{max}$, we use the slope when x_d<1um,

$$\frac{-qN}{\varepsilon_s}=\frac{(-1.6\times10^{-19})(10^{16})}{11.7\varepsilon_o}=1.54\times10^{9}\,\text{V/cm}^2$$

Using this slope, we need x_d=5um for $\mathcal{E}$=0,

$$\mathcal{E}_{\max}=(1.54\times10^{9})(5\times10^{-4})=7.723\times10^{5}\,V/cm$$
$$V=(1.3\mu m)(7.723\times10^{5}\,V/cm)=100.4V\,.$$

(d) Express V (areas under electric field profile) in terms of $W_{dep,}$

$$V = area = 30.9W_{dep}-23.175$$
$$W_{dep}=\sqrt{\frac{V+23.175}{30.9}}$$
$$C=\frac{\varepsilon_s}{W_{dep}}=\frac{\varepsilon_s}{\sqrt{\frac{V+23.175}{30.9}}}\,.$$

Depletion-Layer Analysis for Schottky Diodes

4.20 (a) Φ_B(Platinum to Si) = 5.3 eV, $N_d = 10^{16}$ cm^{-3} and $A = 10^{-5}$ cm^2

Now, $E_c - E_f = kT \ln(N_c/N_d) = 0.205$ eV

$\therefore\ q\Phi_s = 4.05 + 0.205 = 4.255$ eV

$\therefore\ \phi_{bi} = -(\Phi_s - \Phi_M) = 1.045$ V (Si to Pt)

The capacitance is give by

$$C = \text{charge/voltage drop} = \frac{\sqrt{qN_d 2\varepsilon_{Si}(\phi_{bi}-V_A)}}{(\phi_{bi}-V_A)} = \frac{\mathbf{K}}{(\phi_{bi}-V_A)^{1/2}}$$

where $\mathbf{K} = 2.88\times10^{-13}$ F V$^{1/2}$

If $V_A = 0$, $C = 2.82\times10\text{-}13$ F = 0.282 pF

(b) For a 25% reduction $C/C(0) = 0.75 \Rightarrow \left(\dfrac{\phi_{\text{bi}}}{\phi_{\text{bi}} + V_{\text{A}}}\right)^{1/2} = 0.75$

$\therefore\ V_{\text{A}} = \phi_{\text{bi}}(1 - 0.75^{-2}) = -0.813$ V

4.21 The development of relationships for the electrostatic variables in a linearly graded MS diode closely parallels the uniformly doped analysis.

(a) With $N_{\text{d}}(x) = ax$ for $x \geq 0$, invoking the depletion approximation yields

$\rho(x) = qax \qquad\qquad 0 \leq x \leq W_{dep}$.

Substituting into Poisson's equation gives

$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{\varepsilon_{Si}} = \frac{qa}{\varepsilon_{Si}}x \qquad 0 \leq x \leq W_{dep} \quad \text{and}$$

$$\int_{\mathcal{E}(x)}^{0} d\mathcal{E}' = \frac{qa}{\varepsilon_{Si}}\int_{x}^{W} x'dx' \qquad \text{or}$$

$$\mathcal{E}(x) = -\frac{qa}{2\varepsilon_{Si}}\left(W_{dep}^2 - x^2\right) \qquad 0 \leq x \leq W_{dep}.$$

Turning to the electrostatic potential, we can write

$$\frac{dV}{dx} = -\mathcal{E}(x) = \frac{qa}{2\varepsilon_{Si}}\left(W_{dep}^2 - x^2\right) \qquad \text{and}$$

$$\int_{V(x)}^{0} dV' = \frac{qa}{2\varepsilon_{Si}}\int_{x}^{W_{dep}}\left(W_{dep}^2 - x'^2\right)dx' \qquad \text{or}$$

$$V(x) = -\frac{qa}{6\varepsilon_{Si}}\left(2W_{dep}^3 - 3W_{dep}^2x + x^3\right) \qquad 0 \leq x \leq W_{dep}.$$

Finally, $V = -(\phi_{\text{bi}} - V_{\text{A}})$ at $x = 0$, and therefore

$$-\left(\phi_{bi} - V_A\right) = -\frac{qa}{3\varepsilon_{Si}}W_{dep}^3$$

$$W_{dep} = \left[\frac{3\varepsilon_{Si}}{qa}\left(\phi_{bi} - V_A\right)\right]^{1/3}.$$

(b) Paralleling the developments for the linearly graded P-N junction, the expression for ϕ_{bi} will be

$$\phi_{bi} = \frac{1}{q}\left[\Phi_B - \left(E_c - E_f\right)\Big|_{x=W_0}\right]$$

where W_0 is the depletion width when $V_A = 0$. Since approximate charge neutrality applies fro $x > W_0$, it follows that

$$n\big|_{x=W_0} = n_i e^{\left[(E_f - E_i)\big|_{x=W_0}\right]/kT} \cong N_d(x = W_0) = aW_0 \qquad \text{or}$$

$$(E_c - E_f)\big|_{x=W_0} \cong \frac{E_g}{2} - kT\ln\left(\frac{aW_0}{n_i}\right)$$

Thus, to determine ϕ_{bi}, one must simultaneously solve the following two equations employing numerical techniques.

$$W_0 = \left[\frac{3\varepsilon_{Si}}{qa}\phi_{bi}\right]^{1/3} \text{ and } \phi_{bi} = \frac{1}{q}\left[\Phi_B - \frac{E_g}{2} + kTln\left(\frac{aW_0}{n_i}\right)\right]$$

(c) $$C_J = \frac{\varepsilon_{Si}A}{W_{dep}} = \frac{\varepsilon_{Si}A}{\left[\frac{3\varepsilon_{Si}}{qa}(\phi_{bi} - V_A)\right]^{1/3}}$$

4.22 (a) From the graph, the built-in voltage is 0.8V.

(b) From Equation 4.16.7, we know that the slope of the line is inversely proportional to the doping concentration. Hence, for Region I,

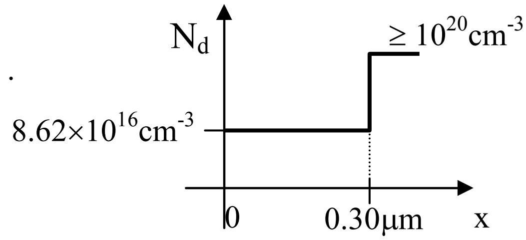
$$\frac{1}{C^2} = \frac{2}{qN_d\varepsilon_s}(\phi_{bi} + V) \Rightarrow Slope \equiv \frac{2}{qN_d\varepsilon_s} = \frac{7\times10^{14}cm^2/F^2}{5V}.$$

Solving for N_d yields
$N_d = 8.62\times10^{16}cm^{-3}$.

For Region II, the slope is 0. This implies that Region II is a very heavily doped N^{++} region. Hence, $N_d \geq 10^{20}cm^{-3}$.

In order to find the width of Region I, we calculate W_{dep} at $V_a = 5V$ since Region I becomes fully depleted. Using Equation 4.3.1,

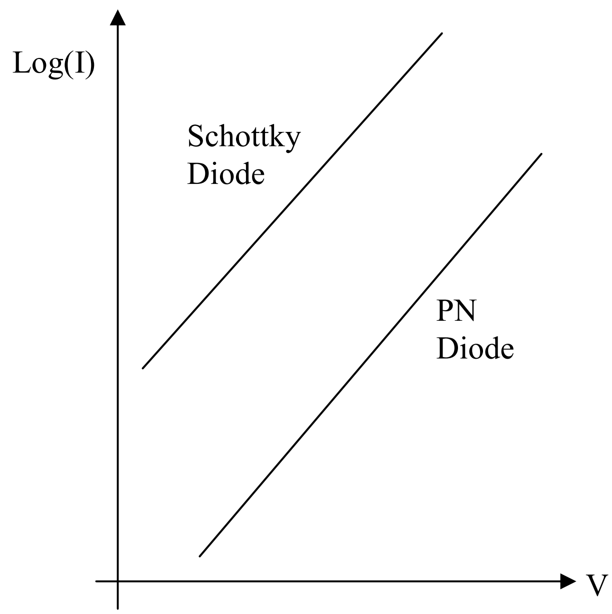
$$W_{dep} = \sqrt{\frac{2\varepsilon_s(\phi_{bi} + 5V)}{qN_d}} = 0.30\mu m.$$



Comparison between Schottky Diodes and PN Junction Diodes

4.23 (a) Similarity:
Both have the same slope of 1 decade / 60mV at room temperature.

Difference:
The schottky diode current is several orders of magnitude larger than the PN diode current.



(b) For Schottky diode, use Equations 4.18.2 and 4.18.6:

$$I_0 = AKT^2 e^{-q\phi_{Bn}/kT} = \left[(0.01)(100)(300^2)e^{-q0.55V/kT}\right]A = 58.5\,\mu A$$

For PN diode, use Equation 4.9.5:

$$I_0 = Aqn_i^2\left(\frac{D_p}{L_p N_d}\right) = (0.01cm^2)q(10^{10}\,cm^{-3})^2\left(\frac{4cm^2/s}{\sqrt{(4cm^2/s)(1\mu s)}}\right) = 0.32\ \mathrm{f}A$$

(c) For Schottky diode,

$$V = \frac{kT}{q}\ln\left(\frac{I}{I_0}+1\right) = 0.36V\,.$$

For PN diode,

$$V = \frac{kT}{q}\ln\left(\frac{I}{I_0}+1\right) = 1.03V\,.$$

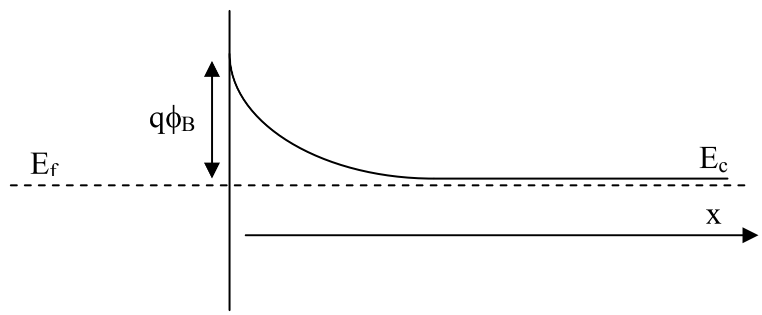
(d) Use a metal or silicide that yields a smaller ϕ_{Bn}.

(e) I_0 may be so large, especially at elevated temperatures, that the reverse leakage current
  a. interferes with the rectification property of the diode, and
  b. causes too large a power (heat) dissipation, $I_0 \times V_r$, under reverse bias.

Ohmic Contacts

4.24 (a) The semiconductor is uniformly doped with a doping concentration of $N_d = 10^{17}cm^{-3}$. Using depletion approximation, the energy band varies as
$E_c(x) = ½ (qN_d/\varepsilon_{Si}) (x - x_0)^2$
where x_0 is a constant and x is positive in the semiconductor and zero at the MS junction.
At $x = 0$, $E_c(0) = 0.65$ eV
(using the equilibrium Fermi-level as the reference)
and at x= 10 nm, $E_c(0) \approx 0 \Rightarrow x_0 = 10$ nm and $N_d = 8.85\times10^{18}$ cm^{-3}

(b)



4.25 (a) $R_c = \dfrac{50mV}{1mA}\times 0.08\mu m^2 = 4\times10^{-8}\ \Omega\cdot cm^2$

(b) Using Eq. 4.21.7,

$$R_c \equiv \frac{V}{J} = \frac{2}{qv_{thx}H\sqrt{N_d}}e^{H\phi_{Bn}/\sqrt{N_d}} = Be^{H\phi_{Bn}/\sqrt{N_d}} \Rightarrow \phi_{Bn} = \frac{\sqrt{N_d}}{H}\ln\left(\frac{R_c}{B}\right)$$

$$\begin{aligned} H &= \left(5.4\times10^{10}\right)\times\sqrt{(m_n/m_0)(\varepsilon_s/\varepsilon_0)} \\ &= \left(5.4\times10^{10}\right)\times\sqrt{0.26\times11.7} \\ &= 9.418\times10^{10}\ cm^{-3/2}V^{-1} \end{aligned}$$

$$v_{thx} = \sqrt{2kT/\pi m_n} = 1.057\times10^7\ cm/s$$

$$B = \frac{2}{qv_{thx}H\sqrt{N_d}} = 1.256\times10^{-9}\ \Omega cm^2$$

$$\therefore\ \phi_{Bn} = \frac{\sqrt{N_d}}{H}\ln\left(\frac{R_c}{B}\right) = 0.3675\ V$$

This is the maximum φ_{Bn} allowed since a larger φ_{Bn} will result in a voltage drop larger than 50 mV at the ohmic contact.

(c) According to the graph, $\phi_{Bn} \approx 0.36\,V$.

4.26

(a) $\phi_{Bn} = 0.67V$ $\quad\therefore R_C \approx 6\times10^{-7}\,\Omega\cdot cm^2$

$\phi_{Bp} = 0.43V$ $\quad\therefore R_C \approx 5\times10^{-8}\,\Omega\cdot cm^2$

(b) $\phi_{Bp} = 0.23V$ $\quad\therefore R_C \approx 7\times10^{-9}\,\Omega\cdot cm^2$

(c) $\phi_{Bn} = 0.28V$ $\quad\therefore R_C \approx 1\times10^{-8}\,\Omega\cdot cm^2$

(d) According to the graph, this contact resistance can be achieved with a barrier height of 0.3V or lower and a doping concentration of $3\times10^{20}\,cm^{-3}$ or higher is required. Looking at Table 4-2, we can see that this cannot be achieved using only one type of metal silicide. So different silicides should be used for contact on N^+ and P^+ silicon.

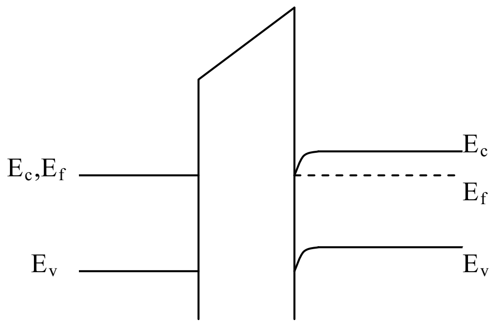
N^+ : $ErSi_{1.7}$ $\quad\ge 3.5\times10^{20}\,cm^{-3}$

P^+ : $PtSi$ $\quad\ge 2.5\times10^{20}\,cm^{-3}$

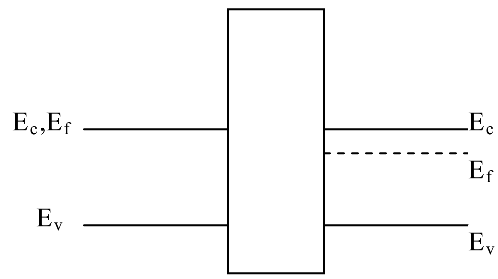
Chapter 5

Energy Band Diagram

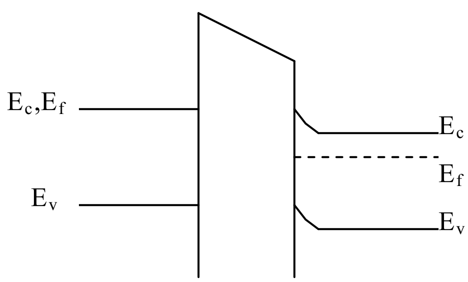
5.1



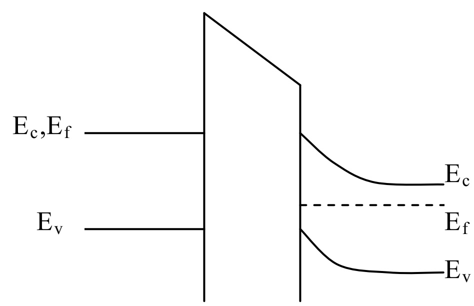
Accumulation



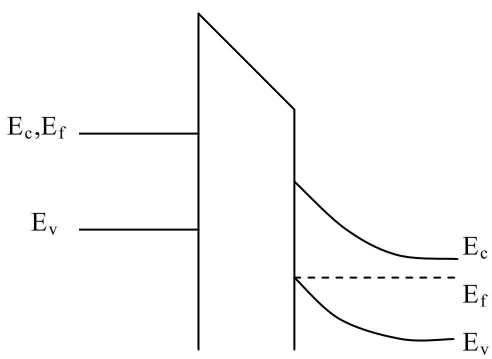
Flat-band



Depletion

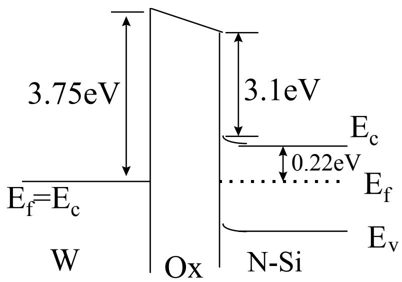


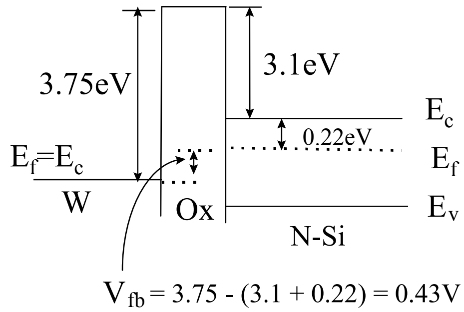
Threshold



Inversion

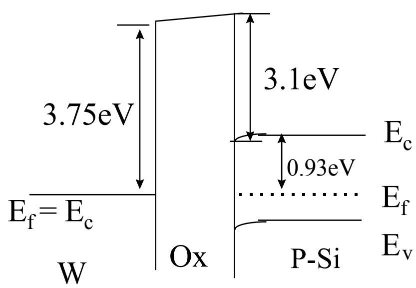
5.2 (a) 1 Ω-cm N-type silicon substrate : $N_d = 5\times10^{15}$ cm^{-3}, $E_f = E_c - 0.223$eV



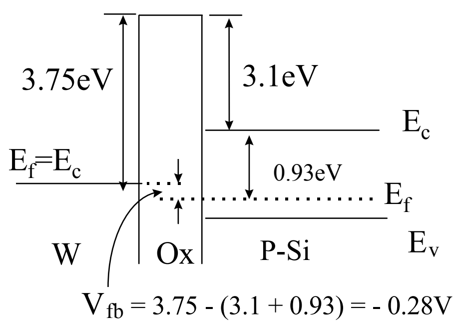


(b) 1 Ω-cm P-type silicon substrate :$N_a = 1.5\times10^{16}$ cm^{-3}, $E_f = E_v + 0.168$eV

Thermal Equil.

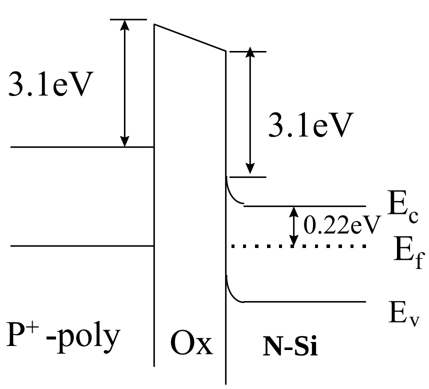


Flat Band

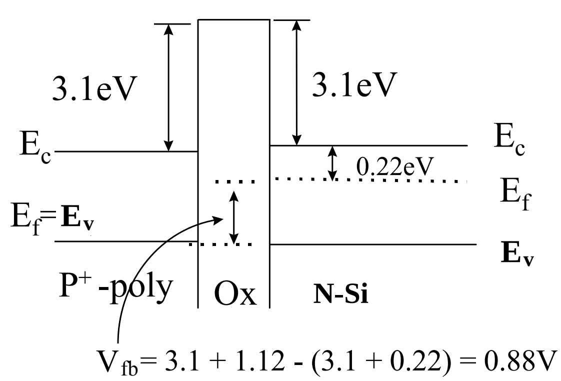


(c) Heavily doped P$^+$-polycrystalline silicon gate with 1 *Ω-cm* N-type silicon substrate.

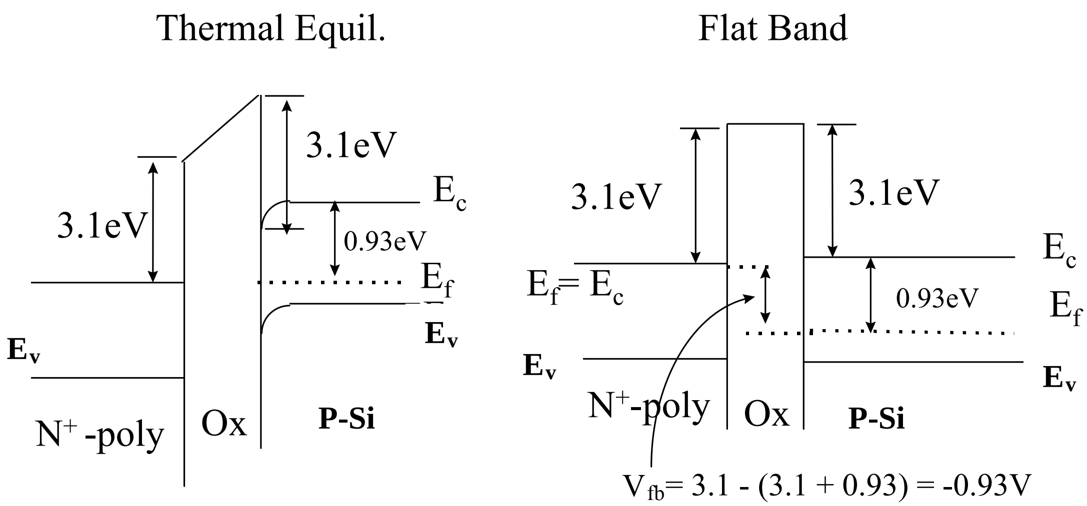
Thermal Equil.



Flat Band



(d) Heavily doped N^+-polycrystalline silicon gate with 1 Ω-*cm* P-type silicon substrate.



MOS System: Inversion, threshold, depletion, and accumulation

5.3 (a) 3
(b) 2
(c) 1
(d) 4
(e) 5

5.4 (a)

$$\phi_B = \left(\frac{kT}{q}\right)\ln\left(\frac{N_a}{n_i}\right) = (0.026V)\times\ln\left(\frac{10^{18}cm^{-3}}{10^{10}cm^{-3}}\right) = 0.479V$$

$$\therefore V_{fb} = \left(\frac{\chi_{si}}{q}\right) - \left[\left(\frac{\chi_{si}}{q}\right) + \frac{1}{2}\left(\frac{E_g}{q}\right) + \phi_B\right] = -\left[\frac{1}{2}\left(\frac{E_g}{q}\right) + \phi_B\right]$$

$$= -\frac{1.12V}{2} - 0.479V = -1.039V$$

(b)

$$W_{d\max} = \sqrt{\frac{2\varepsilon_{si}\times 2\phi_B}{qN_a}} = \sqrt{\frac{2\times(11.7)\times(8.84\times10^{-14})\times 2\times(0.479)}{(1.6\times10^{-19})\times(10^{18})}} = 3.521\times10^{-6}cm$$

(c)

$$V_t = V_{fb} + 2\phi_B + \frac{\sqrt{2\varepsilon_{si} qN_a \times 2\phi_B}}{C_{ox}}$$
$$= (-1.039V) + 2\times(0.479V)$$
$$+ \frac{\sqrt{2\times(11.7)\times(8.85\times10^{-14}F/cm)\times(1.6\times10^{-19}C)\times(10^{18}cm^{-3})\times 2\times(0.479V)}}{(3.9)\times(8.85\times10^{-14}F/cm)/(2\times10^{-7}cm)}$$
$$= 0.2455V$$

(d)
Only the flat-band voltage changes.

$$V_{fb} = \left[\left(\frac{\chi_{si}}{q}\right)+\left(\frac{E_g}{q}\right)\right]-\left[\left(\frac{\chi_{si}}{q}\right)+\frac{1}{2}\left(\frac{E_g}{q}\right)+\phi_B\right]=\frac{1}{2}\left(\frac{E_g}{q}\right)-\phi_B$$
$$= \frac{1.12V}{2} - 0.479V = 0.081V$$
$$\therefore\ V_t = (0.081V) + 2\times(0.479V)$$
$$+ \frac{\sqrt{2\times(11.7)\times(8.85\times10^{-14}F/cm)\times(1.6\times10^{-19}C)\times(10^{18}cm^{-3})\times 2\times(0.479V)}}{(3.9)\times(8.85\times10^{-14}F/cm)/(2\times10^{-7}cm)}$$
$$= 1.3655V$$

5.5 (a) At V_g-V_{fb} = -1V, the MOS capacitor is in accumulation.

$$C_{ox} = \frac{Q_S}{V_{ox}} = \frac{\varepsilon_{ox}}{T_{ox}} = 4\times10^{-7}\frac{F}{cm^2}$$
$$T_{ox} = 8.62nm$$

(b) At threshold,

$$V_g - V_{fb} = 1V = \phi_s + V_{ox} = 2\phi_B + \frac{(2\varepsilon_s qN_a 2\phi_B)^{1/2}}{C_{ox}} \quad \text{where}$$

$$\phi_B = \frac{kT}{q}\ln\left(\frac{N_a}{n_i}\right).$$

Solving iteratively, we get

$$N_a = 2.9\times10^{16}cm^{-3}.$$

(c) Since $\phi_s \approx 0$, $V_{ox} = -1V$.

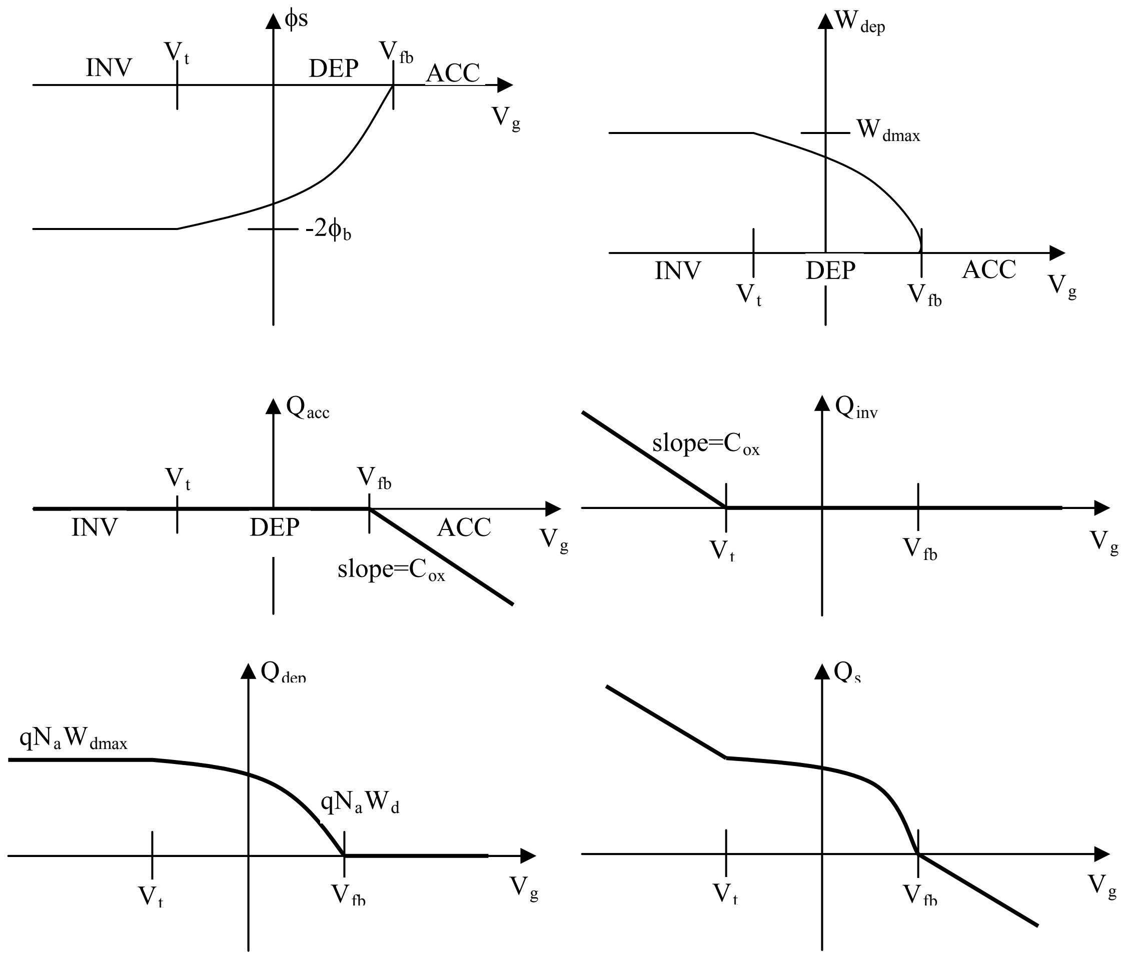
(d) $V_g - V_{fb} = 1V = \phi_s + V_{ox} = \phi_s + \dfrac{(2\varepsilon_s qN_a\phi_s)^{1/2}}{C_{ox}}$

$\phi_s + 0.245(\phi_s)^{1/2} = 0.5$

Solving the equations above, we get

$\phi_s = 0.354V$.

5.6



5.7 (a) From Equation 5.3.2,

$$V_g = V_{fb} + \phi_s + \frac{1}{C_{ox}}\sqrt{2qN_a\varepsilon_s\phi_s} \text{ where } \phi_s = \left(\sqrt{\phi_s}\right)^2.$$

Rearranging the terms, we obtain

$$\left(\sqrt{\phi_s}\right)^2 + \left(\frac{\sqrt{2qN_a\varepsilon_s}}{C_{ox}}\right)\sqrt{\phi_s} + \left(V_{fb} - V_g\right) = 0\,.$$

Solving for $\sqrt{\phi_s}$ yields

$$\sqrt{\phi_s} = \frac{-\frac{\sqrt{2qN_a\varepsilon_s}}{C_{ox}} \pm \sqrt{\frac{2qN_a\varepsilon_s}{C_{ox}^2} - 4\left(V_{fb} - V_g\right)}}{2}\,.$$

Since $\sqrt{\phi_s}$ cannot be less than 0,

$$\sqrt{\phi_s} = \frac{-\sqrt{2qN_a\varepsilon_s}}{2C_{ox}} + \sqrt{\frac{2qN_a\varepsilon_s}{4C_{ox}^2} - \left(V_{fb} - V_g\right)} = \frac{\gamma}{2} + \sqrt{\frac{\gamma^2}{4} - \left(V_{fb} - V_g\right)}$$

where

$$\gamma = \frac{\sqrt{2qN_a\varepsilon_s}}{C_{ox}}\,.$$

Hence,

$$\begin{aligned}\phi_s &= \frac{\gamma^2}{4} + \left[\frac{\gamma^2}{4} - \left(V_{fb} - V_g\right)\right] + \frac{\gamma}{2}\sqrt{\frac{\gamma^2}{4} - \left(V_{fb} - V_g\right)} \\ &= \frac{\gamma^2}{2} - \left(V_{fb} - V_g\right) + \frac{\gamma}{2}\sqrt{\frac{\gamma^2}{4} - \left(V_{fb} - V_g\right)}\,.\end{aligned}$$

Or,

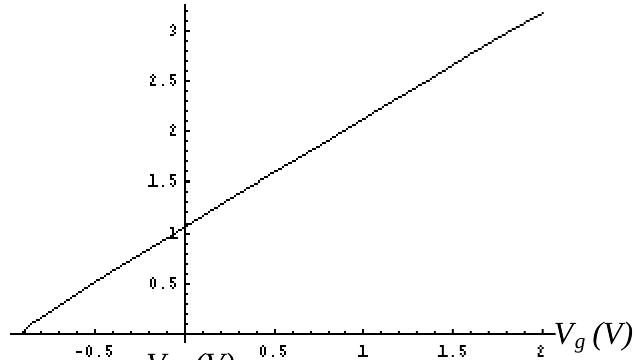
$$\phi_s = \left[\frac{\gamma}{2} + \sqrt{\frac{\gamma^2}{4} - \left(V_{fb} - V_g\right)}\right]^2.$$

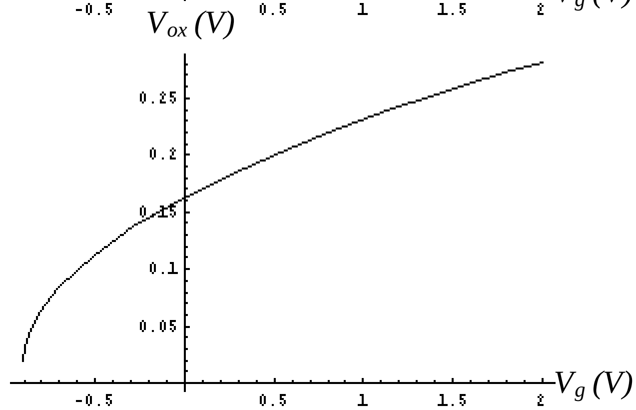
(b) From Equation 5.3.1,

$$V_{ox} = \frac{\sqrt{2qN_a\varepsilon_s}}{C_{ox}}\sqrt{\phi_s} = \gamma\left[\frac{\gamma}{2} + \sqrt{\frac{\gamma^2}{4} - \left(V_{fb} - V_g\right)}\right].$$

(c) We can qualitatively predict that $\phi_s \propto V_g$ since $|\gamma| << 1$. Also, $V_{ox} \propto \sqrt{V_g}$.

$\phi_s\,(V)$





(d) From Equation 4.2.10,

$$W_{dep} = \frac{\sqrt{2q\varepsilon_s}}{C_{ox}}\sqrt{\phi_s} = \frac{\sqrt{2q\varepsilon_s}}{C_{ox}}\left[\frac{\gamma}{2} + \sqrt{\frac{\gamma^2}{4} - \left(V_{fb} - V_g\right)}\right].$$

5.8 (a) $C_{ox} = \dfrac{\varepsilon_{ox}}{T_{ox}} = 3.45\times10^{-7}\,\dfrac{F}{cm^2}$

$$\phi_B = kT\ln\frac{N_a}{n_i} = 0.4V$$

$$V_{fb} = -\frac{E_g}{2} - \phi_B = -0.96V$$

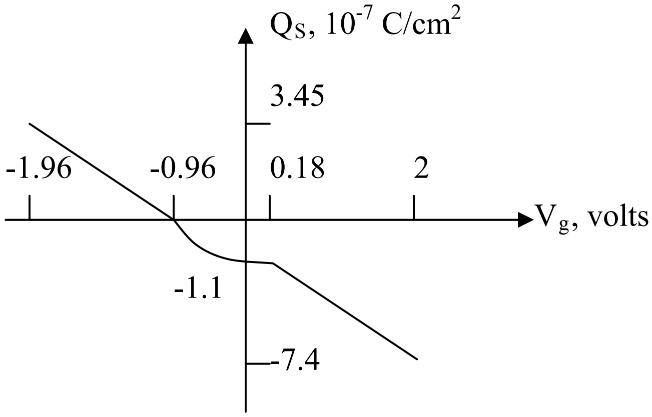
$$V_t = V_{fb} + 2\phi_B + \frac{1}{C_{ox}}\sqrt{2q\varepsilon_s N_a 2\phi_B} = \left(-0.96 + 0.8 + 0.34\right)V = 0.18V$$

(b) $Q_{acc} = -C_{ox}\left(V_g - V_{fb}\right) = 3.45\times10^{-7}\,\dfrac{C}{cm^2}$

(c) $Q_{dep} = -\sqrt{2q\varepsilon_s N_a 2\phi_B} = -1.1\times10^{-7}\,\dfrac{C}{cm^2}$

$$Q_{inv} = -C_{ox}\left(V_g - V_t\right) = -6.3\times10^{-7}\,\frac{C}{cm^2}$$

(d)



5.9

	Parameters	Increase	Decrease	Unchanged
a	Accumulation Region Capacitance			X
b	Flat-band Voltage, Vfb	X		
c	Depletion Region Capacitance		X	
d	Threshold Voltage, Vt		X	
e	Inversion Region Capacitance		X	

(a) At accumulation: Accumulation capacitance

$$C = C_{ox} = \varepsilon_{ox}/T_{ox}$$

(b) At flat-band: Flat-band Voltage

$$V_{fb} = \varphi_g - \left(4.05 + 0.56 + 0.026\ln\left(\frac{N_a}{n_i}\right)\right)$$

$$N_a \downarrow, |\phi_B| \downarrow$$

(c) At depletion: Depletion Capacitance

$$1/C = 1/C_{ox} + W_{dep}/\varepsilon_s$$

$$N_a \downarrow, W_{dep} \uparrow, C \downarrow$$

(d) At threshold: Threshold Voltage

$$V_t = V_{fb} + |2\phi_B| + Q_d / C_{ox}$$
$$N_a \downarrow, |\phi_B| \downarrow, Q_d \downarrow, V_t \downarrow$$

(e) At inversion: Inversion Capacitance

$$1/C = 1/C_{ox} + W_{d\max} / \varepsilon_s$$
$$N_a \downarrow, W_{d\max} \uparrow, C \downarrow$$

5.10 To find the value of the oxide capacitance,

$$C_{ox} = \frac{\varepsilon_{ox}}{T_{ox}} = 1.15 \times 10^{-7} \frac{F}{cm^2}.$$

The capacitance at $V_g = V_{fb}$ is given by

$$C_{fb} = \frac{1}{1/C_{ox} + L_D/\varepsilon_s} = 7.9 \times 10^{-8} \frac{F}{cm^2}$$

where

$$L_D = \left[\frac{\varepsilon_s kT}{q^2 N_a}\right]^{1/2} = 4.09 \times 10^{-6}\,cm.$$

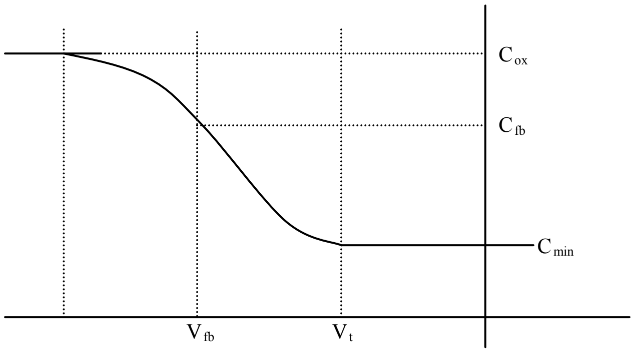
To find the minimum value of the capacitance,

$$W_{d\max} = \sqrt{\frac{4\varepsilon_s \phi_B}{qN_a}} = 3 \times 10^{-5}\,cm \;;\; \phi_B = \frac{kT}{q}\ln\frac{N_a}{n_i} = 0.348V$$
$$C_{\min} = \frac{1}{1/C_{ox} + W_{d\max}/\varepsilon_s} = 2.65 \times 10^{-8} \frac{F}{cm^2}.$$

V_t is given by

$$V_t = 2\phi_B + \frac{1}{C_{ox}}\sqrt{2\varepsilon_s q N_a 2\phi_B} + V_{fb} = 1.11V$$

where $V_{fb} = \varphi_g - \varphi_s = -2.00$ V.



To find the effective oxide charge, we need to find

$$\varphi_{Al} = 4.1V$$
$$\varphi_s = 4.05 + 0.562 + 0.3482 = 4.96V$$
$$\varphi_{Al-s} = -0.86V$$
$$\Delta V_t = V_{fb} - \varphi_{Al-s} = -2 - (-0.86) = -1.14V = \frac{-Q_{ox}}{C_{ox}}$$

Hence,

$$Q_{ox} = 1.31\times10^{-7}\frac{C}{cm^2} = 8.2\times10^{11}\frac{q}{cm^2}.$$

Field Threshold Voltage

5.11 (a) $V_{fb} = \phi_{Al} - \phi_{Si} = 4.1V - (\chi_{Si} + E_g/2q + \dfrac{E_i - E_f}{q})$

$$= 4.1V - (4.05V + (1.12/2)V + (kT/q)\cdot\ln(N_a/n_i)) = -0.80V$$

(b) $\phi_B = \dfrac{E_i - E_f}{q} = kT/q\cdot\ln(N_a/n_i) = 0.290V$

$$W_{d\max} = \sqrt{\frac{2\varepsilon_s 2\phi_B}{qN_a}} = 0.866um$$
$$V_t = V_{fb} + 2\phi_B + qN_aW_{d\max}/C_{ox} \ge 5V$$
$$C_{ox} \le 2.65\times10^{-9}F/cm^2$$

(c) $C_{ox} = \dfrac{\varepsilon}{T_{ox}} = \kappa\varepsilon_0/1\mu m = 2.65\times10^{-9}F/cm^2$ where $K = 2.99$

(d) In accumulation,

$V_g - V_{fb} = -1V$,

$C_{ox} = Q_s/V_{ox}$

$V_t \geq 5V$

$C_{ox} \leq 2.65\times10^{-9}\, F/cm^2$

$K \leq 2.99$

It is the maximum allowable K.

(e) $V_g - V_t = -\dfrac{Q_{inv}}{C_{ox}}$

$Q_{inv} = -C_{ox}(V_g - V_t) = -(2.65\times10^{-9}\, F/cm^2)(2V) = -5.3\times10^{-9}\, C/cm^2$

(f) $1/C = 1/C_{ox} + 1/C_{dep} = 1/C_{ox} + W_{d\max}/\varepsilon_s = 1/2.65\times10^{-9}\, F/cm^2 + 0.866\mu m/\varepsilon_s$

$C = 2.17\times10^{-9}\, F/cm^2$

(g) $V_g = V_{fb} + V_s + V_{ox}$

$(V_g \geq V_t) = V_{fb} + 2\phi_B + V_{ox}$

$7V = -0.8V + 2(0.29V) + V_{ox}$

$V_{ox} = 7.22V$

Oxide Charge

5.12 $\Delta V = -Q/C_{ox}$

$Q = -C_{ox}\Delta V = -\dfrac{C_0}{A}\Delta V = -\dfrac{45\times10^{-12}}{6.4\times10^{-5}}\times 0.05 = -3.52\times10^{-8}\, C/cm^2$

$V_{fb} = \psi_g - \psi_s - Q_f/C_{ox}$

It should be negative since negative charges increase V_{fb}.

5.13 Oxide charge will change the device characteristics. Mobile charges are particularly bad as they give the device instability. That is, as the charge moves from one side of the oxide to the other, it will change the threshold voltage. This is undesirable.

Mobile charges are generally introduced into the oxide during wafer cleaning or oxidation. Strict measures of cleanliness can be achieved by using ultra-clean chemicals. And, the impurities such as sodium can be immobilized by introducing a chlorine compound during oxidation.

C-V Characteristics

5.14 $V_g = V_{fb} + \phi_s + V_{ox}$

Using $V_{ox} = \dfrac{qN_a W_d}{C_{ox}}$ and $\phi_s = \dfrac{W_d{}^2 qN_a}{2\varepsilon_s}$, we solve for W_d

$$V_g - V_{fb} = qN_a\left(\frac{W_d{}^2}{2\varepsilon_s} + \frac{W_d}{C_{ox}}\right)$$

$$W_d = \frac{-2\varepsilon_s + \sqrt{4\varepsilon_s{}^2 - 4C_{ox}\left(2\varepsilon_s C_{ox}\right)\left(\dfrac{V_g - V_{fb}}{qN_a}\right)}}{2C_{ox}}$$

$$\frac{W_d}{\varepsilon_s} = \frac{-1 + \sqrt{1 - \dfrac{2C_{ox}{}^2\left(V_g - V_{fb}\right)}{q\varepsilon_s N_a}}}{C_{ox}}$$

Since $\dfrac{1}{\mathrm{C}} = \dfrac{1}{C_{ox}} + \dfrac{1}{C_{dep}} = \dfrac{1}{C_{ox}} + \dfrac{W_d}{\varepsilon_s}$,

$$\frac{1}{C} = \sqrt{\frac{1}{C_{ox}{}^2} - \frac{2\left(V_g - V_{fb}\right)}{q\varepsilon_s N_a}}\,.$$

5.15 (a) The substrate doping is P-type since the threshold voltage is larger than V_{fb}.

(b) $T_{ox} = A\dfrac{\varepsilon_{ox}}{C_{ox}} = \dfrac{10^{-4} \times 0.345 \times 10^{12}}{50 \times 10^{-12}} = 6.9\ nm$

(c) $V_t - V_{fb} = 2\phi_B + \dfrac{\sqrt{2\varepsilon_s qN_a 2\phi_B}}{C_{ox}} = 2\dfrac{kT}{q}\ln\left(\dfrac{N_a}{n_i}\right) + \dfrac{2\sqrt{\varepsilon_s qN_a \ln\dfrac{N_a}{n_i}}}{C_{ox}}$

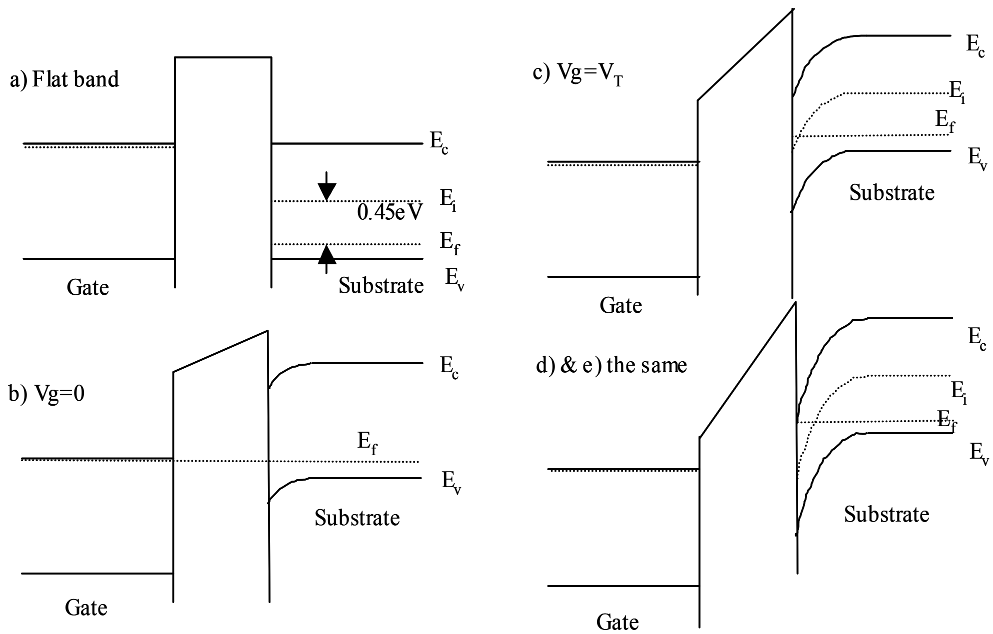
Using V_{fb}=-1V and V_t = 0.5V and solving iteratively, we obtain

$N_a \cong 3 \times 10^{17}\, cm^2$.

(e) At position C, MOS has the minimum capacitance.

$$C_{\min} = \left(\frac{1}{C_{ox}} + \frac{W_{d\max}}{\varepsilon_s}\right)^{-1} = 12.7\, pF \quad \text{where} \quad W_{d\max} = \sqrt{\frac{2\varepsilon_s}{qN_a}\frac{2kT}{q}\ln\frac{N_a}{n_i}}$$

(e)



(f) $V_g = 0 = V_{fb} + \phi_s + \dfrac{1}{C_{ox}}\sqrt{2\varepsilon_s q N_a \phi_s}$

Substituting the numerical values and solving the quadratic equation, we obtain

$$\phi_s + 0.644\sqrt{\phi_s} - 1 = 0$$
$$\phi_s = 0.53V\ .$$

5.16 (a) P-type, since $V_t > V_{fb}$.

(b)

T_{ox} :	A	**B**	($C_{max}=\varepsilon_{ox}/T_{ox}$)
V_{fb} :	A	**B**	(directly from the graph)
W_{dmax} :		**A**	B ($1/C_{min} = 1/C_{ox}+W_{dmax}/\varepsilon_s$ and $C_{ox}^A > C_{ox}^B$)
N_{sub} :	A	**B**	($W_{dmax} \sim (N_{sub})^{-1/2}$)
V_t :	A	**B**	(directly from the graph)

5.17 $A = 100\times100\ \mu m^2 = 10^{-4}\ cm^{-2}$

For the MOS capacitor, the field in the oxide $\mathcal{E}_{max} = 8\times10^6 = \dfrac{10V}{T_{ox}}$

$\Rightarrow T_{ox} = 1.25\times10^{-6}$ cm or 12.5 nm

$\therefore\ C_{ox} = \frac{\varepsilon_{ox}}{T_{ox}} A = 28.32 \text{ pF} = C_{\text{MOS}}$

For P^+N junction,

$$\phi_B = \frac{kT}{q}\ln\left(\frac{N_d}{n_i}\right) = 0.356 \text{ V for } N_d = 10^{16}\ \text{cm}^{-3}$$

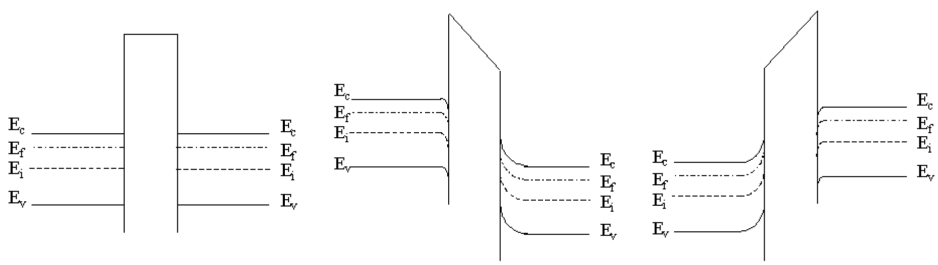
$$W_{dep} = \sqrt{\frac{2\varepsilon_s(\phi_B + V_B)}{qN_d}} = 8.35\times10^{-5} \text{ cm for } V_R = 5 \text{ V.}$$

$$C = \frac{\varepsilon_s}{W_{dep}} A = 1.25 \text{ pF} = C_{\text{P-N jn}}$$

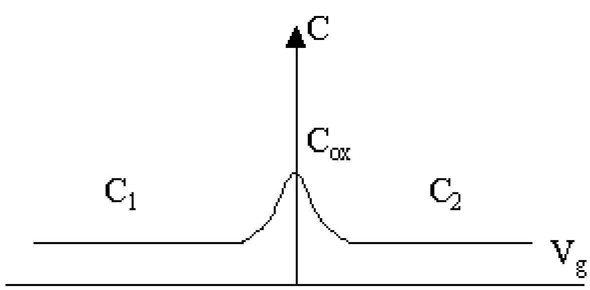
$C_{\text{MOS}} / C_{\text{P-N jn}} = 22.66$

5.18 (a) The flatband voltage is 0V because the 2 silicon sides are equally doped

(i)Vg =0 (ii) Vg<0 and large (iii) Vg>0 and large



(b)



As both sides are equally doped, the values of C_1 and C_2 will be equal.

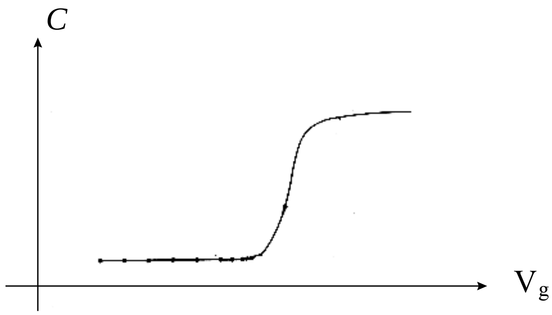
$$C_1 = C_2 = \left(C_{ox}^{-1} + C_{dep}^{-1}\right)^{-1}$$

$$C_{dep} = \frac{\varepsilon_s}{W_{dep}} \quad \text{where}$$

$$W_{dep} = \sqrt{\frac{2\varepsilon_s}{qN_d}\frac{2kT}{q}ln\left(\frac{N_d}{n_i}\right)} \Rightarrow C_{\text{dep}} = 34.5 \text{ nF cm}^{-2} \text{ and } C_{ox} = \frac{\varepsilon_{ox}}{T_{ox}}.$$

(c) When the left side is P-type, both the silicon layers go into accumulation and depletion for the same type of bias.

The flat band voltage will be $(\chi_{Si} + E_g/2 + \phi_{BL}) - (\chi_{Si} + E_g/2 - \phi_{BR}) = \phi_{BL} + \phi_{BR} = 0.713$ V



5.19

Bias Condition	Surface Potential	MOS cap (LF)	MOS Cap (HF)	MOSFET
accumulation	~ 0	C_{ox}	C_{ox}	C_{ox}
flatband	= 0	C_{ox}	C_{ox}	C_{ox}
threshold	$= 2\phi_B$	C_{ox}	$(C_{ox}^{-1} + W_{dep,max}/\varepsilon_S)^{-1}$	C_{ox}
inversion	$\sim 2\phi_B$	C_{ox}	$(C_{ox}^{-1} + W_{dep,max}/\varepsilon_S)^{-1}$	C_{ox}

5.20 (a) Accumulation -1 V < V_g
Depletion -3 V < V_g < -1 V
Inversion V_g < -3 V
The substrate is N-type.

(b) $C_0 = \frac{\varepsilon_{ox}}{T_{ox}} A$

Thus, T_{ox} = 205 nm.

(c) From the high-frequency *C-V* curve, $C_{\min} = \left(C_0^{-1} + C_{dep}^{-1}\right)^{-1}$

Plugging in $C_{\min} = 0.4C_0 \Rightarrow C_{\text{dep}} = 0.67\times C_0 = 54.9$ pF

Now, $C_{\text{dep}} = \frac{\varepsilon_s}{W_{dep}} A$

For uniform doping $W_{dep} = \sqrt{\dfrac{2\varepsilon_s(2\phi_B)}{qN_d}}$ at threshold, where $\phi_B = \dfrac{\mathrm{kT}}{\mathrm{q}}\ln\left(\dfrac{\mathrm{N}_d}{\mathrm{n}_i}\right)$

$$\Rightarrow N_d = \frac{4\varepsilon_s kT}{W_{dep}^2 q^2} ln\left(\frac{N_d}{n_i}\right)$$

which can be solved by iteration and we get the value of $N_{\mathrm{d}} \sim 10^{15}$ cm^{-3}.

(d) $V_{fb} \sim .55V + (kT/q)\ln(10^{15}/10^{10}) = .85V$

Since V_{fb} from the plot is ~ -1V, then Q_{ox} consists of positive charges. The shift in V_{fb} is 1.85V. Thus,

Q_{ox}= 1.85V × C_0 = 1.85V × 82pF / 4.75x10^{-3} cm^2 = 32 nC/cm^2.

Poly-gate Depletion

5.21 (a) Using Gauss Law,

$$Q_{poly} = qN_{poly}W_{dpoly} = \varepsilon_{poly}\mathcal{E}_{poly} = \varepsilon_{ox}\mathcal{E}_{ox},$$

where $\mathcal{E}_{ox}$ is the electric field inside the oxide and

$$W_{dpoly} = \frac{\varepsilon_{ox}\mathcal{E}_{ox}}{qN_{poly}}.$$

(b) $$\phi_{poly} = \frac{qN_{poly}W_{dpoly}^2}{2\varepsilon_{poly}} = \frac{qN_{poly}}{2\varepsilon_{poly}}\frac{\varepsilon_{ox}^2\mathcal{E}_{ox}^2}{q^2N_{poly}^2} = \frac{\varepsilon_{ox}^2\mathcal{E}_{ox}^2}{2q\varepsilon_{poly}N_{poly}}.$$

(c) $V_g = \phi_{poly} + V_{ox} + \phi_s + V_{fb}$.

$$V_{ox}C_{ox} = |Q_s| = Q_{poly} \Rightarrow V_{ox} = \frac{1}{C_{ox}}\sqrt{2q\varepsilon_{poly}N_{poly}\phi_{poly}}$$

$$V_g = \phi_{poly} + V_{fb} + \phi_s + V_{ox} = \phi_{poly} + V_{fb} + 2|\phi_B| + \frac{T_{ox}}{\varepsilon_{ox}}\sqrt{2q\varepsilon_{poly}N_{poly}\phi_{poly}}$$

$$\phi_{poly} + \frac{T_{ox}}{\varepsilon_{ox}}\sqrt{2q\varepsilon_{poly}N_{poly}}\sqrt{\phi_{poly}} - \left(V_g - V_{fb} - 2|\phi_B|\right) = 0$$

$$\sqrt{\phi_{poly}} = \frac{-\dfrac{T_{ox}}{\varepsilon_{ox}}\sqrt{2q\varepsilon_{poly}N_{poly}} \pm \sqrt{\dfrac{T_{ox}^2}{\varepsilon_{ox}^2}2q\varepsilon_{poly}N_{poly} + 4\left(V_g - V_{fb} - 2|\phi_B|\right)}}{2}$$

$$= -\frac{T_{ox}}{2\varepsilon_{ox}}\sqrt{2q\varepsilon_{poly}N_{poly}} \pm \sqrt{\frac{T_{ox}^2}{4\varepsilon_{ox}^2}2q\varepsilon_{poly}N_{poly} + \left(V_g - V_{fb} - 2|\phi_B|\right)}$$

We know that $\sqrt{\phi_{poly}}>0$. Also, if we set

$$\alpha=\frac{T_{ox}}{\varepsilon_{ox}}\sqrt{2q\varepsilon_{poly}N_{poly}} \text{ and } \beta=\left(V_g-V_{fb}-2|\phi_B|\right), \text{ then}$$

$$\sqrt{\phi_{poly}}=-\alpha+\sqrt{\alpha^2+\beta} \text{ and } \phi_{poly}=\beta+2\alpha^2\left(1-\sqrt{1+\frac{\beta}{\alpha^2}}\right).$$

Hence,

$$\varphi_{poly}=\left(V_g-V_{fb}-2|\phi_B|\right)+\frac{T_{ox}^2}{\varepsilon_{ox}^2}q\varepsilon_{poly}N_{poly}\left(1-\sqrt{1+\frac{2\varepsilon_{ox}^2\left(V_g-V_{fb}-2|\phi_B|\right)}{T_{ox}^2q\varepsilon_{poly}N_{poly}}}\right).$$

(d) $$W_{dpoly}=\frac{\varepsilon_{ox}\mathcal{E}_{ox}}{qN_{poly}}=\sqrt{\frac{2\varepsilon_{poly}\phi_{poly}}{qN_{poly}}}=\sqrt{\frac{2\varepsilon_{poly}}{qN_{poly}}}\left(-\alpha+\sqrt{\alpha^2+\beta}\right)$$

$$=\sqrt{\frac{2\varepsilon_{poly}}{qN_{poly}}}\left[\sqrt{\frac{T_{ox}^2}{4\varepsilon_{ox}^2}2q\varepsilon_{poly}N_{poly}+\left(V_g-V_{fb}-2|\phi_B|\right)}-\frac{T_{ox}}{2\varepsilon_{ox}}\sqrt{2q\varepsilon_{poly}N_{poly}}\right].$$

(e) Using the equation derived in part (c), we find $\phi_{poly}=0.28\,V$.

And, using the equation derived in part (d), we find $W_{dpoly}=2.46\,nm$.

$$\left(|\phi_B|=\frac{kT}{q}\ln\left(\frac{N_a}{n_i}\right)=0.41\,V,\quad V_{fb}=-0.97\,V,\quad and\ \varepsilon_{poly}=\varepsilon_s\right).$$

(f) Calculate V_t using Equation 5.4.3:

$$V_t=V_{fb}+2|\phi_B|+\frac{1}{C_{ox}}\sqrt{4q\varepsilon_sN_a|\phi_B|}=(-0.97+0.82+0.095)V=-0.055\,V$$

Using Equation 5.8.3 with $\phi_{poly}=0.28\,V$,

$$Q_{inv}=C_{ox}\left(V_g-V_t-\phi_{poly}\right)=\frac{\varepsilon_{ox}}{T_{ox}}(1.5+0.055-0.28)V=2.20\times10^{-6}\,coul/cm^2.$$

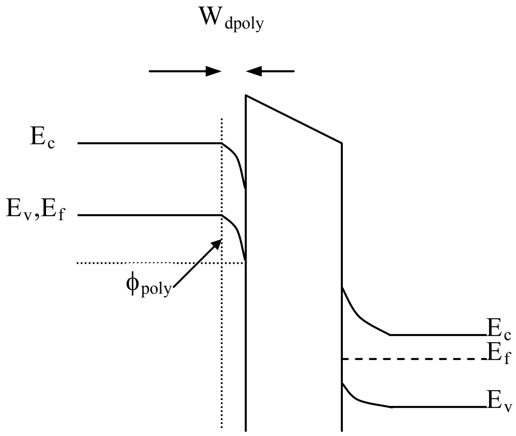
(g) First, we calculate C_{oxe}.

$$C_{oxe}=\frac{\varepsilon_{ox}}{T_{ox}+W_{dpoly}/3}=1.22\times10^{-6}\,F/cm^2.$$

$$Q_{inv} = C_{oxe}\left(V_g - V_t\right) = 1.90\times 10^{-6}\, coul/cm^2 .$$

This value is smaller than what we have found in part (f).

5.22 (a)



V_g is negative, V_{fb} is positive, ϕ_{st} is negative, V_{ox} is negative and ϕ_{poly} is negative

5.23 (a) $Q_{poly} = qN_{poly}W_{dpoly} = qN_{poly}\sqrt{\dfrac{2\varepsilon_s V_{poly}}{qN_{poly}}} = \sqrt{2\varepsilon_s qN_{poly}V_{poly}}$

(b) $C_{poly} = \dfrac{\varepsilon_s}{W_{dpoly}} = \dfrac{\varepsilon_s qN_{poly}}{Q_{poly}} = \sqrt{\dfrac{\varepsilon_s qN_{poly}}{2V_{poly}}}$

(c) $C_{total} = (C_{ox}{}^{-1} + C_{poly}{}^{-1})^{-1} = (\dfrac{1}{C_{ox}} + \sqrt{\dfrac{2V_{poly}}{\varepsilon_s qN_{poly}}})^{-1}$

Threshold Voltage Expression

5.24
$$\begin{aligned} V_t &= V_{fb} + \phi_s + V_{ox} \\ &= V_{fb} + \phi_s + \frac{qN_a W_{dep}}{C_{ox}} \\ &= V_{fb} + \phi_s + \frac{\sqrt{qN_a 2\varepsilon_s \phi_s}}{C_{ox}} \\ &= V_{fb} + 2\phi_B + \frac{\sqrt{qN_a 2\varepsilon_s 2\phi_B}}{C_{ox}} \end{aligned}$$

Chapter 6

MOSFET V_t

6.1 15 Ω-cm $=10^{15}$ cm^{-3}, $\therefore E_f = E_v + 0.26$eV oxide trap density = $8\text{x}10^{10}$ cm^{-2}, $Z = 50$ μm, $L = 2$ μm, $T_{ox} = 5$nm

(a) $V'_{fb} = V_{fb} + \Delta V$. $V_{fb} = 3.1 - (3.1 + 0.86) = -0.86$eV, $\Delta V = Q_f/C_{ox} = 18.5$ mV.
Therefore $V'_{fb} \cong V_{fb} = -0.86$ eV
When oxide thickness is thin, the trap charge effect can be ignored.

(b) $V_t = V'_{fb} + V_{ox} + V_s \cong V_{fb} + 2\phi_B + (2\varepsilon_s q N_a 2\phi_B)^{1/2}/C_{ox}$
$= -0.86 + 0.6 + 0.02\text{V} = -0.24\text{V}$

(c) To make $V_t = 0.5$V, one should implant boron into silicon substrate such that $\Delta V_t = Q_{imp}/C_{ox}$. Therefore ion implant dose should be
$(0.5\text{V}+0.24\text{V}) \times C_{ox} \div q = 3.2\times10^{12}$ cm^{-2}.

6.2 (a) Using Equation 4.16.4 and referring to Table 1-4, we find

$$\phi_{bi} = \phi_{Bn} - kT\ln\left(\frac{N_c(GaAs)}{N_d}\right) = 1V - kT\ln\left(\frac{4.7\times10^{17}}{1\times10^{17}}\right) = 0.96V .$$

Then,

$$W_{dep} = \sqrt{\frac{2\varepsilon_s\phi_{bi}}{q}\frac{1}{N_d}} = \sqrt{\frac{2(13\varepsilon_0)\phi_{bi}}{q}\frac{1}{N_d}} = 0.12\,\mu m .$$

(b) $$W_{dep} = 0.2\,\mu m = \sqrt{\frac{2(13\varepsilon_0)(\phi_{bi}+V)}{q}\frac{1}{N_d}} \Rightarrow V = \frac{qN_dW_{dep}^2}{2(13\varepsilon_0)} - \phi_{bi} = 1.82V .$$

A negative V_g is need to increase W_{dep} and turn-off the channel. (A metal/N-type semiconductor Schottky diode exhibits the same forward/reverse bias properties as an P^+/N diode.)

(c) Yes. If the positive V_g is kept small (say 0.5V), the forward current of the Schottky gate maybe comparable to the subthreshold drain leakage current. A positive V_g would reduce W_{dep} and therefore raise I_{ds}.

(d) The channel thickness or doping concentration must be reduced so that $W_{dep} \geq$ channel thickness at $V_g = 0$.

6.3 $C_{ox} = 6.9\times10^{-7}\,\dfrac{F}{cm^2}$, $\phi_B = \dfrac{\mathrm{kT}}{\mathrm{q}}\ln\dfrac{\mathrm{N_{sub}}}{\mathrm{n_i}} = 0.47\mathrm{eV}$

(a)` $V_t = V_{fb} + 2\phi_B + \dfrac{1}{C_{ox}}\sqrt{2q\varepsilon_s N_{sub} 2\phi_B}$

$$V_t = -\frac{E_g}{2} - \phi_B + 2\phi_B + \frac{1}{C_{ox}}\sqrt{2q\varepsilon_s N_{sub} 2\phi_B} = -0.09 + 0.61 = 0.52V$$

(b) $V_t = V_{fb} - 2\phi_B - \dfrac{1}{C_{ox}}\sqrt{2q\varepsilon_s N_{sub} 2\phi_B}$

$$V_t = -\frac{E_g}{2} + \phi_B - 2\phi_B - \frac{1}{C_{ox}}\sqrt{2q\varepsilon_s N_{sub} 2\phi_B} = -0.56 - 0.47 - 0.61 = -1.64V$$

(c) $V_t = V_{fb} - 2\phi_B - \dfrac{1}{C_{ox}}\sqrt{2q\varepsilon_s N_{sub} 2\phi_B}$

$$V_t = \frac{E_g}{2} + \phi_B - 2\phi_B - \frac{1}{C_{ox}}\sqrt{2q\varepsilon_s N_{sub} 2\phi_B} = -0.52V$$

(d) $V_b = 0V$
$V_s = 0V$
$V_d = 2.5V$
$V_g = 2.5V$

(e) $V_b = 2.5V$
$V_s = 2.5V$
$V_d = 0V$
$V_g = 0V$

(f) $I_{dsat} = \dfrac{\mu_n W C_{ox}}{2L}(V_{gs} - V_t)^2$

$$\frac{I_{dsatc}}{I_{dsatb}} = \frac{(-2.5-(-0.52))^2}{(-2.5-(-1.64))^2} \approx 5.3$$

The transistor with the lower absolute value of threshold voltage has a higher saturation current. That is why P^+ poly-gate PMOSFETs are typically used in IC.

(g) The ratio of the current is the ratio of the mobilities.
To find μ_n,
$(V_{gs} + V_t + 0.2)/6T_{oxe} = (2.5 + 0.52 + 0.2)V/(6\times5\times10^{-7})cm = 1.07MV/cm$
and $\mu_n = 250\mathrm{cm^2V^{-1}s^{-1}}$.

To find μ_p,
$$-\left(V_{gs}+1.5V_t-0.25\right)/6T_{oxe}=-\left(-2.5+1.5\times(-0.52)-0.25\right)V/\left(6\times5\times10^{-7}\right)cm=1.01MV/cm$$
and $\mu_p = 63cm^2V^{-1}s^{-1}$.

$$\frac{I_{dsat(c)}}{I_{dsat(a)}}=\frac{\mu_p}{\mu_n}\approx\frac{1}{4}$$

Basic MOSFET IV Characteristics

6.4 (a) Due to the highly doped regions nearby, transistor C-V always approaches C_{ox} in inversion. Hence, it is impossible to determine the frequency. Either high or low frequency could have been used.

(b) Since $V_t>V_{fb}$, this is a NMOS.

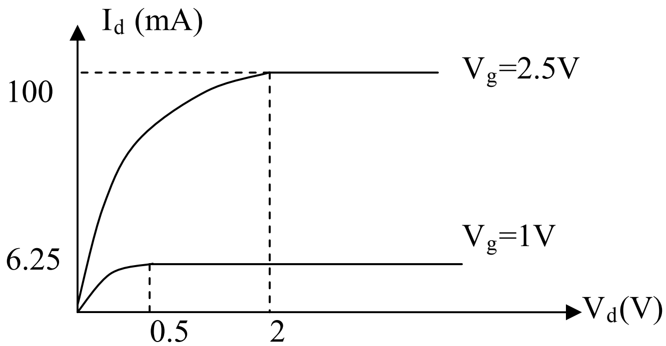
(c) From the I_d-V_g curve, V_t is 0.55V. More precisely,
$$I_d=\mu C_{ox}\frac{W}{L}V_{ds}\left(V_g-V_t-\frac{V_{ds}}{2}\right)\Rightarrow 0.55-V_t-\frac{V_{ds}}{2}=0$$
$$V_t=0.5\text{V}$$

(d) $\textit{Slope of curve } I_d\text{ -}V_g \textit{ line}=\mu C_{ox}\frac{W}{L}V_{ds}=5\times10^{-3}\Omega^{-1}$.

$$V_{ds}=0.1\text{V}$$
From the CV curve,
$$C_{ox}\text{WL}=1\text{pF}\Rightarrow \text{C}_{\text{ox}}\frac{\text{W}}{\text{L}}=10^{-4}\frac{\text{F}}{\text{cm}^2}$$
$$\text{Thus, }\mu=\frac{5\times10^{-3}}{0.1\times10^{-4}}=500\frac{cm^2}{Vs}$$

(e) $V_{dsat}=V_g-V_t$
$$I_{dsat}=\frac{\mu C_{ox}W}{2L}(V_{dsat})^2=0.025\frac{A}{V^2}.V_{dsat}{}^2$$

V_g	1V	2.5V
V_{dsat}	0.5V	2V
I_{dsat}	6.25mA	100mA



6.5 (a) For V_{gs}=4V, $V_{dsat} \sim 3V=(V_{gs}-V_t)$ and V_t=1V

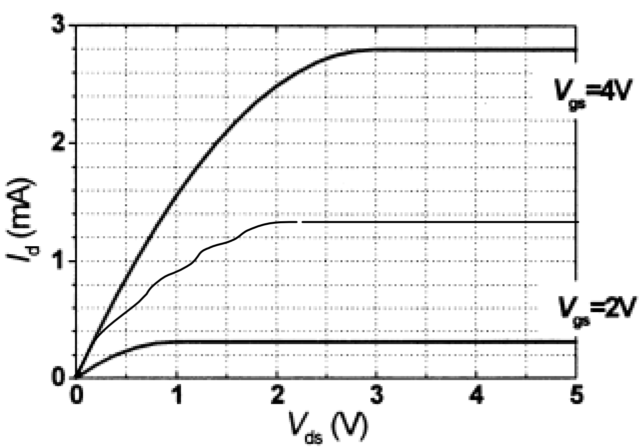
(b) $I_{dsat} = \mu_n C_{ox} W/2L \cdot (V_g - V_t)^2$

$C_{ox} = 3.45\times10^{-7}\, F/cm^2$

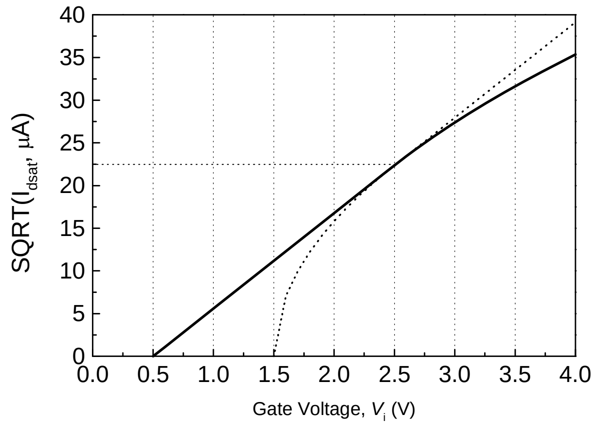
$$\mu_n = \frac{2I_{dsat}}{C_{ox}\left(\frac{W}{L}\right)(V_g - V_t)^2} = 361 cm^2/Vs$$

(c) At V_{gs}=3V, V_{dsat}=(3-1)V=2V

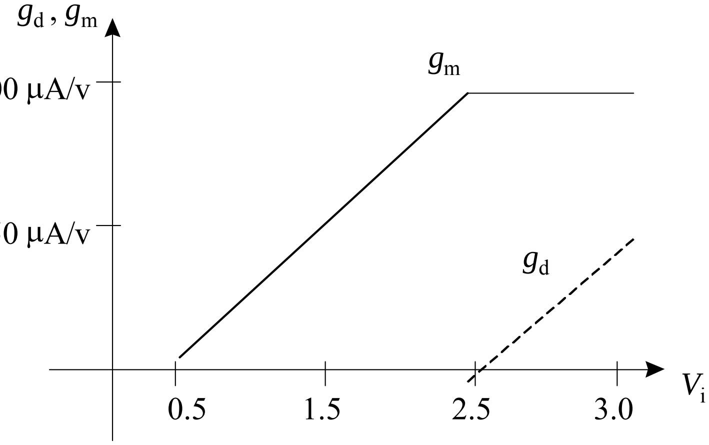
$I_{dsat} = 361\times3.45\times10^{-7}\cdot(10/2)\cdot(3-1)^2/2 = 1.25\times10^{-3}\,A$



6.6 (a) At saturation, $V_d = V_{dsat} = V_g - V_t$. V_{dd}=2V. Therefore the transistor is in saturation mode when V_g<2.5V. $I_{dsat} = 125(V_g\text{-}0.5)^2$ μA. When V_g>2.5V, the transistor is in linear region with $I_d = 500(V_g\text{-}1.5)$ μA.



(b) & (c) Transconductance: solid line, Output Conductance: dotted line



6.7 (a) $V_{gs} - V_t = 2V$

$V_{gs} = 2.5V$

(b) $Q_n = -C_{ox}(V_{gs} - V_t - V_c) = 0$ (Pinch-Off)

(c) $I_{ds} @V_{ds} = 4V$

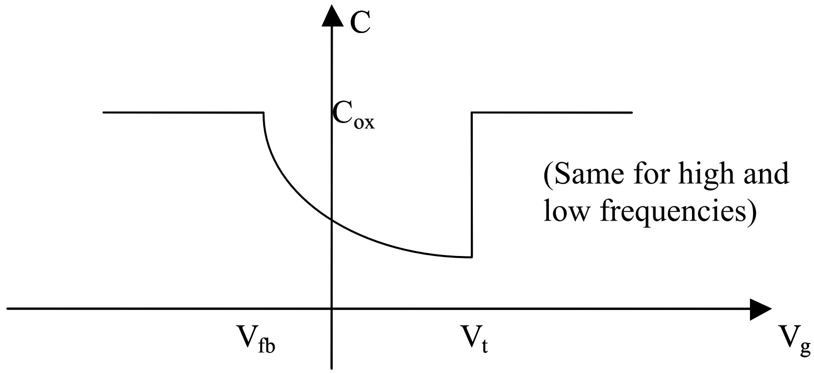
$V_{dsat} = V_{gs} - V_t = 3V$

(in saturation)

$I_{ds} \propto (V_{gs} - V_t)^2$

$I_{ds} = 10^{-3} \times \dfrac{3^2}{2^2} = 2.25 \times 10^{-3} A$

(d)



6.8 (a) $\phi_B = \frac{kT}{q}\ln\frac{N_a}{n_i} = 0.297V$

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = 7.08\times10^{-8}\ F/cm^2$$

$$V_{fb} = \chi_{Si} - (\chi_{Si} + \frac{E_g}{2} + \phi_B) = -0.857\text{V}$$

$$V_g = V_{fb} + V_s + V_{ox} \Rightarrow V_t = V_{fb} + 2\phi_B + \frac{\sqrt{2\varepsilon_s q N_a 2\phi_B}}{C_{ox}} = -0.064V$$

(b) $I_{dsat} = \frac{\mu_n C_{ox} W}{2L}(V_g - V_t)^2 = 1.21\text{mA}$

(c) Since V_d is less than (V_g-V_t), it is in the linear region.

$$I_d = \frac{\mu C_{ox} W}{L}\left[(V_g - V_t)V_d - \frac{V_d^2}{2}\right]$$

$$g_d = \frac{\partial I_d}{\partial V_D} = \frac{\mu C_{ox} W}{L}\left[(V_g - V_t) - V_d\right] = 1.17mS$$

(d) Since V_d is less than (V_g-V_t), it is in the linear region.

$$I_d = \frac{\mu C_{ox} W}{L}\left[(V_g - V_t)V_d - \frac{V_d^2}{2}\right]$$

$$g_d = \frac{\partial I_d}{\partial V_g} = \frac{\mu C_{ox} W}{L} V_d = 1.13mS$$

Potential and Carrier Velocity in MOSFET Channel

6.9 $I_d = -Q_n\mu_n \dfrac{dV_c}{dx}W = (V_g - V_t - V_c)C_{ox}\mu_n \dfrac{dV_c}{dx}W$

$$\therefore \int_0^x \frac{I_d}{\mu_n C_{ox} W}dx = \int_0^{Vc} (V_g - V_t - V_c)dV_c$$

$$\rightarrow I_d \cdot x/(\mu_n C_{ox} W) = (V_g - V_t)V_c - 1/2V_c^2$$

Solving this quadratic equation of V_c, we get

$$\therefore V_c(x) = (V_g - V_t) \pm \sqrt{(V_g - V_t)^2 - \frac{2I_d x}{\mu_n C_{ox} W}}$$

Choosing "-" so that $V_c(0)=0$,

$$\therefore V_c(x) = (V_g - V_t)\left[1 - \sqrt{1 - \frac{2I_d x}{\mu_n C_{ox} W (V_g - V_t)^2}}\right]$$

$$= (V_g - V_t)\left[1 - \sqrt{1 - \frac{2x\mu_n C_{ox}\frac{W}{2L}(V_g - V_t)^2}{\mu_n C_{ox} W (V_g - V_t)^2}}\right]$$

$$= (V_g - V_t)\left[1 - \sqrt{1 - x/L}\right].$$

6.10 (a) $I_{ds} = WC_{oxe}(V_{gs} - mV_{cs} - V_t)\mu_{es} dV_{cs}/dx$

$$\int_0^x I_{ds}dx = WC_{oxe}\mu_s \int_0^{V_{cs}} (V_{gs} - mV_{cs} - V_t)dV_{cs}$$

$$I_{ds}x = WC_{oxe}\mu_s (V_{gs} - V_t - \frac{mV_{cs}}{2})V_{cs}$$

Equating the expression above with

$$\mathrm{I}_{ds} = \frac{W}{L}C_{oxe}\mu_s(V_{gs} - V_t - \frac{m}{2}V_{ds})V_{ds},$$

we get

$$\frac{\mathrm{x}}{\mathrm{L}}\left(V_{gs}-V_t-\frac{mV_{ds}}{2}\right)V_{ds}=(V_{gs}-V_t-\frac{mV_{cs}}{2})V_{cs}$$

$$mV_{cs}{}^2-2(V_g-V_t)V_{cs}+\frac{x}{L}(2(V_g-V_t)-mV_{ds})V_{ds}=0$$

Solving the quadratic equation, we get

$$\mathrm{V_{cs}}=\frac{V_{gs}-V_t}{m}\pm\frac{\sqrt{(V_g-V_t)^2-m\frac{x}{L}(2(V_g-V_t)-mV_{ds})V_{ds}}}{m}$$

$$V_{cs}=\frac{V_{gs}-V_t}{m}(1-\sqrt{1-\frac{x}{L}})$$

(b) $$Q_{inv}(x)=C_{oxe}(V_{gs}-mV_{cs}-V_t)=C_{oxe}\left[V_{gs}-V_t-(V_{gs}-V_t)(1-\sqrt{1-\frac{x}{L}}\right]$$

$$=C_{oxe}\left(V_{gs}-V_t\right)\left[1-\left(1-\sqrt{1-\frac{x}{L}}\right)\right]=C_{oxe}(V_{gs}-V_t)\left(\sqrt{1-\frac{x}{L}}\right)$$

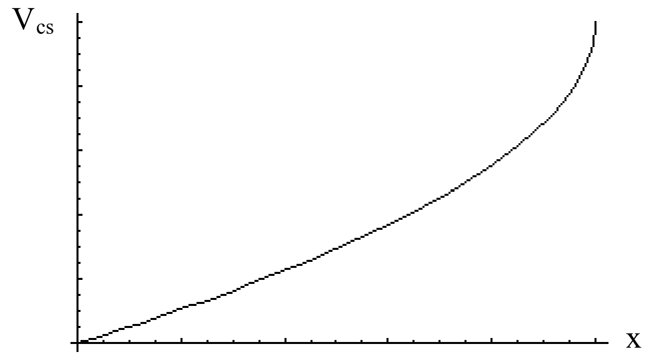
(c) $$\frac{dV_{cs}}{dx}=\varepsilon(x)=\frac{\left(V_g-V_t\right)}{m}\left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{1-x/L}}\right)\left(\frac{1}{L}\right)=\frac{\left(V_g-V_t\right)}{2mL}\left(\frac{1}{\sqrt{1-x/L}}\right)$$

$$\upsilon(x)=\mu_n\frac{dV_{cs}}{dx}=\mu_n\varepsilon(x)=\frac{\mu_n\left(V_g-V_t\right)}{2mL}\left(\frac{1}{\sqrt{1-x/L}}\right)$$

(d) $$WQ_{inv}\mu_n\varepsilon=W.C_{oxe}(V_{gs}-V_t)\left(\sqrt{1-\frac{x}{L}}\right).\frac{\mu_n\left(V_g-V_t\right)}{2mL}\left(\frac{1}{\sqrt{1-x/L}}\right)$$

$$=\frac{WC_{oxe}\mu_n}{2mL}\left(V_{gs}-V_t\right)^2=I_{dsat}$$

(e)



IV Characteristics of Novel MOSFET

6.11 (a) $I_d = -Q_n \mu_n \dfrac{dV_c}{dx} W = (V_g - V_t - V_c) C_{ox}(x) \mu_n \dfrac{dV_c}{dx} W$

$$= (V_g - V_t - V_c) \frac{\varepsilon_{ox}}{Ax^2 + B} \mu_n \frac{dV_c}{dx} W$$

$$\therefore \int_{-L/2}^{L/2} I_d \cdot (Ax^2 + B) dx = \int_0^{V_{ds}} (V_g - V_t - V_c) \varepsilon_{ox} \mu_n W dV_c$$

$$\rightarrow I_d \cdot [\frac{A}{3} x^3 + Bx]_{-L/2}^{L/2} = \varepsilon_{ox} \mu_n W [(V_g - V_t) V_c - 1/2 V_c^2]_0^{V_{ds}}$$

$$\therefore I_d = \frac{W}{L} \cdot \frac{\varepsilon_{ox} \mu_n}{\frac{AL^2}{12} + B} \cdot [(V_g - V_t) V_{ds} - 1/2 V_{ds}^2]$$

(b) $V_{dsat} = V_{ds} @ \dfrac{\partial I_d}{\partial V_{ds}} |_{V_{gs}} = 0 \qquad \therefore V_{dsat} = V_g - V_t$

(c) It suggests a large W_{dmax}. $V_{ox} = Q_n / C_{ox}$

6.12 (a) $I_d = -Q_n \mu_n \dfrac{dV_c}{dx} W(x) = (V_g - V_t - V_c) C_{ox} \mu_n \dfrac{dV_c}{dx} W(x)$

$$\therefore \int_0^L I_d / W(x) dx = \int_0^{V_{ds}} (V_g - V_t - V_c) \mu_n C_{ox} dV_c$$

$$\rightarrow I_d \cdot \ln[(W_0 + L)/W_0] = \mu_n C_{ox} [(V_g - V_t) V_{ds} - 1/2 V_{ds}^2]$$

$$\therefore I_d = \frac{\mu_n C_{ox}}{\ln(1 + L/W_0)} [(V_g - V_t) V_{ds} - 1/2 V_{ds}^2]$$

(b) $V_{dsat} = V_{ds} @ \frac{\partial I_d}{\partial V_{ds}}|_{V_{gs}} = 0 \qquad \therefore V_{dsat} = V_g - V_t$

$$\therefore I_{dast} = \frac{\mu_n C_{ox}}{\ln(1 + L/W_0)} \cdot \frac{(V_g - V_t)^2}{2}$$

CMOS

6.13 (a) $V_{fb,NMOS}$ = -(E_g/2) – (kT/q * ln(5e15/1e10)) = - 0.55 -0 .4 = -0.95V
$V_{fb,PMOS}$ = -0.55 + 0.4 = -0.15V
Not symmetrical

(b) $V_{fb,NMOS}$ = 0.55 - 0.4 = 0.15V
$V_{fb,PMOS}$ = 0.55 + 0.4 = 0.95V
Not symmetrical

(c) Since V_{ox} and V_s will be symmetrical, I would use a mid-gap gate material such as tungsten.
So the workfuction will be 4.05 eV + $E_{g,Si}$/2 = 4.6eV. However, processing issues makes tungsten (or any metal gates for that matter) a challenge to implement.

(d) In the same process, the NMOS and PMOS will have same oxide thickness. If the substrate doping levels for n and p flavors are the same, then I would use P^+ gates for PMOS devices and N^+ gates for NMOS devices. In this way, the flatband voltages will be symmetrical and the resulting $|V_t|$ small.

6.14 (a) PMOS, N-type substrate:

$$\phi_n = kT \ln \frac{N_d}{n_i} = 0.38V$$

$$\mathrm{V_{fb}} = \phi_m - \chi_{si} - \frac{E_g}{2} + \phi_n = 4.1 - 4.05 - 0.55 + 0.38 = -0.12\mathrm{V}\,.$$

NMOS, N-type substrate:

$$\phi_p = kT \ln \frac{N_a}{n_i} = 0.42V$$

$$\mathrm{V_{fb}} = \phi_m - \chi_{si} - \frac{E_g}{2} - \phi_p = 4.1 - 4.05 - 0.55 - 0.42 = -0.92\mathrm{V}$$

(b) $C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = 6.9 \times 10^{-7} \frac{F}{cm^2}$

PMOS:

$$V_t = V_{fb} - 2\phi_n - \frac{1}{C_{ox}}\sqrt{2\varepsilon_s qN_d 2\phi_n} = -0.12 - 0.76 - 0.10 = -0.98V$$

NMOS:

$$V_{fb} + 2\phi_p + \frac{1}{C_{ox}}\sqrt{2\varepsilon_s qN_d 2\phi_p} = -0.92 + 0.84 + 0.24 = 0.16V$$

(c) The threshold voltage must be changed by

$$\Delta V_t = -\frac{Q_{impl}}{C_{ox}} = 0.82\text{V}.$$

Hence,

$$Q_{impl} = -5.7\times10^{-7}\frac{C}{cm^2}.$$

6.15 $$I_{ds} = WC_{oxe}(V_{gs} - mV_{cs} - V_t)\frac{\mu_s\, dV_{cs}/dx}{1 + \frac{dV_{cs}}{dx}\Big/\mathcal{E}_{sat}}$$

$$\int_0^L I_{ds}\,dx = \int_0^{V_{ds}}\left[WC_{oxe}\mu_s\left(V_{gs} - mV_{cs} - V_t\right) - I_{ds}/\mathcal{E}_{sat}\right]dV_{cs}$$

$$I_{ds}L + I_{ds}\frac{V_{ds}}{\varepsilon_{sat}} = WC_{oxe}\mu_s(V_{gs} - V_t - \frac{m}{2}V_{ds})V_{ds}$$

$$I_{ds} = WC_{oxe}\mu_s(V_{gs} - V_t - \frac{m}{2}V_{ds})V_{ds}\Bigg/\left(L + \frac{V_{ds}}{\mathcal{E}_{sat}}\right)$$

$$I_{ds} = \frac{W}{L}C_{oxe}\mu_s(V_{gs} - V_t - \frac{m}{2}V_{ds})V_{ds}\Bigg/\left(1 + \frac{V_{ds}}{L\mathcal{E}_{sat}}\right) = \frac{I_{ds}\text{(Long channel)}}{1 + V_{ds}/\mathcal{E}_{sat}L}.$$

6.16

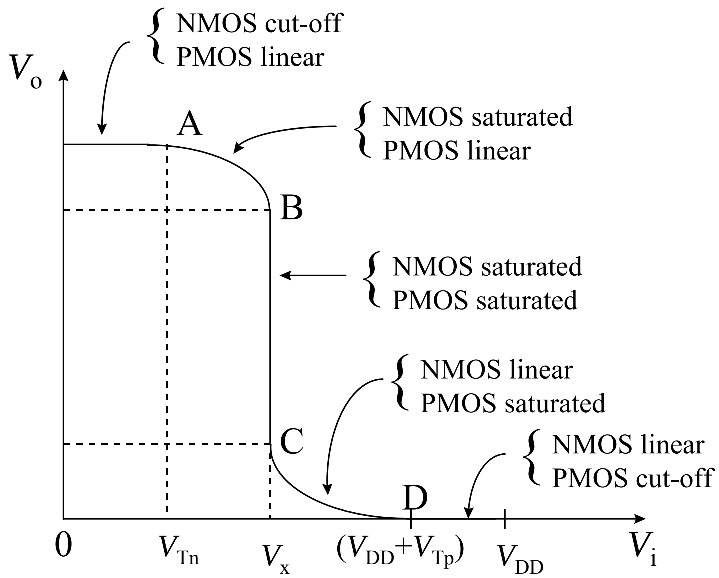
	NFET Operation Mode	PFET Operation Mode
A	**Cut-off**	**Linear**
B	**Saturation**	**Linear**
C	**Linear**	**Saturation**
D	**Linear**	**Cut-off**

A: Vgs<Vth for NFET, therefore it is cut off. For PFET |Vgs| > |Vth| and |Vds|<|Vdsat| (|Vds|~0V, |Vdsat| ~ 1.05V), so it operates in linear mode.

B: For NFET Vgs > Vth and Vds>Vdsat (Vds~1.75V, Vdsat ~ 0.3V), so it operates in saturation mode. For PFET |Vgs| > |Vth| and |Vds|<|Vdsat| (|Vds|~0.25V, |Vdsat| ~ 0.6V), so it operates in linear mode.

The answers to C and D can be worked out through the same procedure.

6.17 (a)



(b) At the point B where $V_i = V_x$, the NMOS is just becoming saturated from the linear region. Since NMOS is in the linear region

$$I_{dn} = K_N \left[(V_x - V_{tn})(V_x - V_{tn}) - \frac{(V_x - V_{tn})^2}{2} \right]$$

Since PMOS is saturated

$$I_{dp} = \frac{K_P}{2}(V_{dd} - V_x + V_{tp})^2$$

But $I_{DN} = I_{DP}$

$$\therefore \quad 40\left[(V_x - 1)^2 - \frac{(V_x - 1)^2}{2} \right] = \frac{35}{2}(5 - V_x - 1)^2$$

$$40(V_x - 1)^2 = 40(4 - V_x)^2 \Rightarrow V_x = 2.45 \text{ V}$$

Thus,

Point	V_i(V)	V_o(V)
A	1	5
B	2.45	3.45
C	2.45	1.45
D	4	0

Body Effect

6.18 For a P-channel MOSFET, we have

$$V_t = V_{fb} + 2\phi_B + \frac{\sqrt{2\varepsilon_s q N_d (2\phi_B + V_{bs})}}{C_{ox}}$$

$$\Delta V_t = \frac{\sqrt{2\varepsilon_s q N_d}}{C_{ox}}\left(\sqrt{2\phi_B + V_{bs}} - \sqrt{2\phi_B}\right)$$

(a) For 100 nm oxide, $C_{ox} = 3.45\times10^{-8}$ F/cm^2.
If V_{bs} = 5V, ΔV_t = -0.8V.
By iteration, using initial guess of ϕ_B = 0.3V, we obtain
$N_d = 8.9\times10^{14}$ /cm^3 and ϕ_B = 0.284V.

(b) If V_{sb} is -2.5 V, ΔV_t = -0.497V.
V_t = -1.5 - 0.497 = 2.0 V

Velocity-Saturation Effect

6.19 In all 3 cases, use the general equation $I = WQ_{inv}v_{drift}$.

Case A:
The NMOS is in the triode region.
On source side, $Q_{inv}=C_{ox}(V_g-V_t)$ = 138e-9(5-.7) = 593 nC/cm^2.
So $\upsilon_{drift} = I/(WQ_{inv})$ = 1.5e-3/(15e-4 × 593e-9) = 1.7 x 10^6 cm/sec.
On drain side, $Q_{inv} = C_{ox}(V_g-V_t-V_d)$ = 138e-9(5-.7-.5) = 524 nC/cm^2.
Thus, υ_{dr} = 1.5e-3/(15e-4 × 524e-9) = 1.9 x 10^6 cm/sec.

Case B:
The NMOS enters saturation region.
On source side, υ_{dr} = 3.75e-3/(15e-4×593e-9) = 4.2 x 10^6 cm/sec.
On drain side, the electron velocity is saturated.
Thus, $\upsilon_{dr} = \upsilon_{sat}$ = 8 x 10^6 cm/sec.

Case C:
Similar to case B.
On source side, υ_{dr} = 4e-3/(15e-4×593e-9) = *4.5 x 10^6 cm/sec.*
On drain side, $\upsilon_{dr} = \upsilon_{sat}$ = *8 x 10^6 cm/sec*.

6.20

	T_{ox}	W	L	V_t	V_g
V_{dsat}	↑	No change	↓	↑	↓
I_{dsat}	↑	↓	↑	↑	↓

Reducing T_{ox} means smaller V_t => larger V_{dsat} $(1/(V_g-V_t) + 1/(E_{sat}L))^{-1}$ & larger I_{dsat} ($Q_{inv} \propto C_{ox}$).
Reducing W has no effect on V_{dsat} and decreases I_{dsat} since $I_{dsat} = WQ_{inv}v_{sat}$.
Reducing L reduces V_{dsat} (as discussed in lecture) and increases I_{dsat}. If you want to consider very short-channel length devices (L => 0), then essentially I_{dsat} is independent of L.
Reducing V_t => larger V_{dsat} & larger I_{dsat}.
Reducing V_g => smaller V_{dsat} & smaller I_{dsat}.

6.21 $$I_d = \mu_s C_{ox} W\left(V_{gs} - V_t - m\frac{V_{ds}}{2}\right)V_{ds} \Big/ \left(L + \frac{V_{ds}}{\varepsilon_{sat}}\right)$$
$$= WC_{ox}(V_{gs} - V_t - mV_{ds})V_{sat}$$
$$= WC_{ox}(V_{gs} - V_t - mV_{ds})\mu_s \frac{\varepsilon_{sat}}{2}$$

$$\left(V_{gs} - V_t - m\frac{V_{ds}}{2}\right)V_{ds} = (V_{gs} - V_t - mV_{ds})\varepsilon_{sat}/2\left(L + \frac{V_{ds}}{\varepsilon_{sat}}\right)$$
$$= (V_{gs} - V_t - mV_{ds})\frac{\varepsilon_{sat}L}{2} + (V_{gs} - V_t - mV_{ds})\frac{V_{ds}}{2}$$

$$V_{gs}V_{ds} - V_tV_{ds} = (V_{gs} - V_t - mV_{ds})\varepsilon_{sat}L$$
$$V_{ds}(V_{gs} - V_t - m\varepsilon_{sat}L) = (V_{gs} - V_t)\varepsilon_{sat}L$$

$$V_{ds} = \frac{(V_g - V_t)\varepsilon_{sat}L}{(V_{gs} - V_t - m\varepsilon_{sat}L)} = \left[\frac{m}{(V_{gs} - V_t)} + \frac{1}{\varepsilon_{sat}L}\right]^{-1}$$

6.22 $$I_{ds} = \frac{\frac{W}{L}C_{oxe}\mu_{ns}\left(V_{gs} - V_t - \frac{m}{2}V_{ds}\right)V_{ds}}{1 + \frac{V_{ds}}{\varepsilon_{sat}L}} = \frac{\frac{W}{L}C_{oxe}\mu_{ns}\left((V_{gs} - V_t)\left(\frac{1}{V_{ds}}\right) - \frac{m}{2}\right)}{\frac{1}{V_{ds}^2} + \left(\frac{1}{\varepsilon_{sat}L}\right)\frac{1}{V_{ds}}}$$

$$\frac{1}{V_{dsat}} = \frac{m}{V_{gs} - V_t} + \frac{1}{\varepsilon_{sat}L}$$

6.23 (a) We know that

$$V_{dsat} = |\varepsilon_c L| \left[\left(1 + 2 \cdot \frac{(V_g - V_t)}{|E_c L|} \right)^{\frac{1}{2}} - 1 \right]$$

$$|\varepsilon_c L| = 0.1 \text{ V} \;\Rightarrow\; V_{\text{dsat}} = 0.1 \cdot \left[\left(1 + 2 \cdot \frac{4}{0.1} \right)^{\frac{1}{2}} - 1 \right] = 0.54 \text{ V}$$

(b) $$|\varepsilon_c L| = 10 \text{ V} \;\Rightarrow\; V_{\text{dsat}} = 10 \cdot \left[\left(1 + 2 \cdot \frac{4}{10} \right)^{\frac{1}{2}} - 1 \right] = 1.83 \text{ V}$$

(c) We know that

$$I_{\text{dsat}} = \frac{\mu_{\text{n0}} C_{\text{ox}} Z}{2L} V_{\text{dsat}}{}^2 \quad \text{and} \quad C_{\text{ox}} Z = \frac{10\text{fF}}{L}.$$

$$7\text{mA} = \frac{\mu_{\text{n0}} 10\text{fF}}{2L^2} 0.54^2$$

$$\mu_{\text{n0}} = 480 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$$

6.24 $V_{dsat} = \varepsilon_{sat} L(V_g - V_t) / (V_g - V_t + \varepsilon_{sat} L)$

What is ε_{sat}? $2v_{sat}/\mu_s = \varepsilon_{sat}$. μ_s is given by the universal mobility curve.

At T_{ox}=60A, $(V_g+V_t+0.2)/6T_{ox} = \varepsilon_{eff}$ = .9MV/cm.
From the curve, $\mu_s \sim 250$ cm^{2}V^{-1}s^{-1}.

This yields $\varepsilon_{sat} \sim 2(8\text{x}10^6)/250$ V/cm = $6.4\text{x}10^4$V/cm. Plug this back into the expression for V_{dsat} to get L ~ 0.19um.

$I_{dsat}/W = Q_{inv} v_{sat} = C_{ox}(V_g - V_t - V_{dsat}) v_{sat}$
$= (3.9\varepsilon_o/60\text{e-}8) \times (2.5\text{-}.5\text{-}.75) \times 8\times10^6 =$ *575uA/um width.*

Note: You will often find in literature that the saturation current is stated in units of uA/um instead of amperes. Also, notice that the Q_{inv} at $V_c = V_{dsat}$ is not zero. That is, I_{dsat} is limited by velocity saturation instead of pinch-off.

6.25 (a)

$$1+\frac{V_{gs}-V_t}{mE_{sat}L}=2$$

$$V_{gs}-V_t=mE_{sat}L=\frac{2mv_{sat}L}{\mu_{ns}}=\frac{2\cdot 1.2\cdot 8\times 10^6 cm/s\cdot 1\times 10^{-5}cm}{300cm^2/V-s}=0.64V$$

(b)

$$V_{gs}-V_t=mE_{sat}L$$

$$L=\frac{V_{gs}-V_t}{mE_{sat}}=\frac{\mu_{ns}(V_{gs}-V_t)}{2mv_{sat}}=\frac{300cm^2/Vs\cdot 0.2V}{2\cdot 1.2\cdot 8\times 10^6 cm/s}=3.13\times 10^{-6}cm=31.3nm$$

Effective Channel Length

6.26 (a) For very small V_{ds},

$$R_{channel}=\frac{V_{ds}}{I_{ds}}=\frac{L}{\mu_s C_{ox}W(V_g-V_t)}$$

In a short-channel device, S/D resistance can seriously degrade saturation current. Note that series resistance is worse for higher currents because $R_{channel}$ is the lowest under these bias conditions.

(b) $$R_{total}=R_s+R_d+R_{channel}=R_{sd}+R_{channel}$$

$$=R_{sd}+\left[\frac{L_{eff}}{\mu_s C_{ox}W(V_g-V_t)}\right]=R_{sd}+(\frac{L_{gate}-\Delta L}{\mu_s C_{ox}W(V_g-V_t)})$$

Think of R_{total} as the y-value, L_{gate} as the x-value, and $(\mu_s C_{ox}W(V_g-V_t))^{-1}$ as the slope. This fits nicely into the standard equation of the line: y = mx + b. You can choose devices with several gate lengths and measure the current from these devices at discrete gate voltages. Remember, that you are assuming V_{ds} is small (<100mV) in these measurements.

From the current, you can plot R_{total} versus L_{gate}. One sample data line is taken with the same V_g at different gate lengths. For example, if you measure your current at 5 different V_g's, you will get 5 separate curves. Ideally, all the lines will intersect at the same point on your plot. This intersection point occurs at $L_{gate}=\Delta L$ and $R_{total}=R_{sd}$.

In practice, it is not always straightforward to make such a plot. For instance, V_t can be difficult to determine accurately. Also, there is a strong dependence of mobility on gate voltage for thin-oxide MOSFETs. It's a good idea to check your data by taking measurements at several different V_g instead of at 2 or 3 gate voltages.

(c) $I_{dsat} = k(V_{gs}-I_{dsat}R_s-V_t)$, where k is a constant of proportionality

$I_{dsat}(1+kR_s) = k(V_{gs}-V_t) = I_{dsat0}$ *, notice here that* $k = I_{dsat0}/(V_{gs}-V_t)$

$I_{dsat} = I_{dsat0}/(1+kR_s) = I_{dsat0}/(1+I_{dsat0}R_s/(V_{gs}-V_t))$

(d) $\mathcal{E}_{sat} = (V_{gs}+V_t+0.2)/6T_{ox} = 1.1$ MV/cm.

$\mu_s \sim 225 cm^2V^{-1}s^{-1}$ is picked out from the universal mobility plot.

$\mathcal{E}_{sat} = 2v_{sat}/(\mu_s) = 7x10^4$V/cm.

I_{dsat0} = (long channel I_{dsat}) / (1+ $(V_{gs}-V_t)/(\mathcal{E}_{sat}L)$)
= 1.6mA / (1 + (1.1/.7)) = 0.622mA.

Plug in 0.622mA into the expression derived in part c and get the following:

@ R_s = 0ohms, I_{dsat} = .62mA

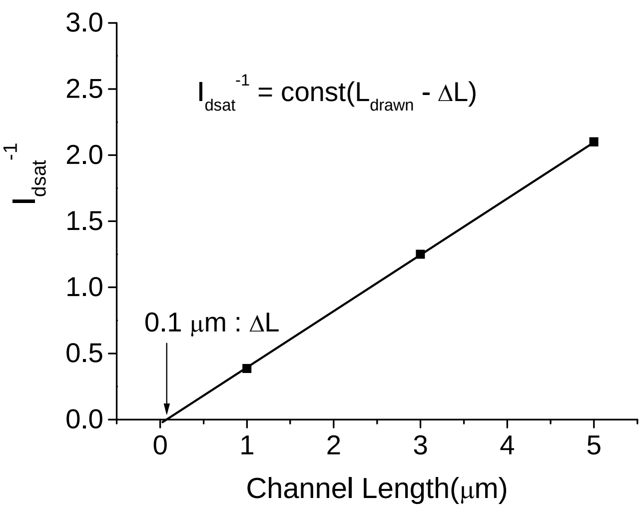
@ R_s = 100ohms, I_{dsat} = .59mA

@ R_s = 1000ohms, I_{dsat} = .40mA

6.27 (a) Choose three transistors with same channel width, Z, and different channel length, L_1, L_2,and L_3. Measure I_{dsat} at saturation condition for the 3 transistors to

get I_{d1}, I_{d2}, and I_{d3}. Solve the 3 equations to get μ, C_{ox}, and L_{eff}.

(b) $\Delta L = L - L_{eff}$ when gate oxide thickness is 4.5nm. Z =10 µm, μ = 300 cm^2/Vs. Using approach of (a), $\Delta L \simeq$ 0.1 µm.



(c) 2.59mA(L_1- L_{eff}) = $Z\mu C_{ox}(V_{gs}-V_t)$, V_t = 0.5V.

Chapter 7

Subthreshold Leakage Current

7.1 a) $Cox=3.45E\text{-}13/11E\text{-}8 = 3.138uF/cm^2$
$Vt = -1.04 + 0.95 + Qdep/Cox = -1.04 + 0.95 + 0.180 = 0.109V$

b) $S = 60(1+Cdep/Cox) = 60(1+2.953E\text{-}7/3.138E\text{-}6) = 65.64mV/dec$

c) $1/S = [\log(W/L*100nA) - \log(I_{leakage})]/Vt$
$I_{leakage} = 121nA$

Field Oxide Leakage

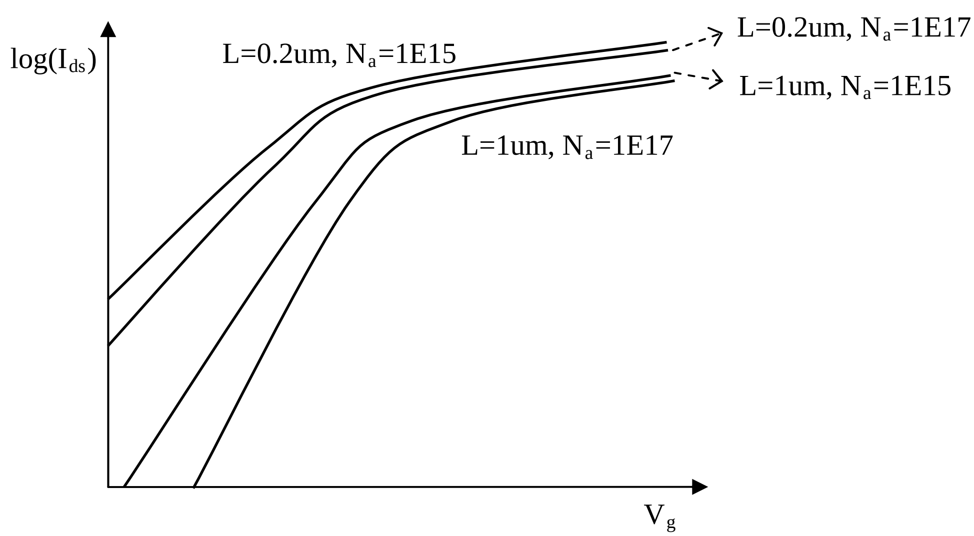
7.2 a) $Cox=3.45E\text{-}13/0.3E\text{-}4 = 1.151E\text{-}8F/cm^2$
$Vt = -1.01 + 0.92 + Qdep/Cox = -1.01 + 0.92 + 33.96 = 33.87V$

b) $S = 60(1+Cdep/Cox) = 60(1+2.119E\text{-}7/1.151E\text{-}8) = 1164.6mV/dec$

c) $1/S = [\log(W/L*100nA) - \log(I_{leakage})]/Vt$
$I_{leakage} = 2.81E\text{-}26nA$

Vt Roll-off

7.3



Trade-off between I_{off} and I_{on}.

7.4 i) Larger V_t : Decrease I_{off} and decrease I_{on}

ii) Larger L : Decrease I_{off} and decrease I_{on}

iii) Shallower junction : Decrease I_{off} and somewhat decrease I_{on}

iv) Smaller V_{dd} : Decrease I_{off} and decrease I_{on}

v) Smaller T_{ox} : Decrease I_{off} and increase I_{on}

A smaller T_{ox} contributes to leakage reduction and increases the precious I_{on}.

7.5

There is a lot of concern that we will soon be unable to extend Moore's Law. In your own words explain this concern and the concerns for high I_{on} and low I_{off}.

(a) Answer this question using 1 paragraph of less then 50 words. : Methods of manufacturing very small objects such as photolithography will run into limitations. Transistors may not be able to provide circuit functions with high speed and low power. Specifically, means of increasing speed (I_{on}) tend to worsen leakage (I_{off}). Other relevant comments include the following. V_t must not be too low because of subthreshold current but small L reduces V_t. Since I_{on} (speed) is roughly proportional to V_{gs}-V_t (V_{dd}-V_t), I_{on} and speed can not be raised by lowering V_t especially because V_{dd} needs to be reduced to reduce the power consumption.

(b) Support your description in (a) with 3 hand drawn sketches of your choice. : Possible choices are Log(I_{ds}) vs. V_{gs} , Vt vs. L and Vds, Ion vs. Ioff, gate leakage vs. Tox, etc.

(c) Why is it not possible to achieve high I_{on} and small I_{off} by picking optimal T_{ox}, X_j, W_{dep} etc? Please explain in your own words.: Decreasing T_{ox} is good for Ion but bad for I_{off}. So is increasing X_j. W_{dep} is basically fixed by the choice of T_{ox} and V_t.

(d) Provide three equations that help to quantify the issues discussed in part (c). : Some relevant equations include Eq. (6.9.14), Eq.(6..10.1), (7.2.8), Eq.(7.3.3), Eq.(7.5.2).

7.6 (a) Final equation:

$$l_d \propto \sqrt[3]{\frac{\varepsilon_{ox}\left(V_t - V_{fb} - 2\phi_B\right)}{qN_a} X_j} = \sqrt[3]{\frac{X_j T_{ox} T_{oxe} 2\varepsilon_s 2\phi_B}{\varepsilon_{ox}\left(V_t - V_{fb} - 2\phi_B\right)}}$$

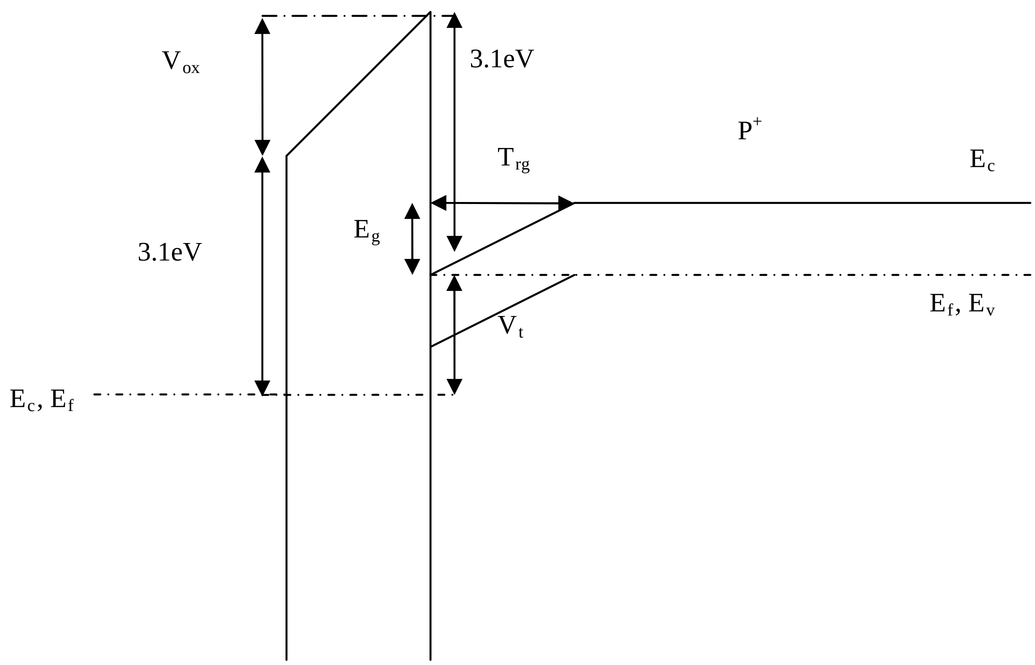
(b) The minimum acceptable L is several times of *ld***.** In order to support the reduction of L at each new technology node, ld must be reduced in proportion to L. According to the derived equation, in order to reduce the minimum acceptable channel length, $\left(V_t - V_{fb} - 2\phi_B\right)$ should be increased by gate workfunction engineering, Xj can be decreased, and the oxide permittivity can be increased .

7.7 (a) A smaller W_{dep}
Reduces DIBL (V_t is not as low) → decreases I_{off}

(b) $I_{dsat} = \frac{W}{2mL} C_{oxe} \mu_{ns} (V_{gs} - V_t)^2$ where $m \equiv 1 + \alpha = 1 + 3T_{oxe} / W_{dmax}$
Reducing W_{dep} increases *m* which decreases I_{dsat}.

MOSFET with Ideal Retrograde Doping Profile

7.8 (a)



(b) The figure above assumes that the gate is N+ polysislicon and the substrate (below the undoped layer) is P+ silison. In that case , it can be seen that $V_t = V_{ox}$. In general, V_t is determined by Eq. (5.2.2) and

$$V_t = V_{fb} + \varphi_{st} + V_{ox}$$

Note – You need to apply a positive gate bias to reach inversion. Thus the Fermi level of the poly gate is below the Fermi level of the substrate and the difference is V_t.

(c) $\varepsilon_{ox}\mathcal{E}_{ox} = \varepsilon_{si}\mathcal{E}_{si}$

$$\varepsilon_{ox}\frac{V_{ox}}{T_{ox}} = \varepsilon_{si}\frac{\phi_{st}}{T_{rg}} \approx \varepsilon_{si}\frac{E_g / q}{T_{rg}}$$

The last equality is the result of a further approximation for ϕ_{st}. Using the result of part (b),

$$\therefore\ V_t = V_{fb} + \phi_{st} + \phi_{st}\frac{\varepsilon_{si}T_{ox}}{\varepsilon_{ox}T_{rg}}$$

(d) First derive Eq. (7.5.2) using Eq. (7.5.1) and eq. (5.5.1). By comparing Eq. (7.5.2) and Eq. (7.5.3) one concludes that, for the same Vt, the depletion region width of the "ideal retrograde doping" MOSFET (Trg) is half of that of a MOSFET with uniformly doped substrate.

(e) A small W_{dep} reduces the V_t roll-off caused by DIBL which in effect decreases I_{off}.

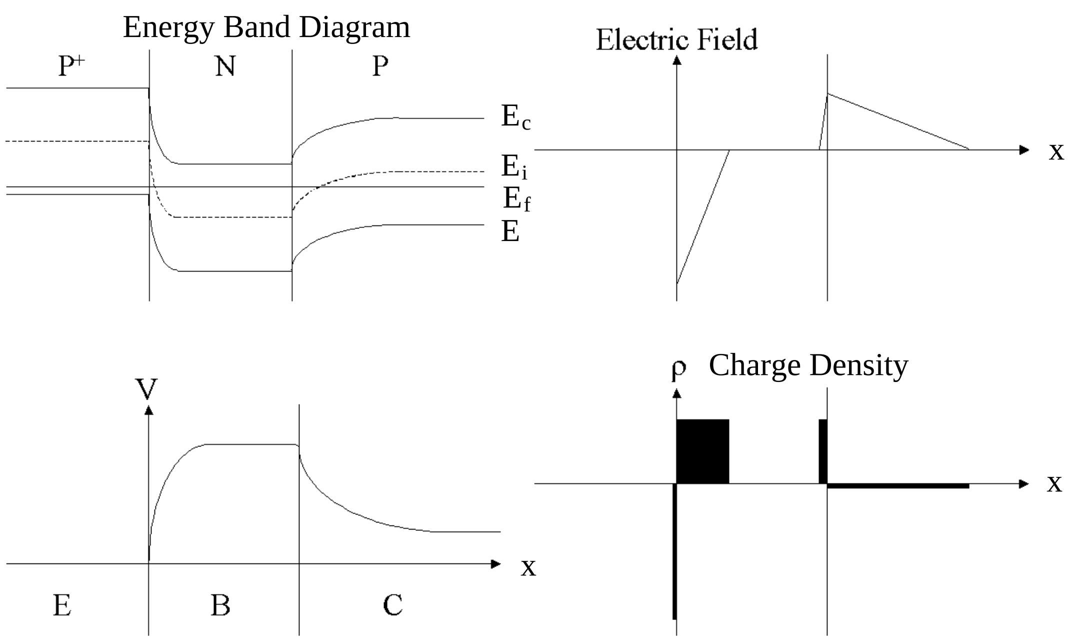
(f) $I_{dsat} = \dfrac{W}{2mL} C_{oxe} \mu_{ns} (V_{gs} - V_t)^2$ where $m \equiv 1 + \alpha = 1 + 3T_{oxe} / W_{dmax}$

Reducing W_{dep} increases *m* which decreases I_{dsat}.
A smaller I_{dsat} causes a longer inverter delay. However, it is important to reduce W_{dmax} so that L can be reduced to increase the device integration density and to reduce chip cost. I_{dsat} and speed still get improved due to the reduction in L.

Chapter 8

Energy Band Diagram of BJT

8.1 (a) & (b)
For the given doping concentrations, one computes E_f-E_i = -0.521 eV, 0.419 eV, and –0.299 eV in the emitter, base and collector, respectively. Also with N_{aE}>>N_{dB}, the E-B depletion width will lie almost exclusively in the base. Likewise, the majority of the C-B depletion width will lie in the collector.



(c) The built-in potential between the collector and emitter is

$$\Delta V_{CE} = \frac{kT}{q}\ln\left(\frac{N_{aE}}{N_{aC}}\right) = 0.026\times\ln\left(\frac{5\times10^{18}}{1\times10^{15}}\right) = 0.221V\ .$$

(d) $W = W_n - x_{nEB} - x_{nCB}$

$$x_{nEB} \cong \left[\frac{2\varepsilon_{Si}}{qN_{dB}}V_{biEB}\right]^{1/2} = 0.112\mu m$$

$$x_{nCB} = \left[\frac{2\varepsilon_{Si}}{qN_{dB}}\frac{N_{aC}}{N_{aC}+N_{dB}}V_{biCB}\right]^{1/2} = 0.01\mu m$$

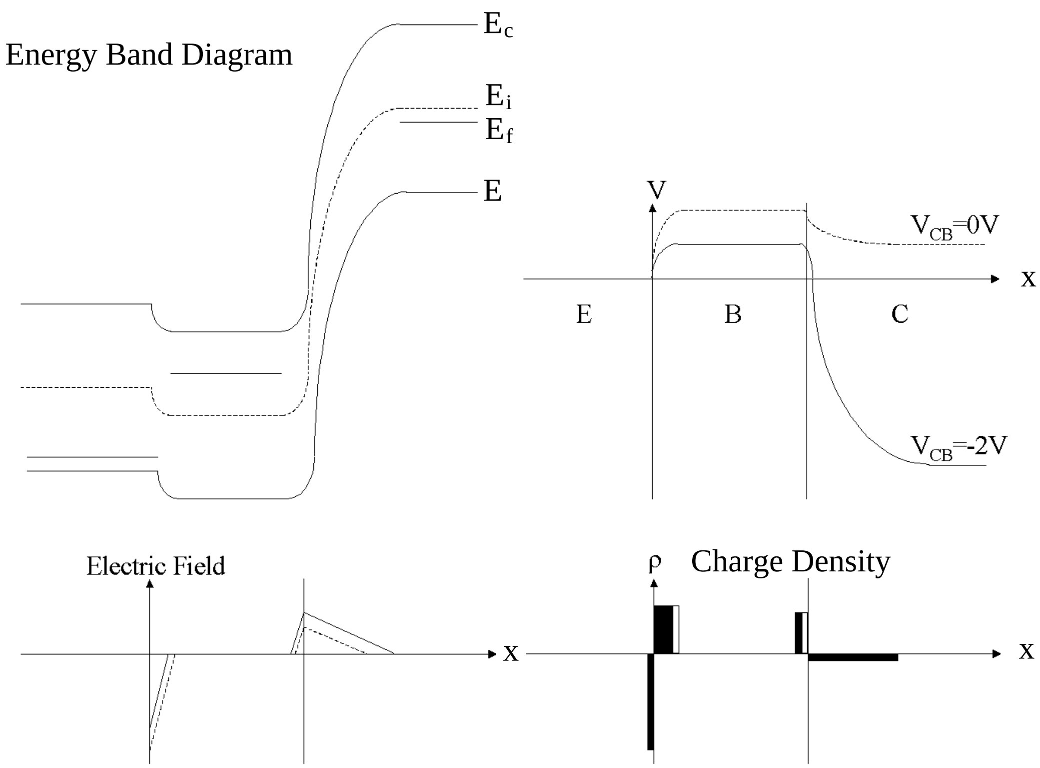
Therefore,

$W = 2.878\mu m$.

(e) $\left|Electric\ Field\right|_{\max}(E-B)=\left(\dfrac{qN_{dB}}{\varepsilon_{Si}}\right)x_{nEB}=1.72\times10^{5}\ Vcm^{-1}.$

$\left|Electric\ Field\right|_{\max}(C-B)=\left(\dfrac{qN_{dB}}{\varepsilon_{Si}}\right)x_{nCB}=1.54\times10^{4}\ Vcm^{-1}.$

(f)



IV Characteristics and Current Gain

8.2 From Eq. 8.4.1 and Eq. 8.4.2,

$$\beta_F \equiv \frac{I_C}{I_B} \Rightarrow I_B = \frac{I_C}{\beta_F}$$

$$I_C = \alpha_F I_E = \alpha_F\left(I_C + I_B\right) = \alpha_F I_C + \alpha_F I_B.$$

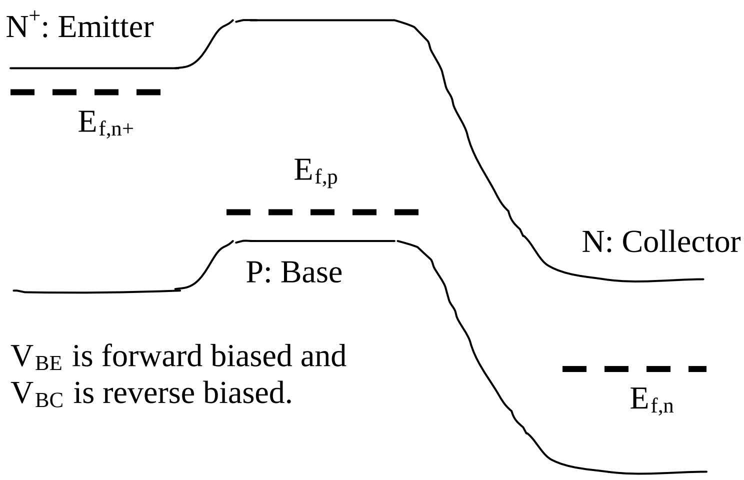
Substituting I_C/β_F into I_B, we obtain

$$I_C = \alpha_F I_C + \alpha_F \frac{I_C}{\beta_F} \Rightarrow 1 = \alpha_F + \frac{\alpha_F}{\beta_F}.$$

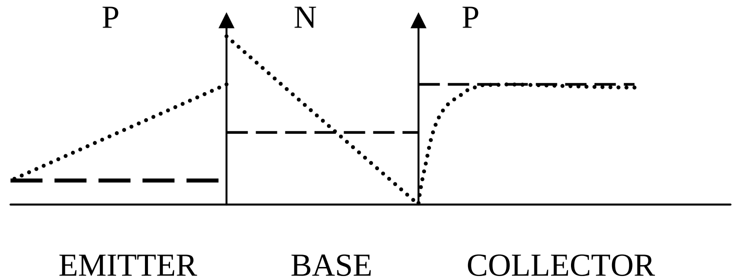
Solving for β_F yields

$$I_C = \frac{\alpha_F}{1-\alpha_F}.$$

8.3 (a)



(b)



The P,N,P on the diagram refer to the minority carrier type in each region. The horizontal dotted lines refer to the equilibrium minority concentration (i.e. p_{N0}, n_{P0}). The remaining dotted curves correspond to the excess minority carrier concentrations.

Assumptions made here: W_E & W_B are shorter than the diffusion lengths of the holes & electrons respectively, resulting in a linear decay of excess minority carriers in the emitter and base. You should also notice that the scale for the y-axis differs for each region.

(c) Base current consists of injection of holes into the emitter and recombination with a very small part of the collector current (remember that $I_E \sim I_C$). The collector current consists almost entirely of electrons emitted from the forward-biased BE junction which travel across the CB junction. Incidentally, an easy way to remember how a BJT works is to associate the names emitter and collector with the physical emission and collection of the minority electrons in the base.

8.4 $n_B(x) = n_B(0)\left(1 - \frac{x}{x_B}\right)$ *and* $p_E(x') = p_E(0')\left(1 - \frac{x'}{x_E}\right)$.

In the base region,

$$J_n(x) = qD_B\left(\frac{dn}{dx}\right) = \frac{qD_B n_B(0)}{x_B}.$$

In Emitter region,

$$J_p(x) = -qD_E\left(\frac{dp}{dx}\right) = \frac{qD_E p_E(0)}{x_E}.$$

$n_B(0) = 10^{13}$ cm^{-3} , $p_E(0') = 10^{11}$ cm^{-3}, and $p_C(x_B^+) = p_{CO} = 10^5$ cm^{-3}
$D_B = 20.8$ cm^{2}s^{-1}, D_E=1.8 cm^{2}s^{-1}, and D_C=11.9 cm^{2}s^{-1}.

(a) $J_{n,B}(x=0) = qD_B n_B(0)/x_B = 0.666\ A/cm^2$,

$J_{p,E}(x'=0') = qD_E p_E(0)/x_E = 3.6\times10^{-4}\ A/cm^2$,

and

$$J_{p,C}(x = 0.5\mu m^+) = qD_C p_{CO}/x_C = 8.7\times10^{-10}\ A/cm^2.$$

Therefore,

$$J_{TOTAL,\,BE} = 0.666 A/cm^2 + 3.6\times10^{-4} A/cm^2 = 0.666 A/cm^2$$

and

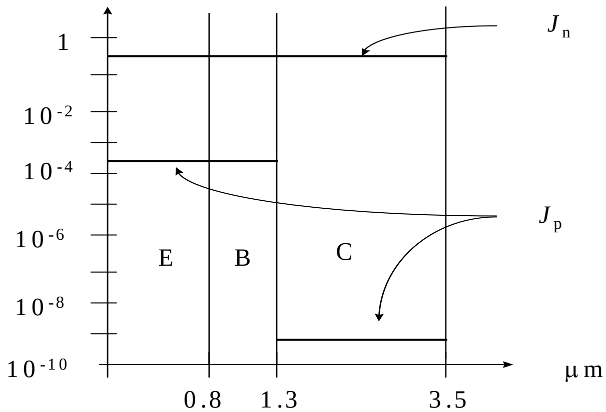
$$J_{TOTAL,\,BC} = J_n(x_B^-) + J_p(x_B^+).$$

For short diode approximation, we assume $J_n(x_B^-) = J_n(x=0)$. More accurate relationship is $J_n(X_B^-) = \alpha_T J_n(x=0)$. However, since $\alpha_T \approx 1$, we can still say $J_n(X_B^-) = J_n(x=0)$.

Hence,

$$J_{TOTAL,\,BC} = 0.666\ A/cm^2$$
$$J_{n,E} = J_{TOTAL,\,BE} - J_p(x'=0') = 0.666\ A/cm^2$$

$$J_{n,C} = J_{TOTAL,\,BC} - J_p(x'=x_B^+) = 0.666\ A/cm^2$$
$$J_{p,E} = J_{p,B} = 3.6\times10^{-4}\ A/cm^2$$



(b) $$\alpha_T = \frac{1}{1+\dfrac{1+W_B^2}{2L_B^2}} = 0.99875 \quad (L_B = 10\mu m)$$

$$\gamma_E = \frac{J_n(0)}{J_n(0)+J_p(0)} = \frac{0.666}{0.666+3.6\times10^{-4}} = 0.999325$$

$$\beta_F = \frac{\alpha_F}{1-\alpha_F} = 519 \quad (\alpha = \alpha_T \gamma_E)$$

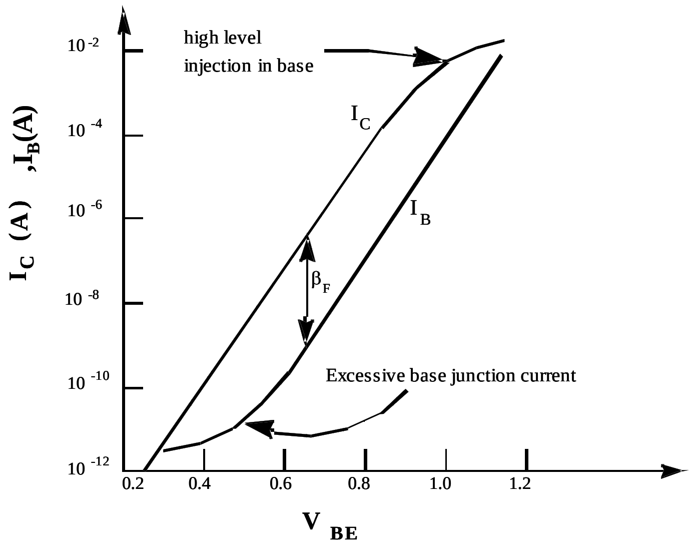
8.5 (a) To improve the emitter injection efficiency, and to reduce the back injected carriers from B-E.

(b) "Small" means $W_B << L_B$, typically $\leq 1\mu m$

(c) ΔW_B would become a larger percentage of W_B. Therefore, it would increase the slope of output characteristics.

(d) At very small values of I_C, the recombination current (excessive base junction current) in the E-B depletion region becomes a less significant part of I_B as I_C increases.

(e) At larger values of I_C, due to the high level injection in base, I_C does not increase exponentially as it does in the moderate level injection region.



(f)

Region	V_{EB}	V_{CB}
Active	+	-
Saturation	+	+
Cutoff	-	-

Schottky Emitter and Collector

8.6 (a) As you will find out in the latter part of this problem the injection of majority carrier of the semiconductor into the metal is much higher than the injection of minority carrier into the semiconductor region from the metal. This would make the emitter efficiency in the BJT very small! Hence, it would not be desirable to use a metal as an emitter in a BJT.

(b) We know that the hole diffusion current is given by

$$I_{\text{diff}} = I_{diff\,0}\left(e^{qV_A/kT} - 1\right) = qA\frac{D_P}{L_P}\frac{n_i^2}{N_d}\left(e^{qV_A/kT} - 1\right).$$

From Section 4.17 and 4.18,

$$I_{\text{te}} = I_{\text{te0}}\left(e^{qV_A/kT} - 1\right) = AKe^{q\Phi_B/kT}\left(e^{qV_A/kT} - 1\right).$$

Noting that $\dfrac{D_P}{L_P} = \sqrt{\dfrac{D_P}{\tau_p}} = \sqrt{\dfrac{(kT/q)\mu_p}{\tau_p}}$,

we have

$$\frac{I_{\text{diff}}}{I_{\text{te}}}=\frac{I_{diff\,0}}{I_{\text{te0}}}=\frac{q\sqrt{\frac{(kT/q)\mu_{\text{p}}}{\tau_{\text{p}}}}\frac{n_{\text{i}}^{2}}{N_{\text{d}}}}{Ke^{-\Phi_{\text{B}}/kT}}=\frac{\left(1.6\times10^{-19}\right)\left[\frac{(0.0259)(437)}{10^{-6}}\right]^{1/2}\left(\frac{10^{20}}{10^{16}}\right)}{(140)e^{-(0.72)/(0.0259)}}$$

$$=5.05\times10^{-7}.$$

(c) A Schottky base-collector junction (the collector being the metal) would be functional in a BJT. The energy diagram of the base-collector junction would be similar to Fig. 4-34(b). It would be effective collecting the electrons arriving at this junction from the P type base to the metal. The field at the Schottky junction sweeps the electrons into the metal collector just as in the PN base-collector junction shown in Fig. 8-1(b).

Gummel Number and Gummel Plot

8.7 (a) $\beta = J_C/J_B = 100.$

(b) Intercept of J_C is 10^{-10} A/cm^2= $qn_i^2 D_n / N_B W_B$

$$N_B=\frac{qn_i^2 D_n}{10^{-10}\cdot W_B}=8\times10^{16}\,cm^{-3}.$$

(c) Peak concentration

$$\approx\frac{n_i^2}{N_B}e^{\frac{qV_{BE}}{kT}}=10^{17}$$

$$e^{\frac{qV_{BE}}{kT}}=8\times10^{13}$$

$$V_{BE}=\ln(8\times10^{13})\times26mV=832mV.$$

(d) $\tau_n=\dfrac{W_B^2}{2D_n}=\dfrac{(0.2\times10^{-4})^2}{2\times10}=2\times10^{-11}\sec.$

Ebers-Moll Model

8.8 (a) Consider that n’ = $n_{P0}(e^{qVa/kT}-1)$ and p’ = $p_{N0}(e^{qVa/kT}-1)$. Next, take a look at the BC junction ,

⇨ n’/p’ = N_B/N_C (1)

Similarly, at the BE junction,

⇨ p’/n’ = N_E/N_B (2)

Multiply (1) and (2) to get $N_E/N_C = (8/4)\times(10/2) = 10$.

Thus, $N_C = 0.1N_E = 10^{17} cm^{-3}$.

(b) The BJT is operating in saturation because both BE and BC junctions are forward-biased and the resulting minority carrier concentrations are larger than the equilibrium values.

(c) The stored minority charge is equal to the area under the curve in the base.

$$Q = \frac{Aq}{2}\left[p'(x=1\mu m) + p'(x=2\mu m)\right]W_B =$$
$$= 10^{-5} \times 1.6\times10^{-19} \times 0.5 \times 14\times10^{14} \times 10^{-4} = 1.12\times10^{-13}\ coul.$$

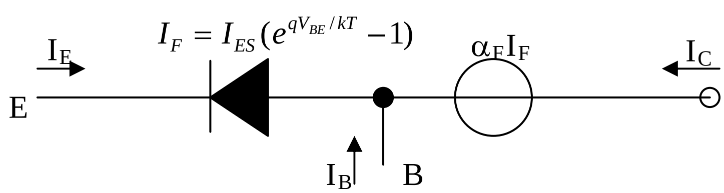
(d) $I_E = I_{nE} + I_{pE} =$

$$= Aq[D_n \frac{n'(x=1\mu m) - n'(x=0\mu m)}{W_E} + D_p \frac{p'(x=1\mu m) - p'(x=2\mu m)}{W_B}] =$$
$$= 10^{-5} \times 1.6\times10^{-19} \times \left[30\left(\frac{2\times10^{14}}{1\times10^{-4}}\right) + 10\left(\frac{6\times10^{14}}{1\times10^{-4}}\right)\right] mA = 0.192\ mA.$$

(e) $\beta = \dfrac{D_B W_E N_E}{D_E W_B N_B} = \dfrac{10\times10^{-4}\times N_E}{30\times10^{-4}\times N_B} = \dfrac{1}{3}\times1\times5 = \dfrac{5}{3} = 1.7$ (Not much gain.)

8.9 If the NPN BJT is biased at the boundary between active mode and saturation mode, then forward-biased emitter-base junction (V_{BE}>0) and unbiased collector-base junction (V_{BC}=0). So I_R=0.

(a) At the given operating point, we can simplify the Ebers-Moll model as follows:



(b) Since $I_C + I_B + I_F = 0 \Rightarrow I_B = I_F(1-\alpha_F)$

$\Rightarrow I_B = I_{ES}(e^{qV_{BE}} - 1)(1-\alpha_F)$

$\Rightarrow V_{BE} = \dfrac{kT}{q}\ln\left[\dfrac{I_B}{I_{ES}(1-\alpha_F)}\right]$

$\therefore V_{EC} = V_{BC} - V_{BE} = 0 - \dfrac{kT}{q}\ln\left[\dfrac{I_B}{I_{ES}(1-\alpha_F)}\right] = \dfrac{kT}{q}\ln\left[\dfrac{I_{ES}(1-\alpha_F)}{I_B}\right].$

Drift-Base Transistors

8.10 (a) I_B is independent of changes in base parameters. I_C is dependent on base parameters. You should convince yourself that this is the case by referring to the current equations in the reader. If a SiGe base is used, I_c increases as a result of the base bandgap narrowing. Coupled with a graded base that shortens base transport time, SiGe-base BJTs are a simple and attractive alternative to conventional Si-base BJTs.

* This solution ignores the case of increased hole barrier between base and emitter. If you wish to include this effect, then I_B(SiGe) is increased by $e^{\Delta Eg/kT}$ over I_B (Si)

(b) In this case, n_{iB}(SiGe) varies along the base region. This requires that we do an integration to find J_{c0}(SiGe).

$$J_{c0}(\text{SiGe}) = \frac{qD_B n_i^2}{N_B} \frac{1}{\int_0^{W_B} \exp\left(\frac{-\Delta E_{g,SiGe}}{kT}\frac{x}{W_B}\right)dx}$$

$$= \frac{qD_B n_i^2}{N_B W_B} \frac{\Delta E_{g,SiGe}/kT}{1-\exp(-\Delta E_{g,SiGe}/kT)}.$$

Divide J_{c0}(SiGe) by J_{c0}(Si) and you get the following:

$$\beta(\text{SiGe})/\beta(\text{Si}) = \frac{\Delta E_{g,SiGe}/kT}{1-\exp(-\Delta E_{g,SiGe}/kT)} = 4.$$

8.11 (a) Find where $N_{dE}(x) = N_{aB}(x)$ to obtain the first junction.

$$10^{20} e^{-\frac{x}{0.106}} = 4\times10^{18} e^{-\frac{x}{0.19}} \quad \Rightarrow \quad x_1 = 0.77\,\mu m.$$

To obtain the second junction, equate $N_{aB}(x)$ to the background concentration.

$$4\times10^{18} e^{-\frac{x}{0.19}} = 5\times10^{19} \quad \Rightarrow \quad x_2 = 1.27\ \mu m.$$

Therefore, the base width is x_2-x_1 = 0.5 μm.

(Please note that the depletion widths have been ignored in this case. In general, you must subtract the depletion region widths in the base in both the junctions from the metallurgical base width.)

(b) Base Gummel Number:

$$\int_{x_1}^{x_2} N_{aB}(x)dx = -\left(4\times10^{18}\right)\left(0.19\times10^{-4}\right)\left[\exp\left(-x/0.19\right)\right]_{x_1}^{x_2} cm^{-2} = 1.23\times10^{12}\,cm^{-2}.$$

From Eq. 8.2.12, the base Gummel number is the number above divided by the electron diffusivity in the base, which we shall assume to be around 30 cm^2/s. The result is 4×10^{10} s-cm^{-4}

Emitter Gummel Number:

$$\int_0^{x_1} N_{dE}(x)dx = -\left(10^{20}\right)\left(0.106\times10^{-4}\right)\left[\exp\left(-x/_{0.106}\right)\right]_0^{x_1} cm^{-2} = 1.06\times10^{15}\, cm^{-2}.$$

From Eq. 8.3.2, the emitter Gummel number is the number above divided by the hole diffusivity in the base, which we shall assume to be around 2 cm^2/s. The result is 2×10^{14} s-cm^{-4}

(c) Since the doping level is not constant, we use the average doping densities to estimate the diffusivities.

Average base doping density:

$$\frac{GN_B}{W_B} = \frac{1.23\times10^{12}}{0.5\times10^{4}} cm^{-3} = 2.46\times10^{16}\, cm^{-3}.$$

Average emitter doping density:

$$\frac{GN_E}{x_E} = \frac{1.06\times10^{19}}{0.77\times10^{4}} = 1.38\times10^{19}\, cm^{-3}.$$

Average electron diffusivity in the base:

$$D'_{nB} = \mu_n\left(2.46\times10^{16}\right)kT/q = \left(1150\times0.026\right)cm^2s^{-1} = 29.9\, cm^2s^{-1}.$$

Average hole diffusivity in the emitter:

$$D'_{pE} = \mu_p\left(1.38\times10^{19}\right)kT/q = \left(70\times0.026\right)cm^2s^{-1} = 1.82\, cm^2s^{-1}.$$

$$\gamma = \frac{1}{1+\dfrac{GN_B D'_{pE}}{GN_E D'_{nB}}} = \frac{1}{1+\dfrac{1.23\times10^{12}\times1.82}{1.06\times10^{15}\times29.9}} = 0.99993.$$

(d) Diffusion current:

$$J_{diff} = -q\mu\upsilon = -qD\, dp/_{dx} \quad\Rightarrow\quad \upsilon = -\frac{D}{n}\frac{dn}{dx} = -D\frac{d\ln n}{dx}.$$

$$p \approx N_B = 4\times10^{18} e^{-x/\lambda}, \quad \lambda = 0.19\mu m \quad \Rightarrow \quad \upsilon = +\frac{D}{\lambda} \qquad (+x \quad direction).$$

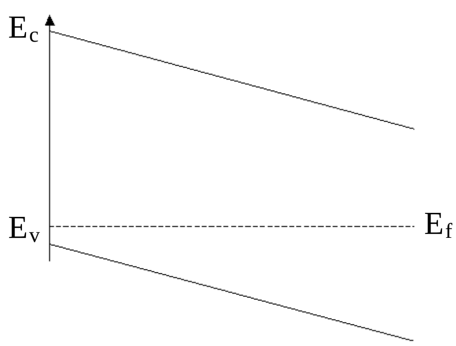
Drift current:

$$J_{drift} = q\mu_p pE = qp\upsilon = J_{diff}$$
$$\mu_p = \left(\frac{q}{kT}\right)D \quad and \quad \upsilon = +D/\lambda = \mu_p E = E\left(\frac{q}{kT}\right)D$$

Therefore,

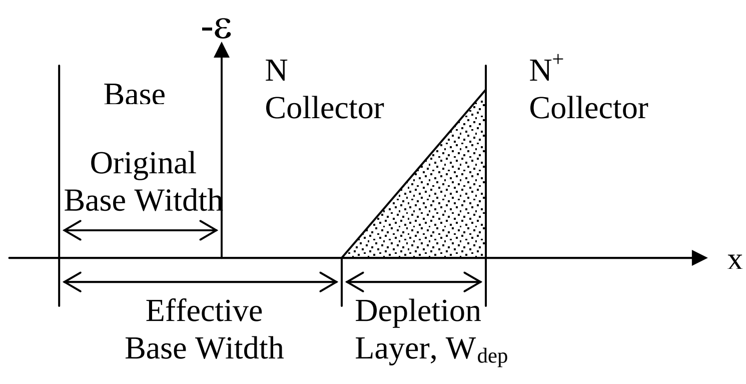
$$E_{bi} = \frac{kt}{q\lambda} = \frac{0.06\,V}{0.19\ \mu m} = 3.16\times10^5\ V/m.$$

Note: E_{bi} should be "-x" direction so that diffusion current and drift current will balance.



Kirk Effect

8.12



Clearly, $W_{B\_Effective} = W_{B\_Original} + (W_C - W_{dep})$. $W_{B\_Original}$ and W_C are assumed to be known. So, in order to find $W_{B\_Effective}$, we need to calculate W_{dep}.

$\rho = qN_C - \frac{I_C}{A_E v_{sat}}$ where v_{sat} is the saturation velocity. The length of the depletion region becomes

$$W_{dep} = \sqrt{\frac{2\varepsilon_s (V_{BC} + \phi_i)}{\frac{I_C}{A_E v_{sat}} - qN_C}}.$$

Therefore,

$$W_{B\_Effective} = W_{B\_Original} + \left(W_C - \sqrt{\frac{2\varepsilon_s (V_{BC} + \phi_i)}{\frac{I_C}{A_E v_{sat}} - qN_C}} \right).$$

Charge Control Model

8.13 The equation describing the system is

$$\frac{dQ_F(t)}{dt} = I_B(t) - \frac{Q_F(t)}{\tau_F \beta_F}.$$

Since $I_B(t) = 0$ for $t<0$ and $I_B(t) = I_{B0}$ for $t \geq 0$, the equation becomes

$$\frac{dQ_F(t)}{dt} = \left(-\frac{1}{\tau_F \beta_F} \right) Q_F(t) + I_{B0}$$

with the initial condition $Q_F(0) = 0$. Solving this equation yields

$$Q_F(t) = \tau_F \beta_F I_{B0} \left(1 - e^{-t/\tau_F \beta_F} \right).$$

Hence,

$$I_C(t) = Q_F(t)/\tau_F = \beta_F I_{B0} \left(1 - e^{-t/\tau_F \beta_F} \right).$$

8.14 $\frac{dQ_B}{dt} = i_B - \frac{Q_B}{\beta \tau_F} \Rightarrow Q_B = A + Be^{-t/\beta \tau_F}$

Boundary Conditions: F

$$Q_B(\infty) = A = \beta \tau_F i_{B2}, \quad Q_B(0^-) = \beta \tau_F i_{B1} = \beta \tau_F i_{B2} + B$$

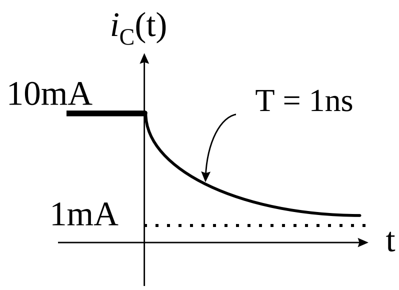
Hence, B= $\beta \tau_F (i_{B1} - i_{B2})$

Therefore,

$$Q_B = \beta\tau_F i_{B2} + \beta\tau_F \left(i_{B1} - i_{B2}\right)e^{-t/\beta\tau_F}$$
$$i_C = \frac{Q_B}{\tau_F} = \beta i_{B2} + \beta \left(i_{B1} - i_{B2}\right)e^{-t/\beta\tau_F},$$
$$\beta = \frac{\alpha_F}{1-\alpha_F} = 100, \quad \beta\tau_F = 1ns(Chracteristic\ time\ T)$$

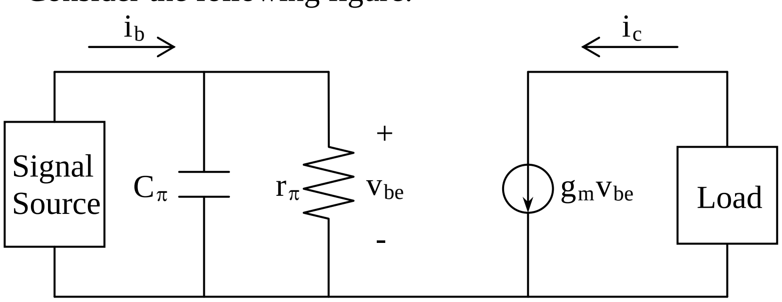
And,

$$i_C = 1\ mA + (9\ mA)e^{-t/1ns} \quad \text{for } t \ge 0, \text{ and}$$
$$i_C = 10\ mA \quad \text{for } t < 0.$$



Cutoff Frequency

8.15 Consider the following figure:



i_b and i_c are given by

$$i_b = v_{be} / input\ impedance = v_{be} \times input\ admittance = v_{be}\left(1/r_\pi + j\omega C_\pi\right)$$
$$i_c = g_m v_{be}.$$

The gain is

$$\beta(\omega) \equiv \left|\frac{i_c}{i_b}\right| = \frac{g_m}{\left|1/r_\pi + j\omega C_\pi\right|} = \frac{1}{\left|1/g_m r_\pi + j\omega\tau_\pi + j\omega C_{dBE}/g_m\right|}$$
$$= \frac{1}{\left|1/\beta_F + j\omega\tau_\pi + j\omega C_{dBE} kT/qI_C\right|}.$$

If β_F >>1 so that $1/\beta_F$ becomes negligible, the equation above shows that $\beta_F(\omega) \propto 1/\omega$, and β becomes 1 at

$$f_T = \frac{1}{2\pi\left(\tau_F + C_{dBE} kT / qI_C\right)}.$$